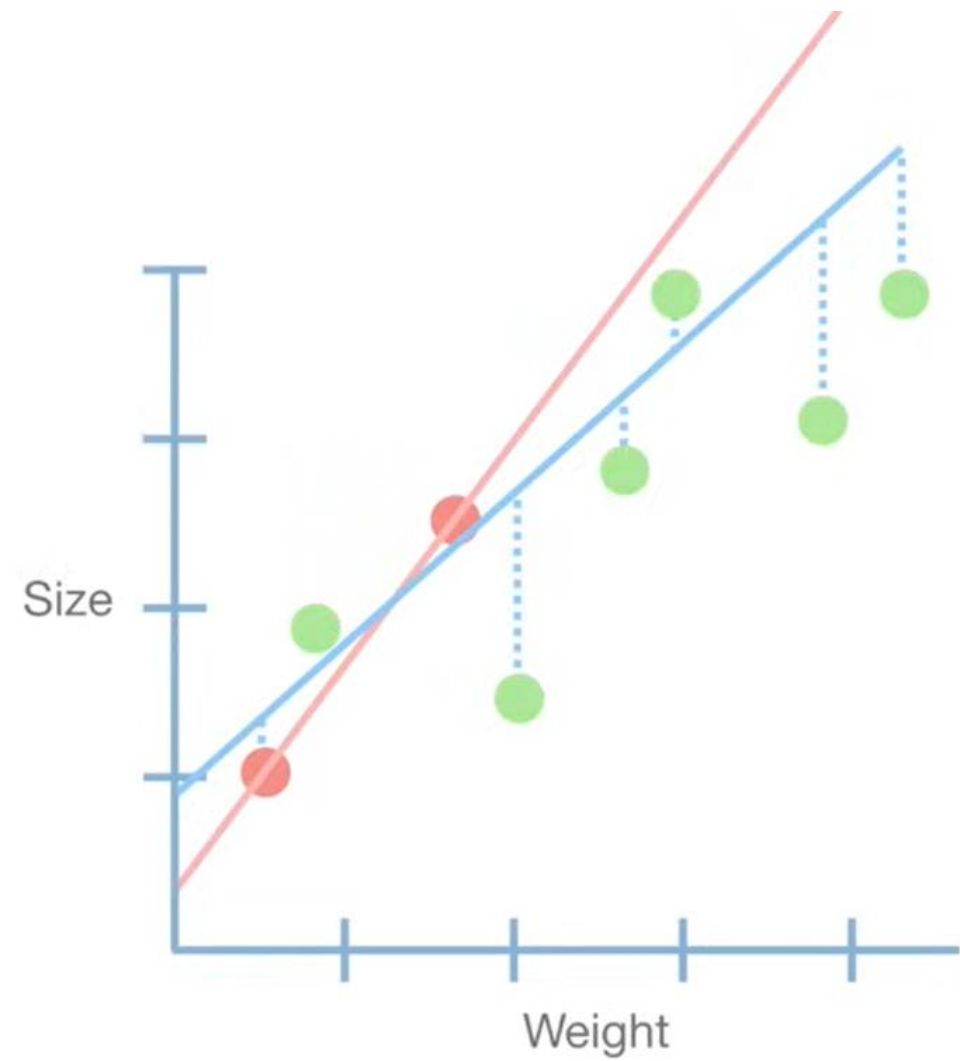


Ridge Regression

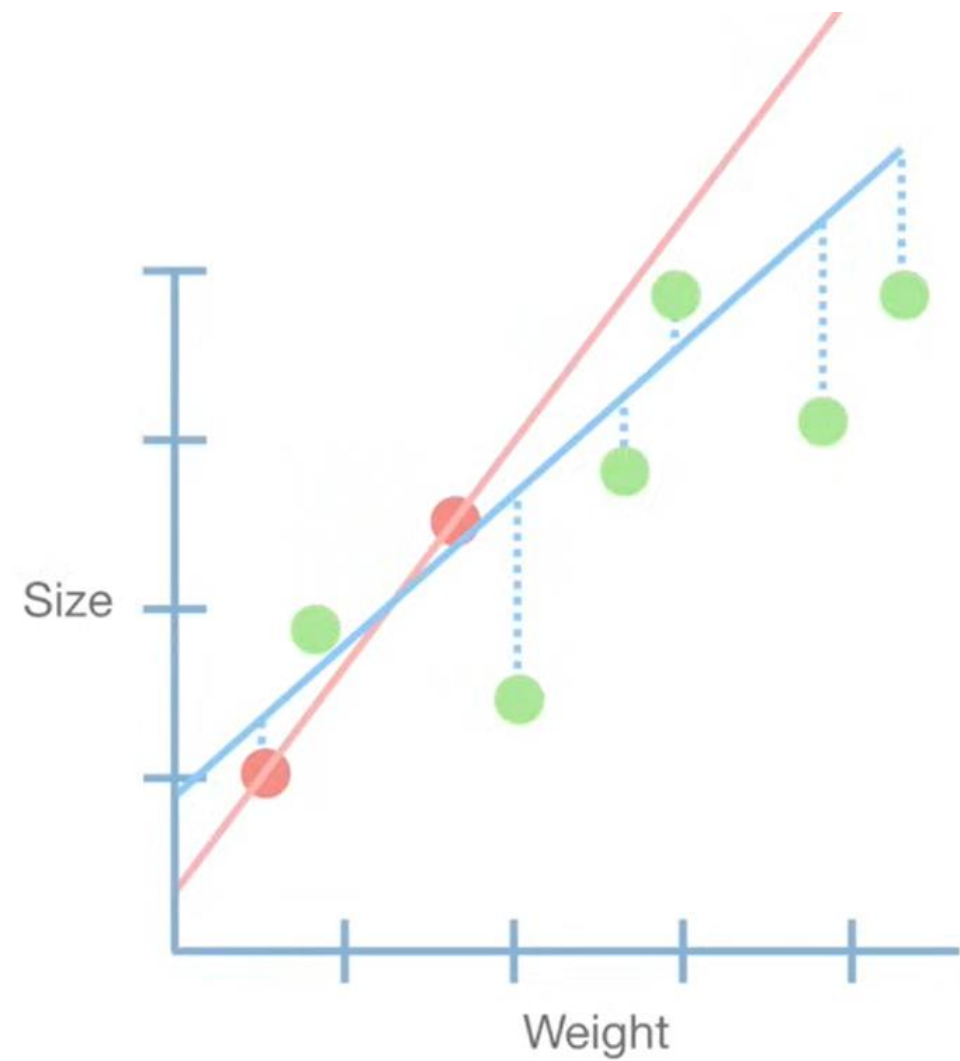
Asst. Prof. Dr. Waranyu Wongseeree

Faculty of Digital Technology

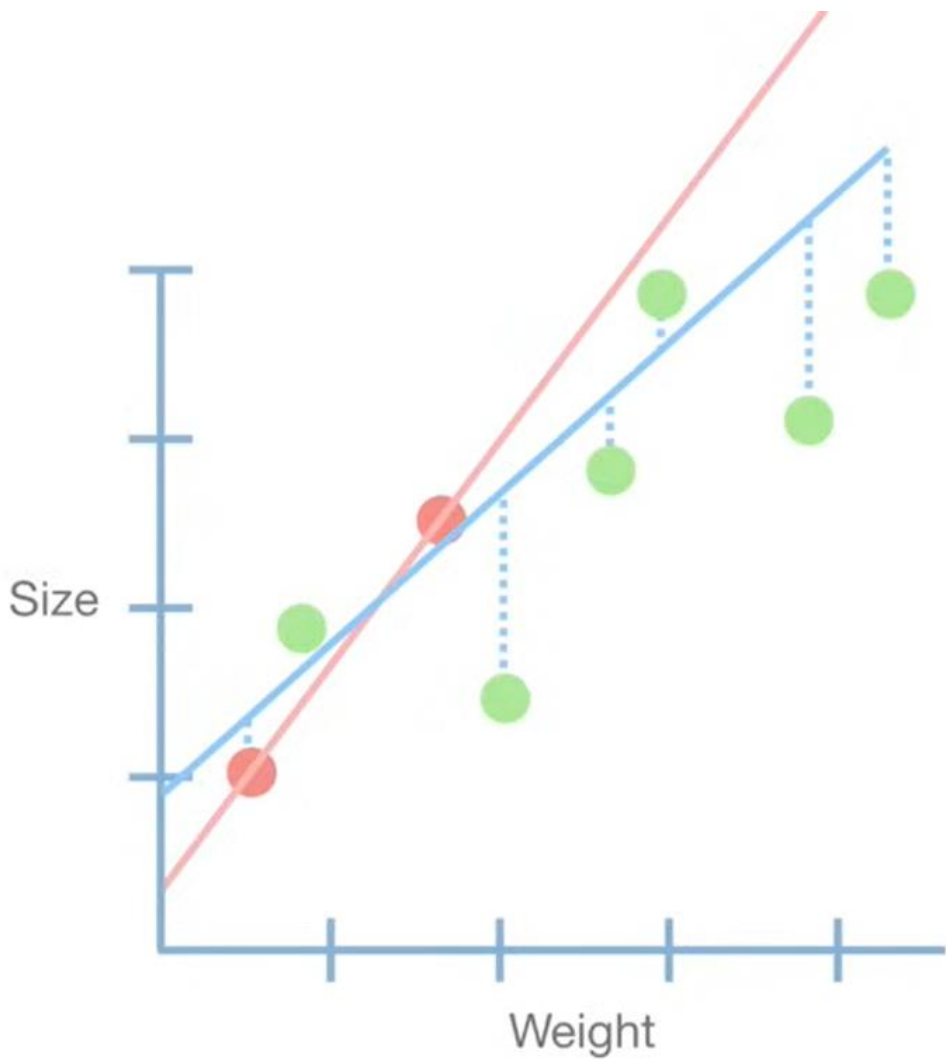
Chitralada Technology Institute



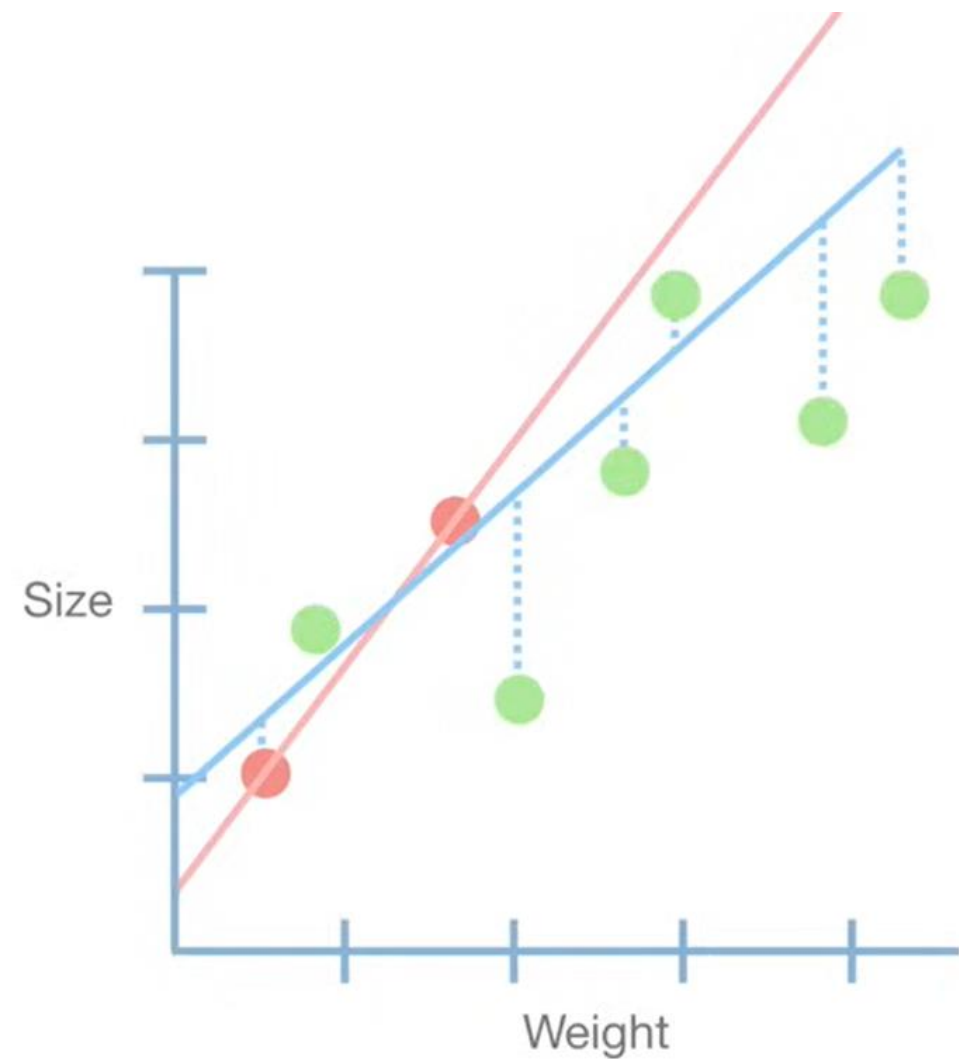
1. Look at a simple example that shows the main ideas behind Ridge Regression...



1. Look at a simple example that shows the main ideas behind Ridge Regression...
2. Go into details about how Ridge Regression works...

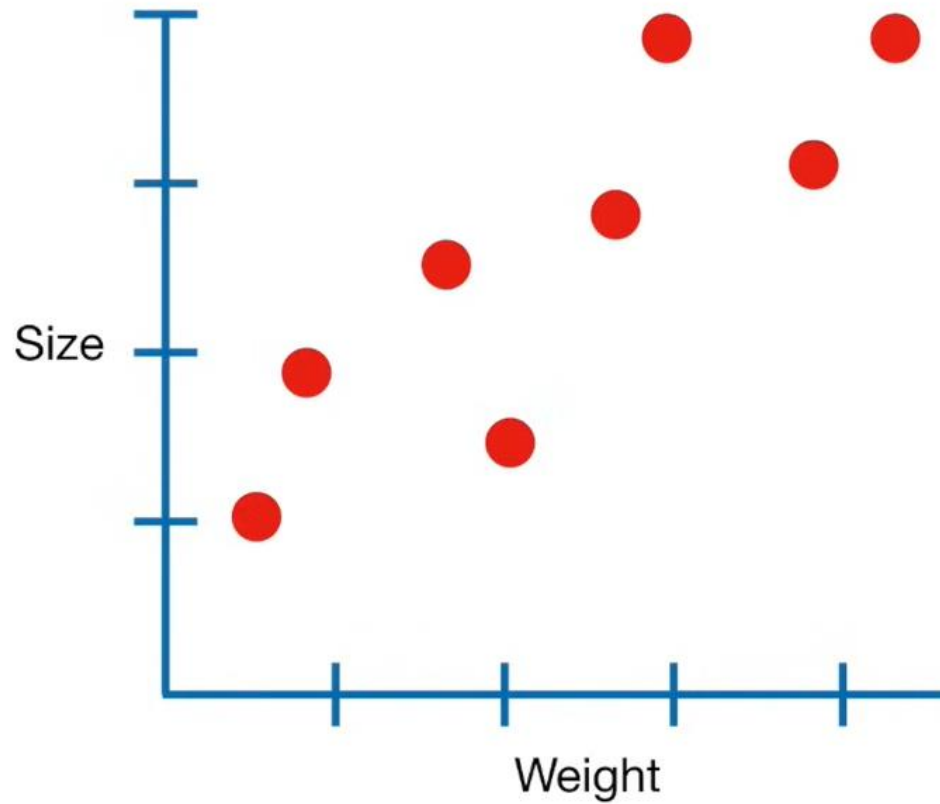


1. Look at a simple example that shows the main ideas behind Ridge Regression...
2. Go into details about how Ridge Regression works...
3. Show how Ridge Regression works in a variety of situations...

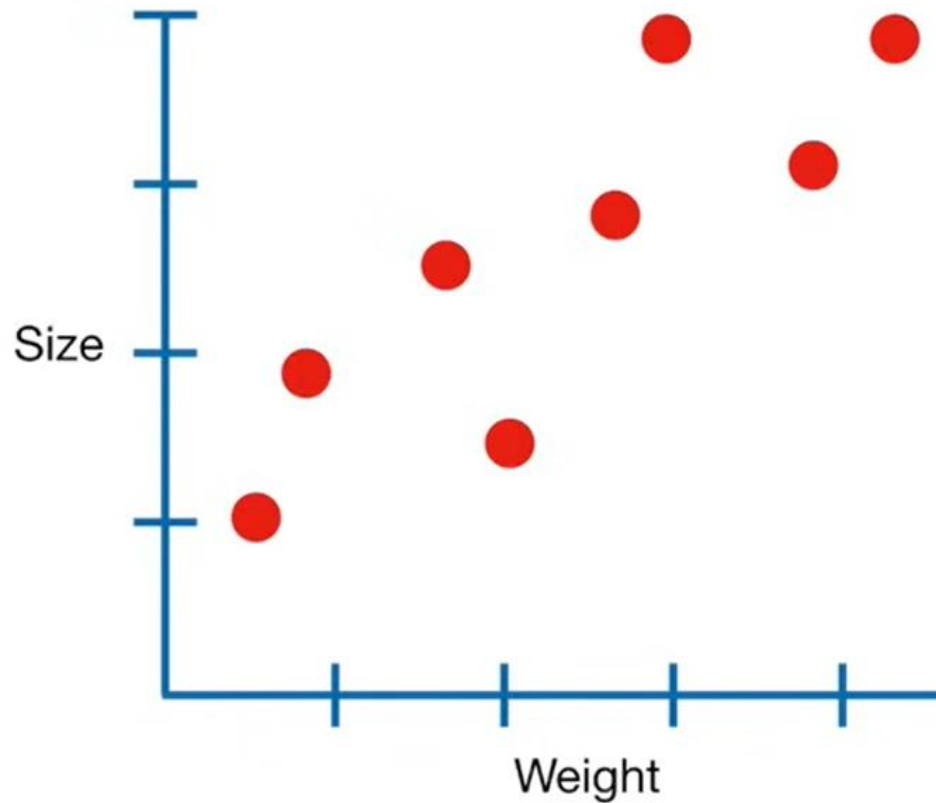


1. Look at a simple example that shows the main ideas behind Ridge Regression...
2. Go into details about how Ridge Regression works...
3. Show how Ridge Regression works in a variety of situations...
4. Lastly, we'll talk about how Ridge Regression can solve the unsolvable!!!!

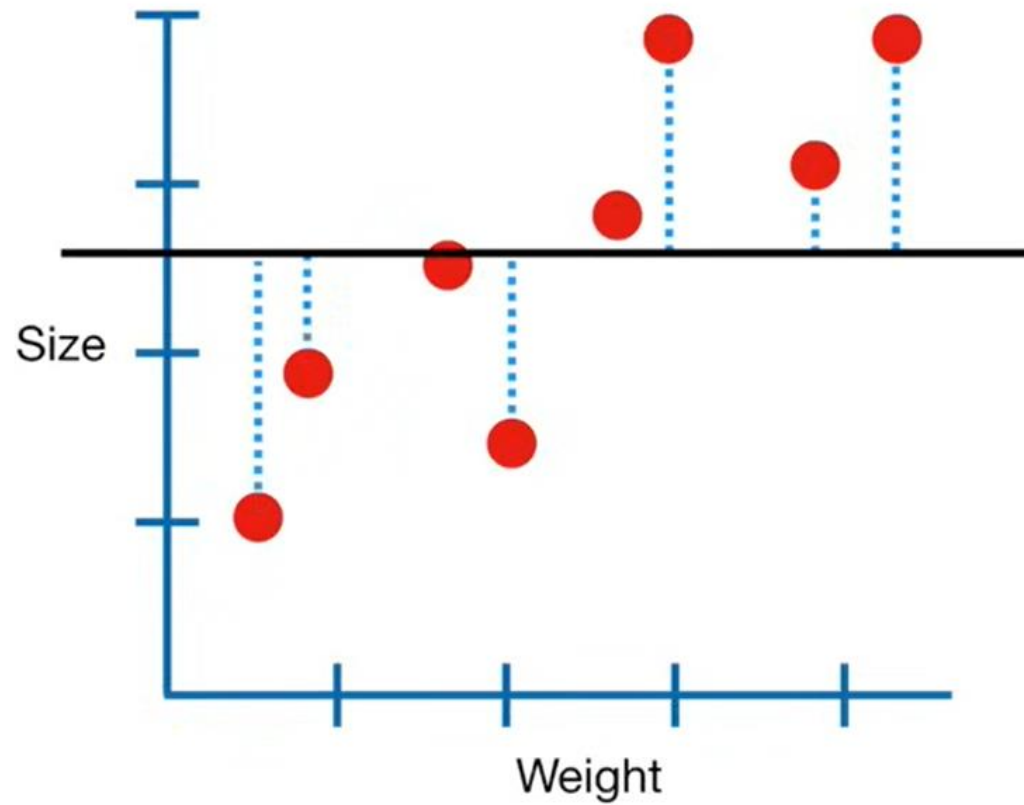
Let's start by collecting **Weight** and **Size** measurements from a bunch of mice...



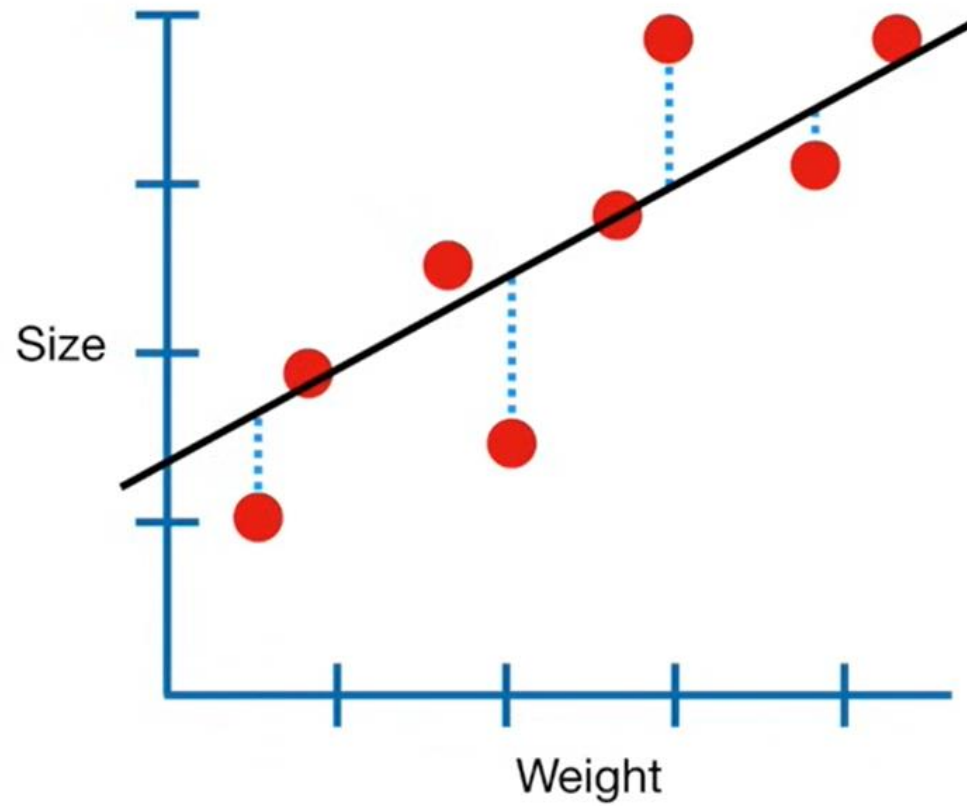
Since these data look relatively linear, we will use **Linear Regression**, AKA **Least Squares**, to model the relationship between **Weight** and **Size**.



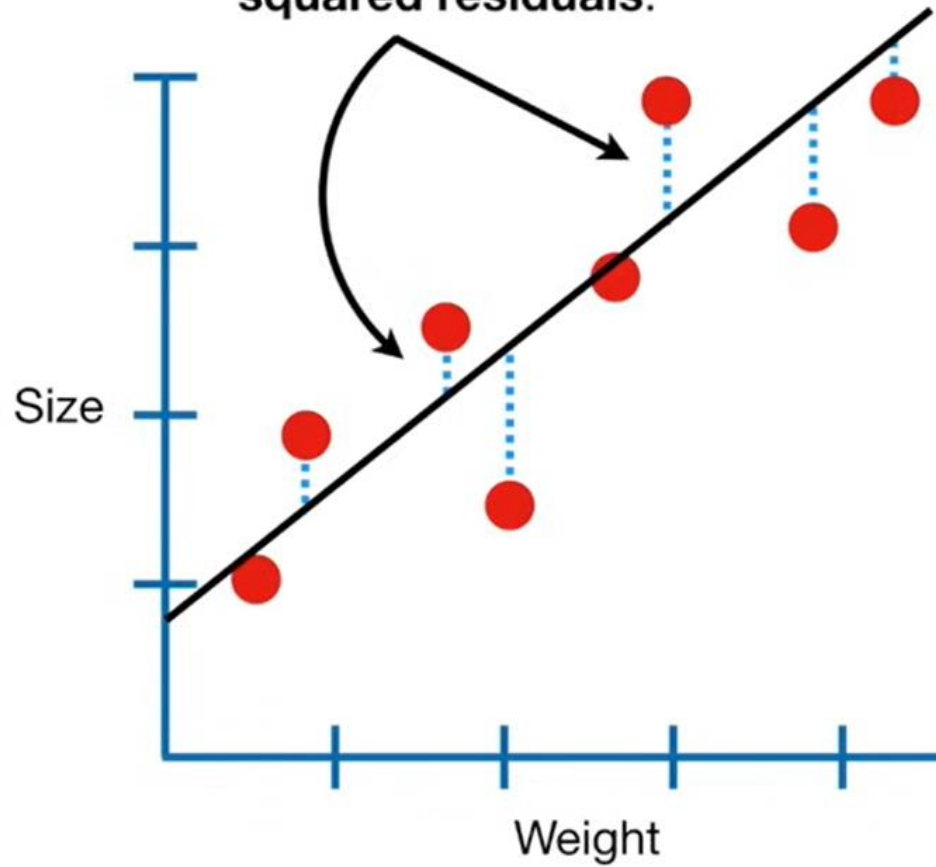
So we'll fit a line to the data using
Least Squares.



So we'll fit a line to the data using
Least Squares.

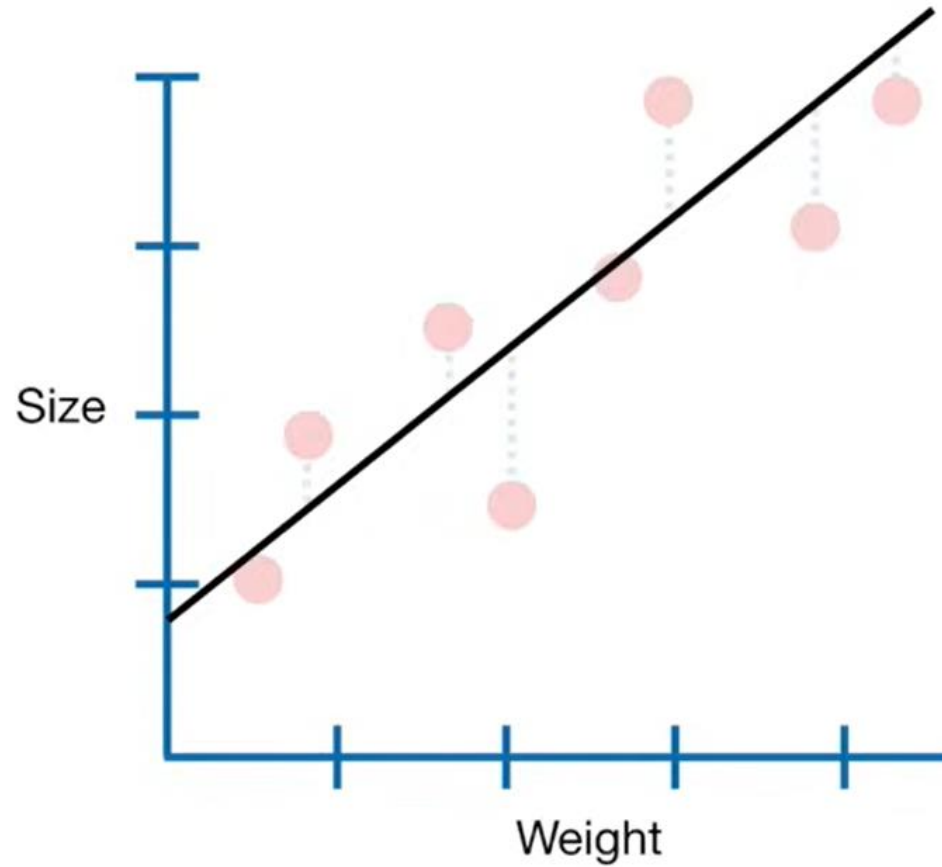


In other words, we find the line that results in the **minimum sum of squared residuals**.



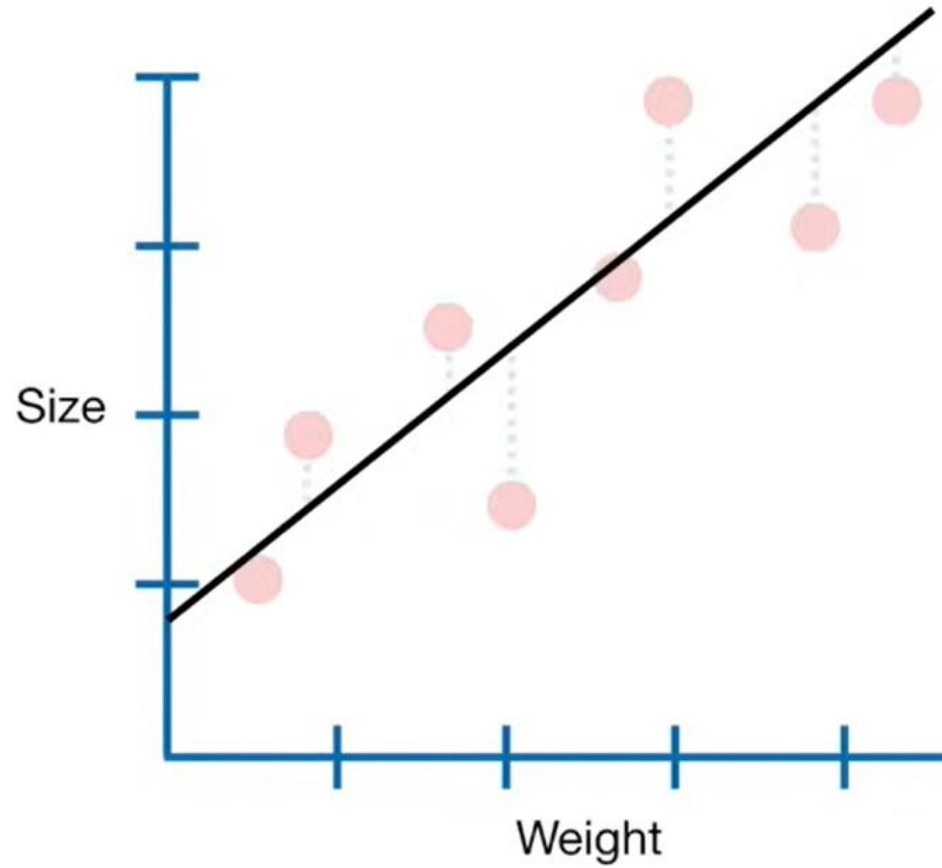
Ultimately, we end up with
this equation for the line:

$$\text{Size} = 0.9 + 0.75 \times \text{Weight}$$

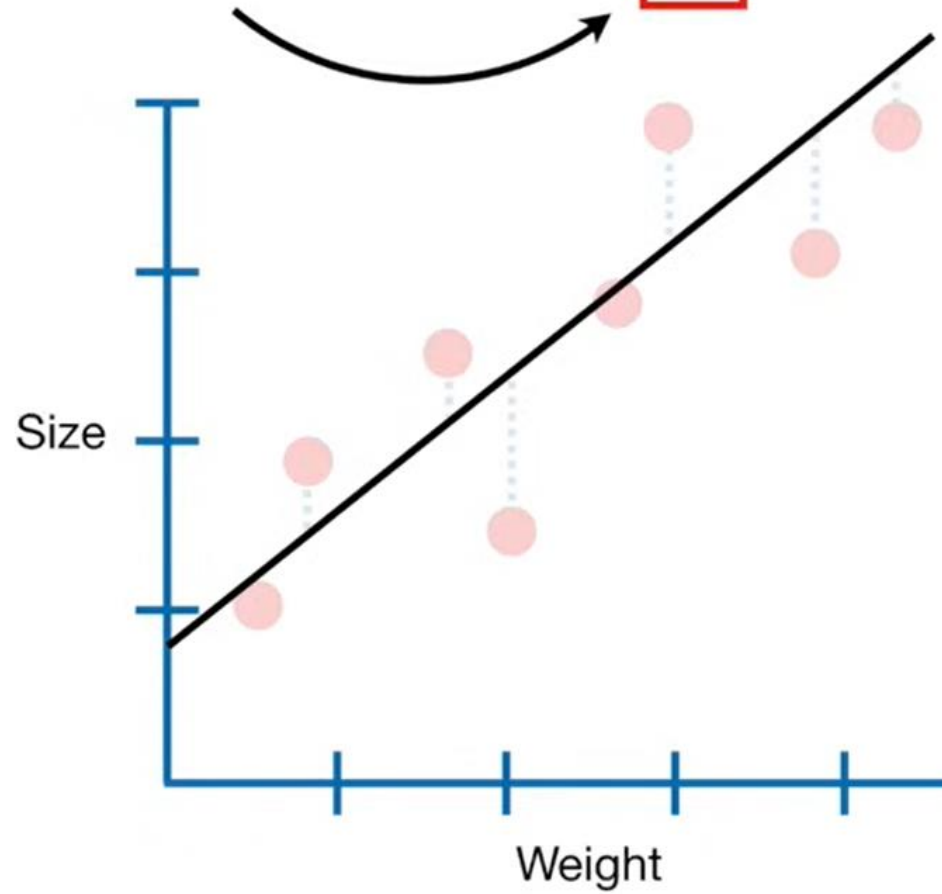


The line has two
parameters...

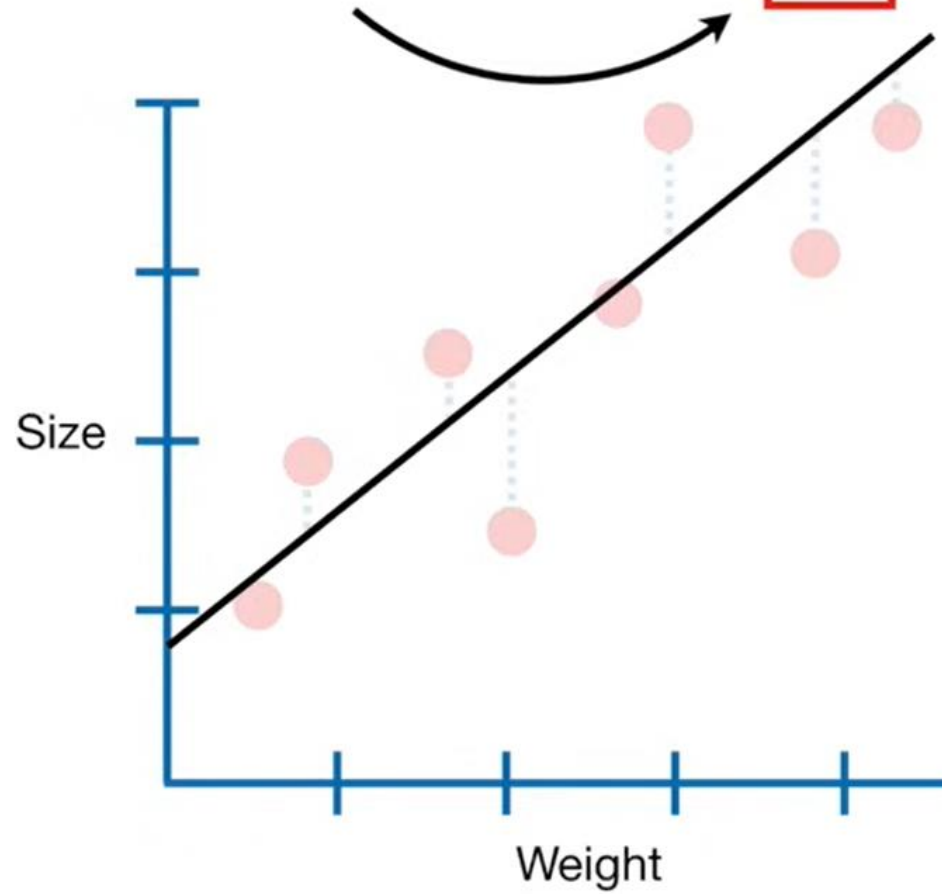
$$\text{Size} = 0.9 + 0.75 \times \text{Weight}$$



...a y-axis intercept... $\text{Size} = 0.9 + 0.75 \times \text{Weight}$

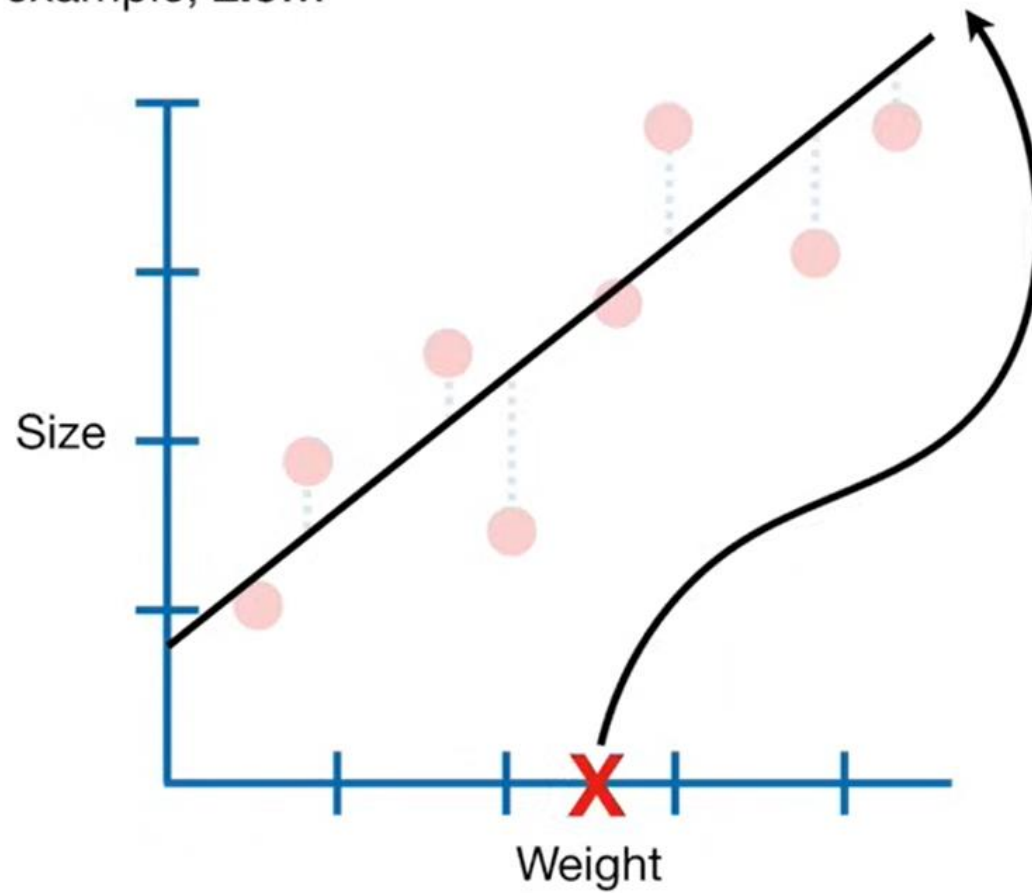


...and a **slope**. $\text{Size} = 0.9 + 0.75 \times \text{Weight}$



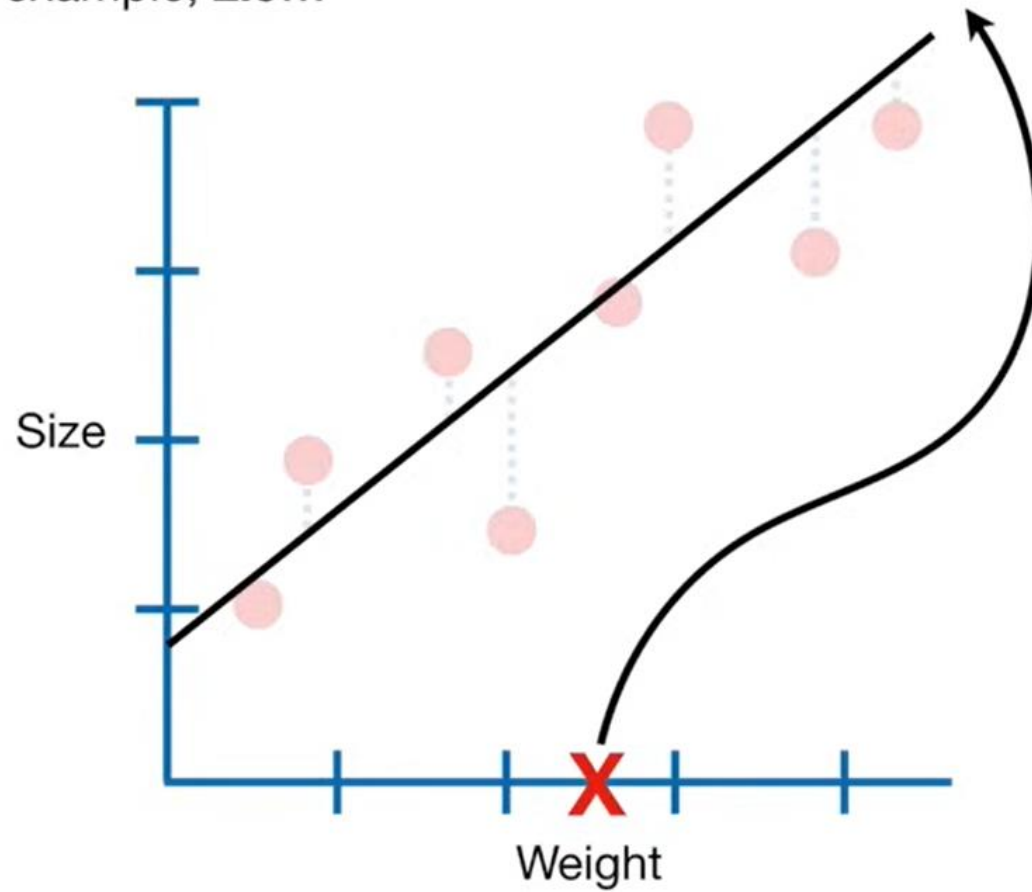
We can plug in a value for **Weight**, for example, **2.5...**

$$\text{Size} = 0.9 + 0.75 \times \text{Weight}$$



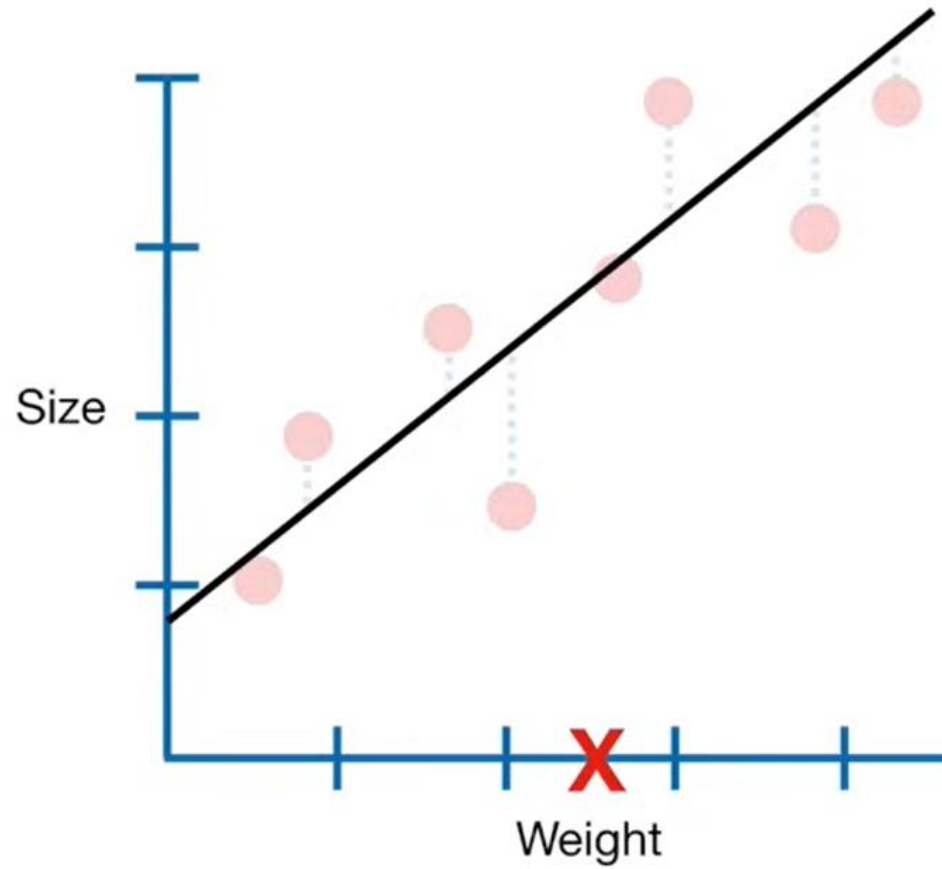
We can plug in a value for **Weight**, for example, **2.5**...

$$\text{Size} = 0.9 + 0.75 \times 2.5$$



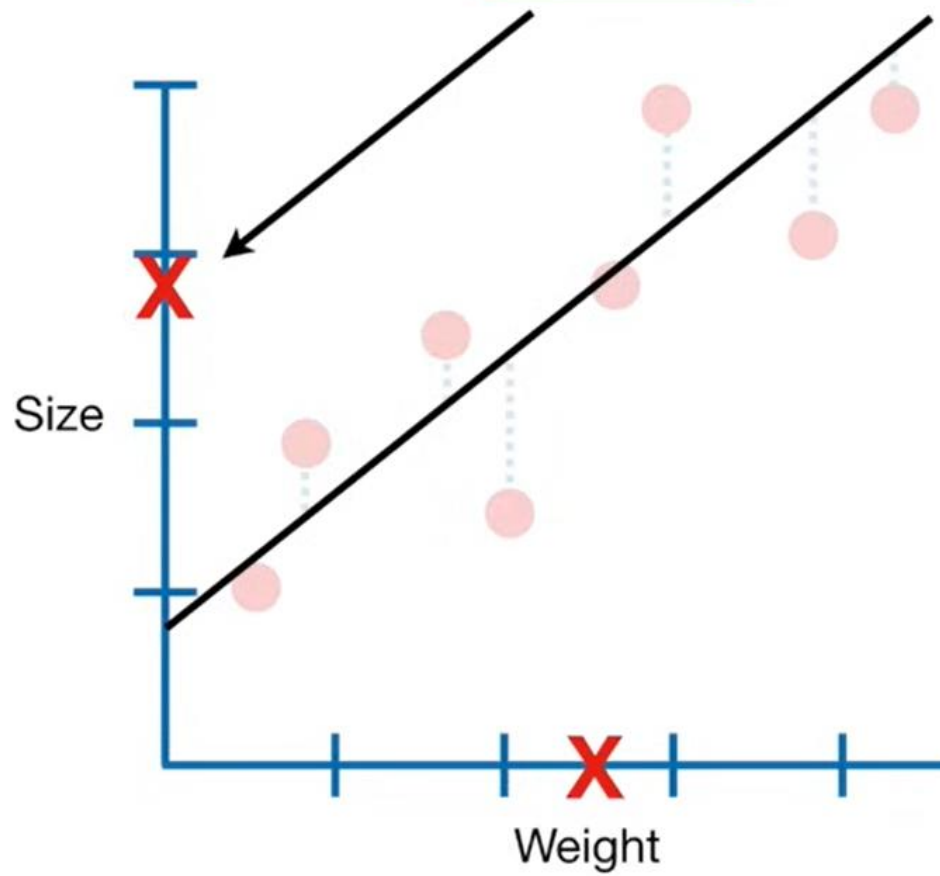
...and do the math...

$$\text{Size} = 0.9 + 1.88$$

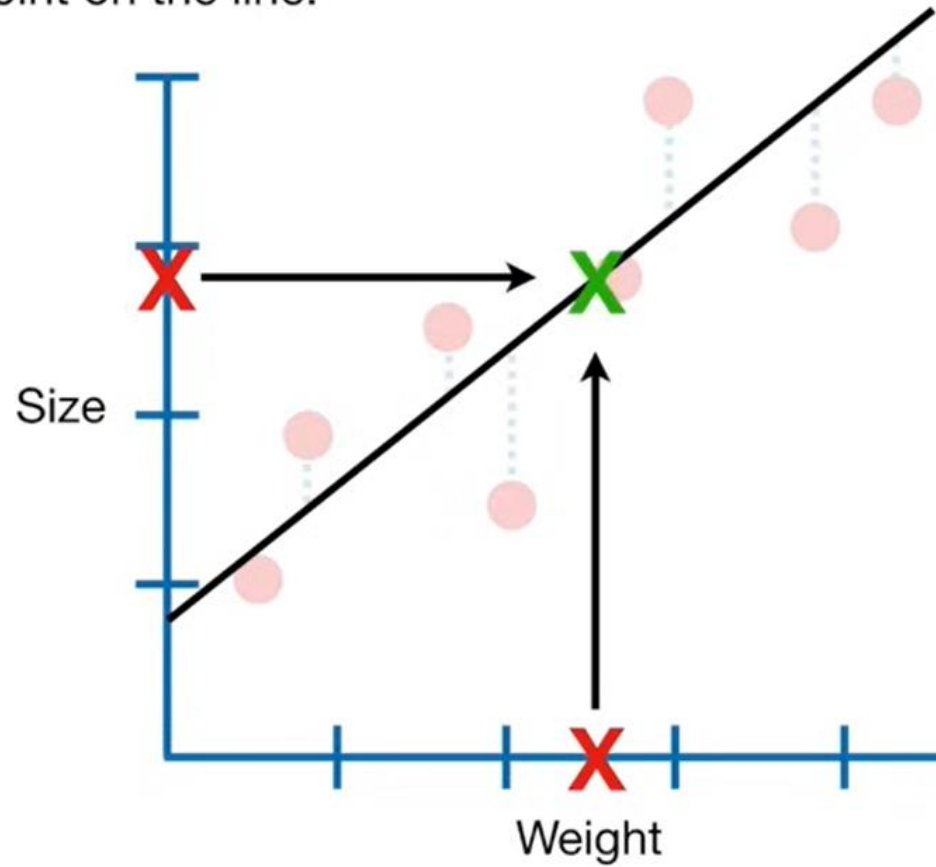


...and get a value for **Size**...

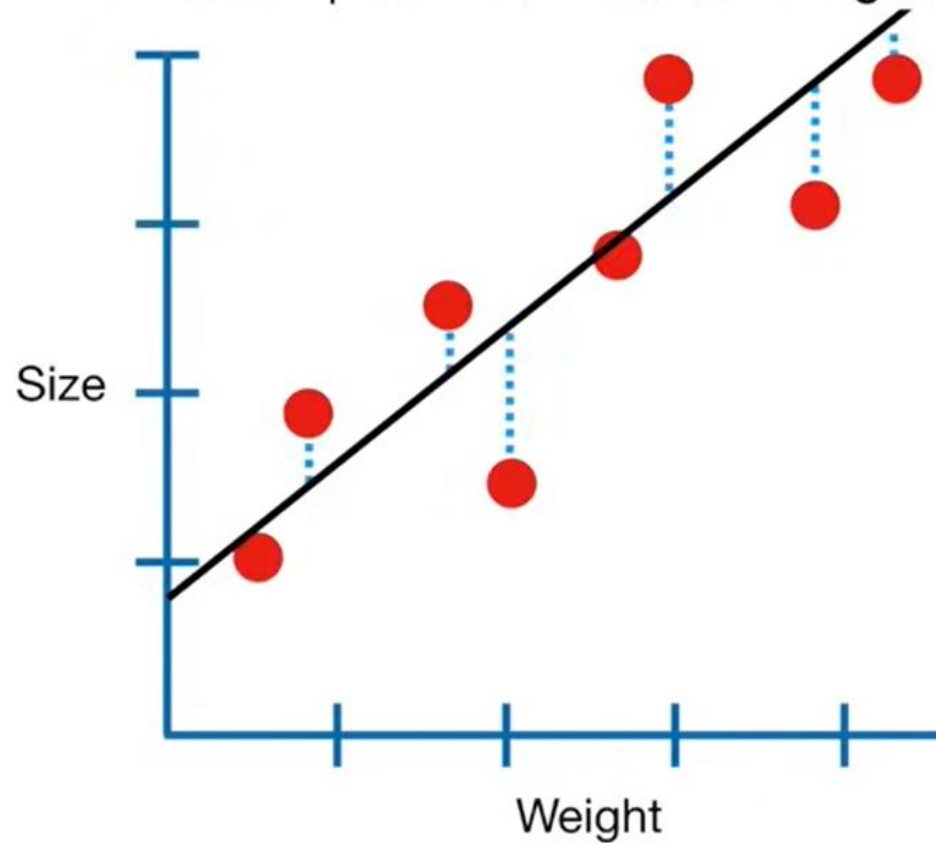
Size = 2.8



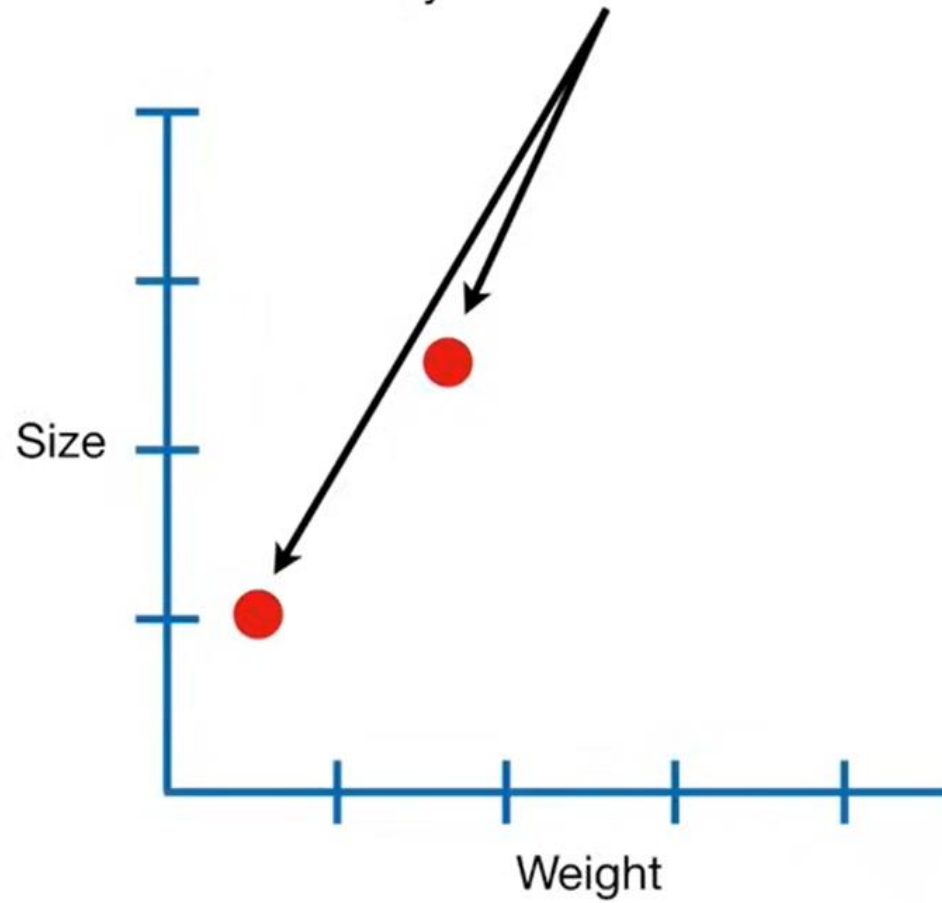
Together, the value for **Weight**,
2.5, and the value for **Size**, **2.8**,
give us a point on the line.



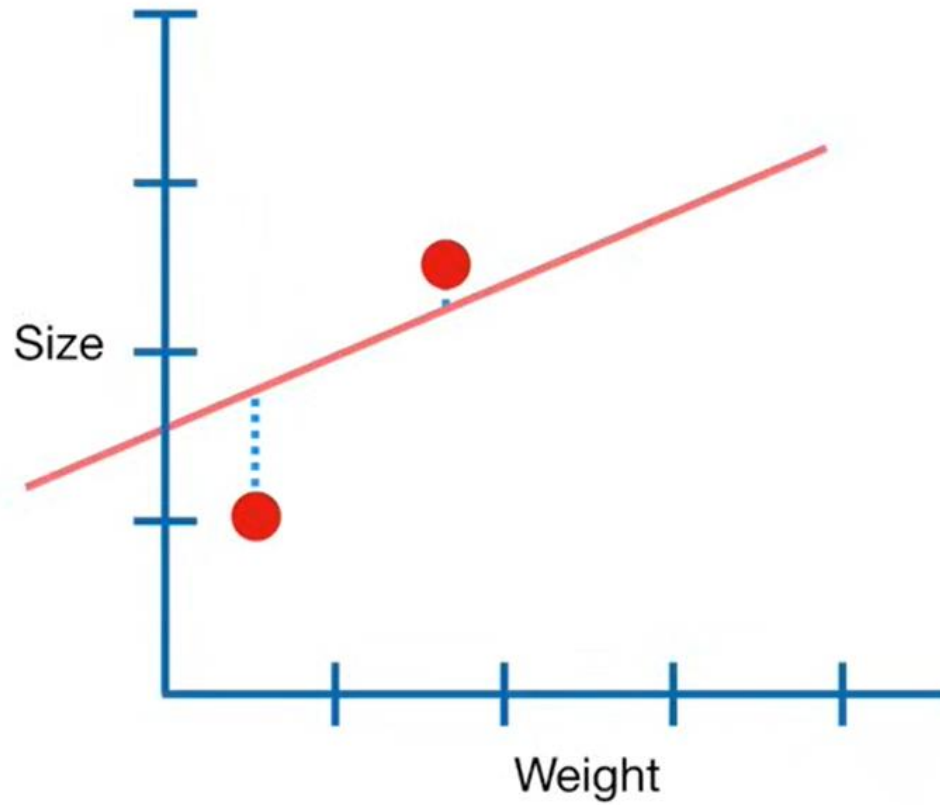
When we have a lot of measurements, we can be fairly confident that the **Least Squares** line accurately reflects the relationship between **Size** and **Weight**.



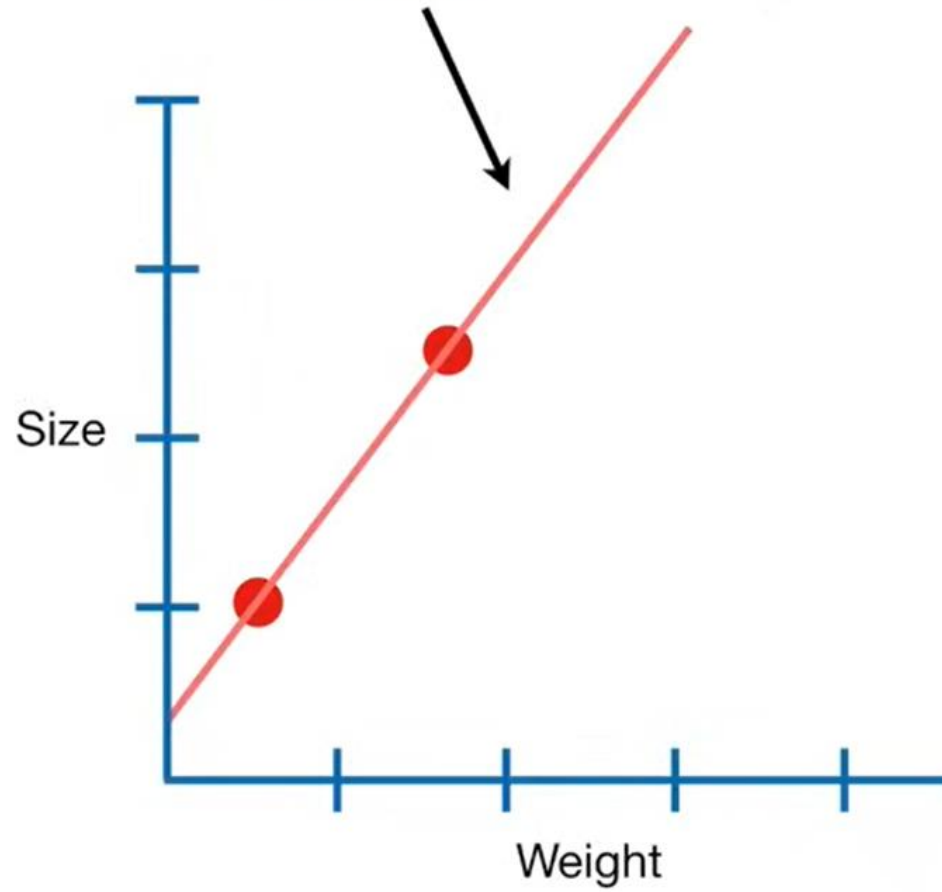
But what if we only have two measurements?

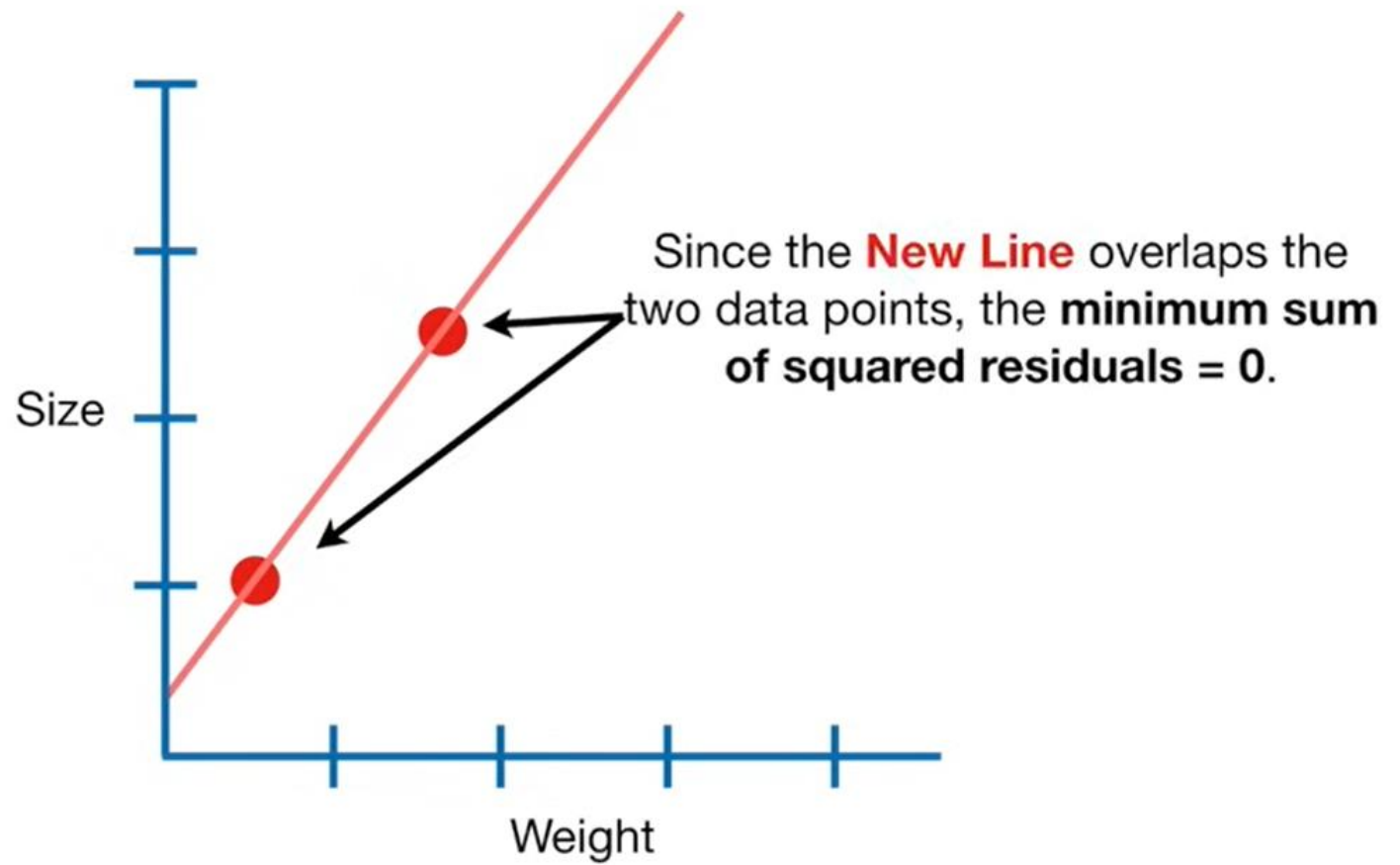


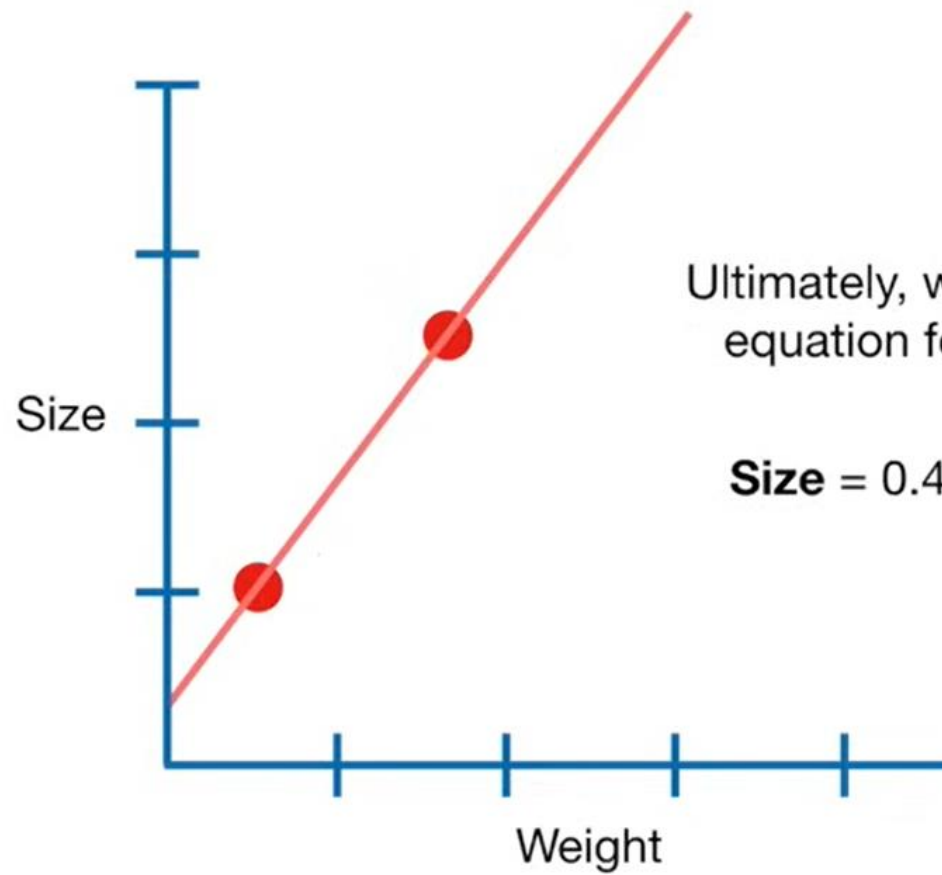
We fit a **New Line** with **Least Squares...**



We fit a **New Line** with **Least Squares**...

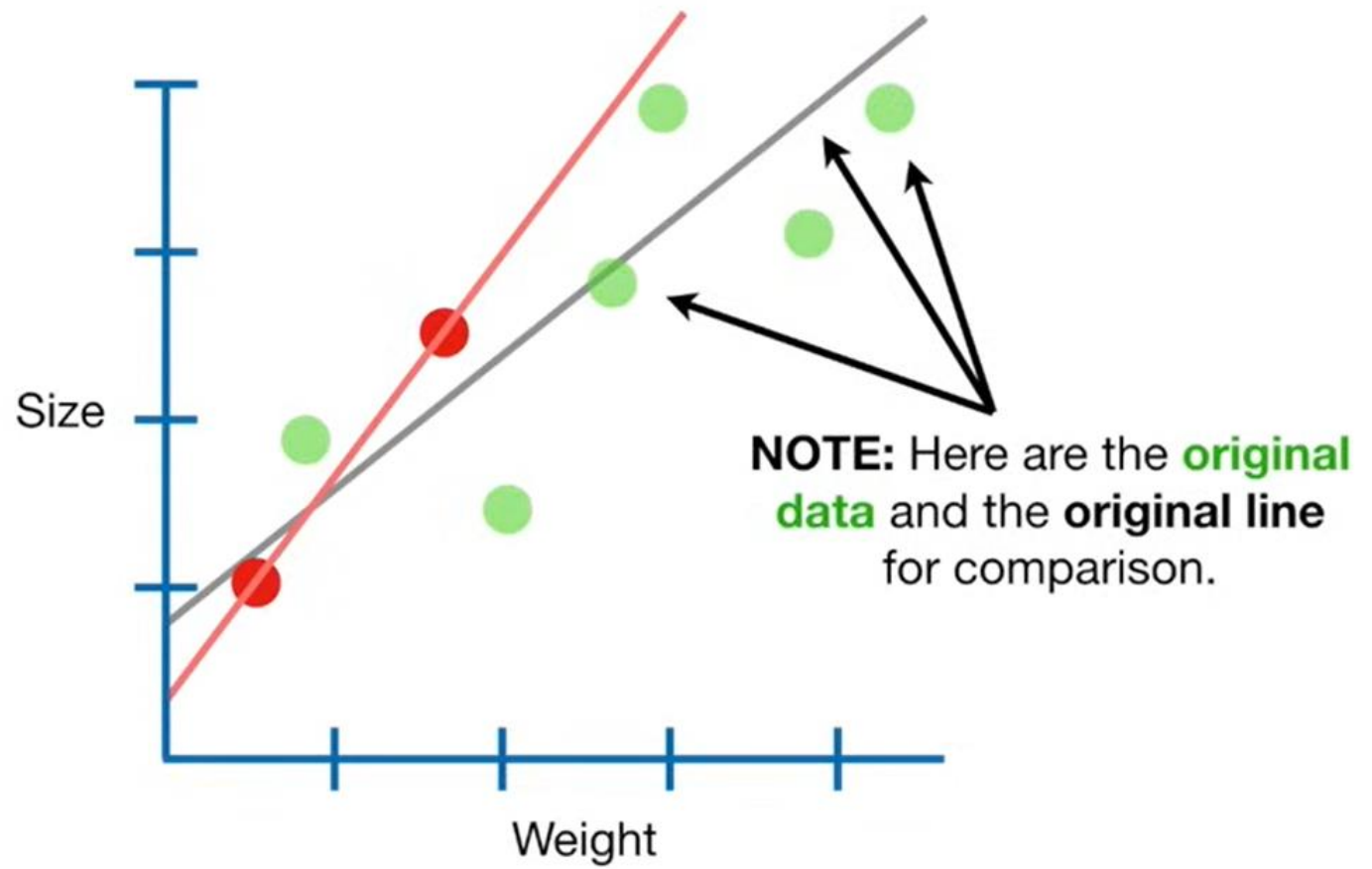




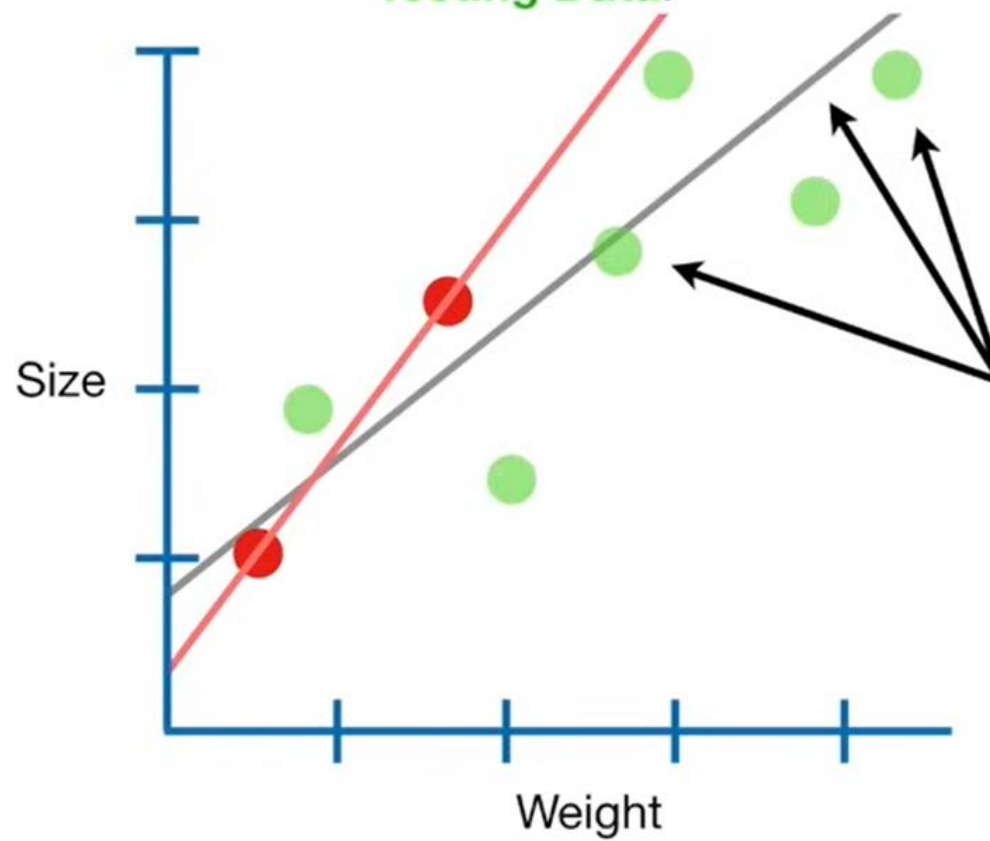


Ultimately, we end up with this equation for the **New Line**:

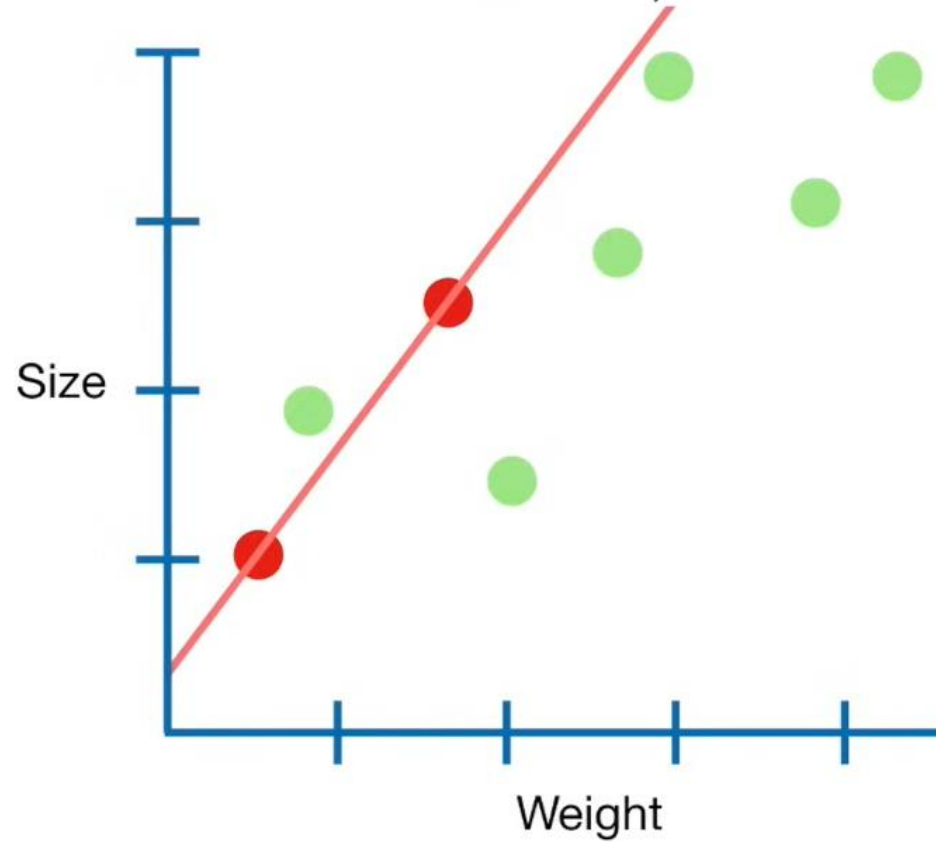
$$\mathbf{Size = 0.4 + 1.3 \times Weight}$$



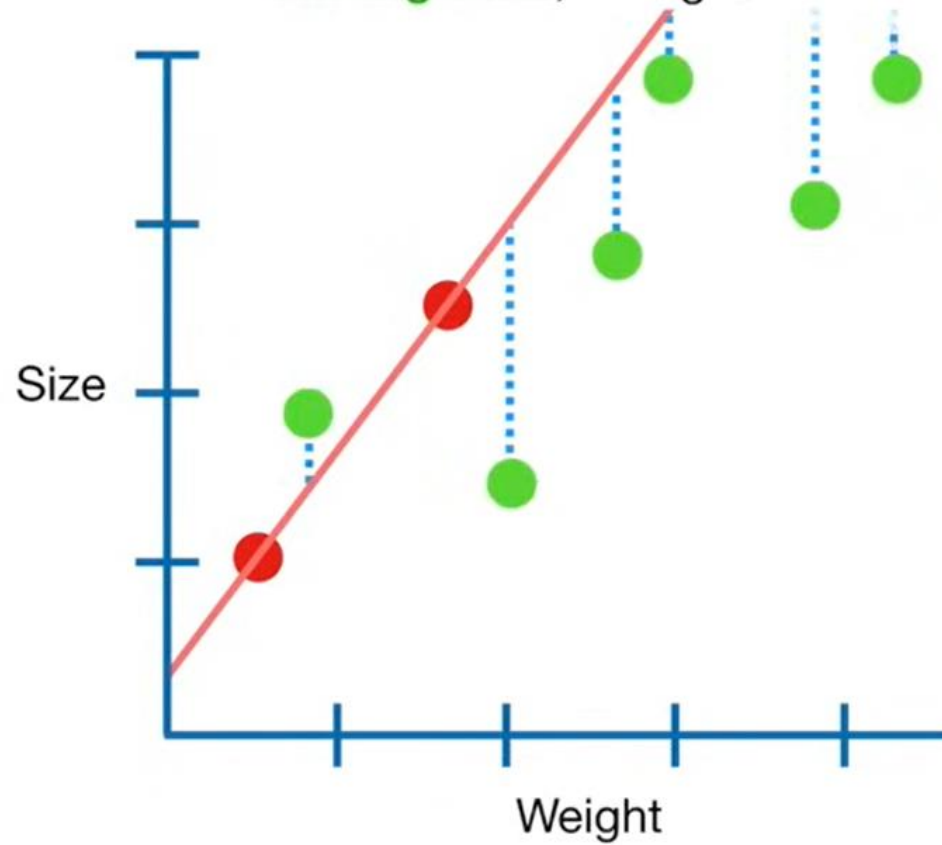
Let's call the **Two Red Dots** the **Training Data**, and the remaining **Green Dots** the **Testing Data**.



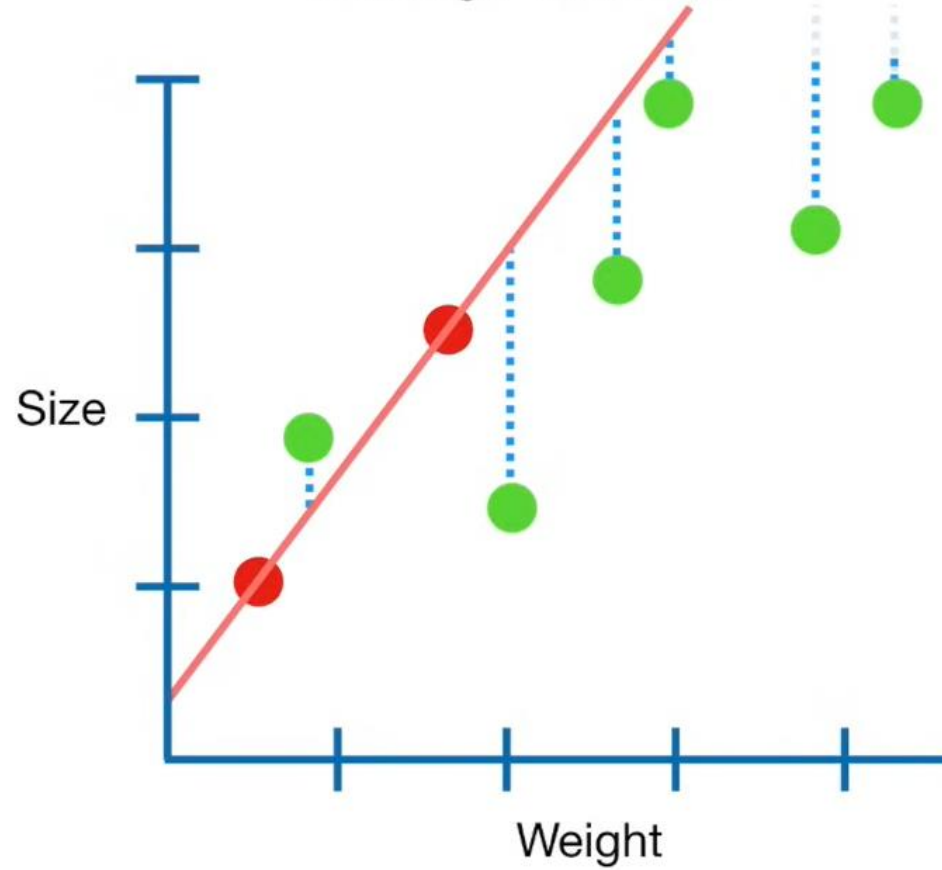
The sum of the squared residuals for just the **Two Red Points**, the **Training Data**, is small (in this case it is 0)...



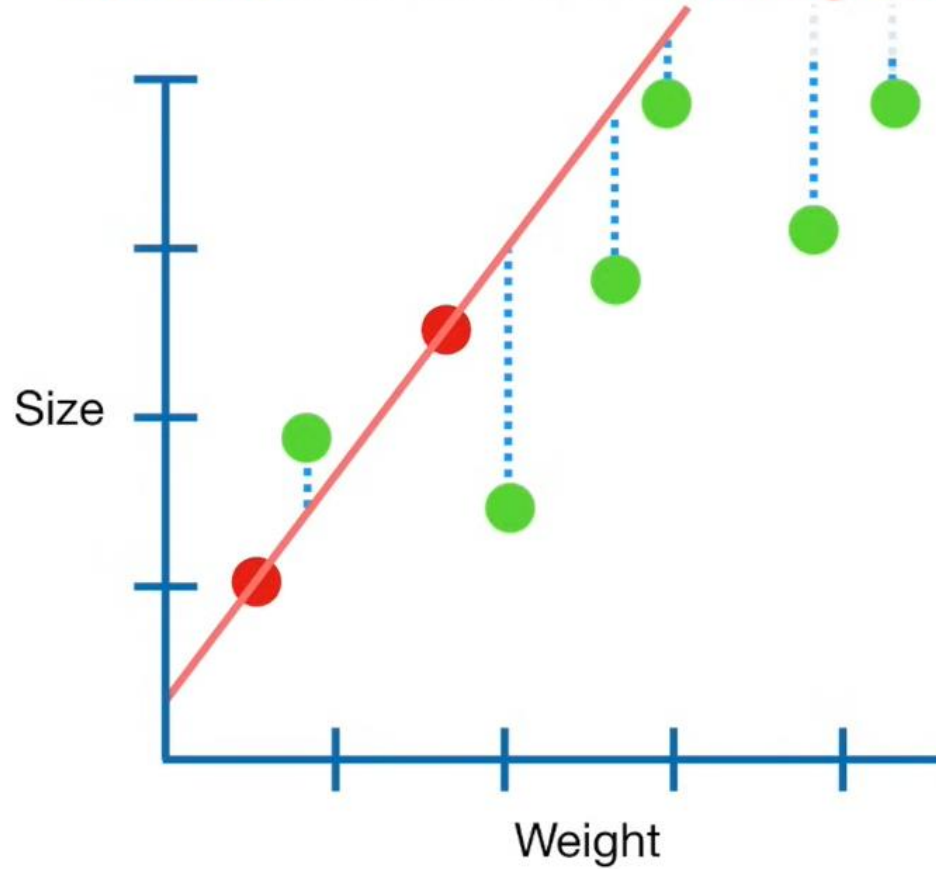
...but the sum of the squared residuals for the **Green Points**, the **Testing Data**, is large...



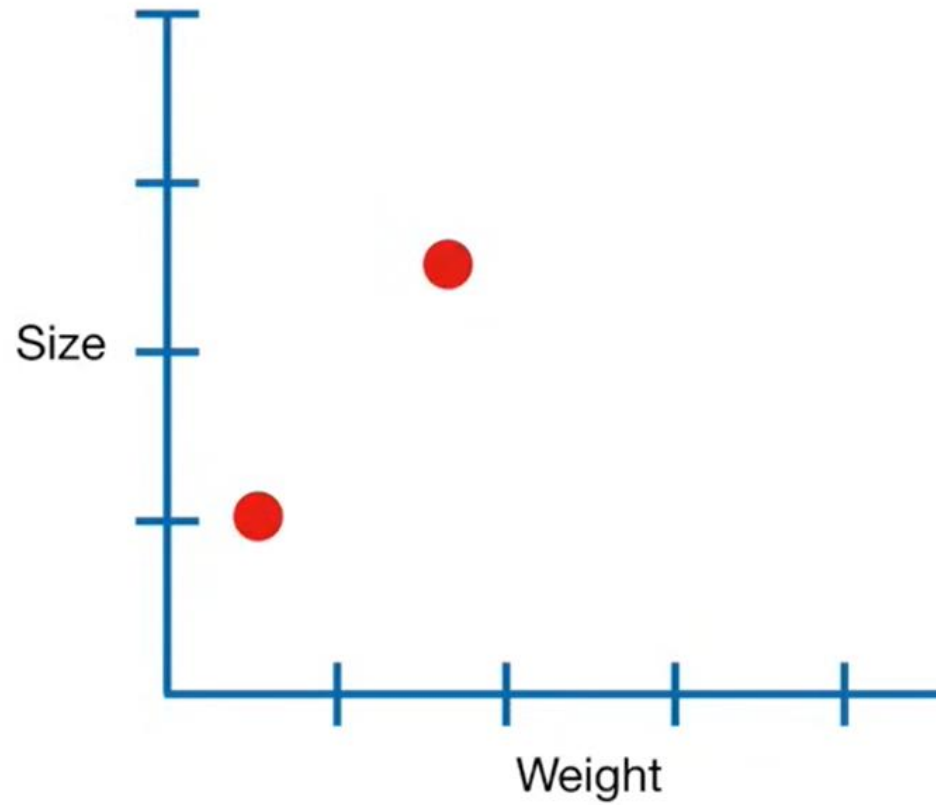
...and that means that the **New Line**
has **High Variance**.



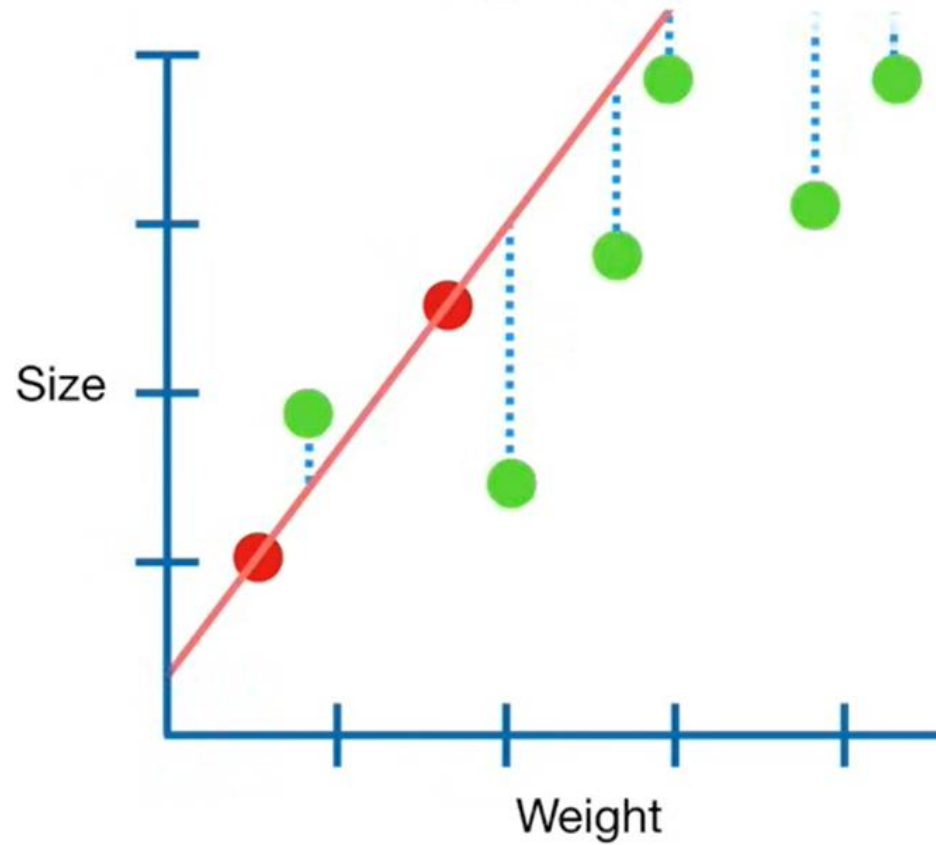
In machine learning lingo, we'd say that the **New Line** is **Over Fit** to the **Training Data**.



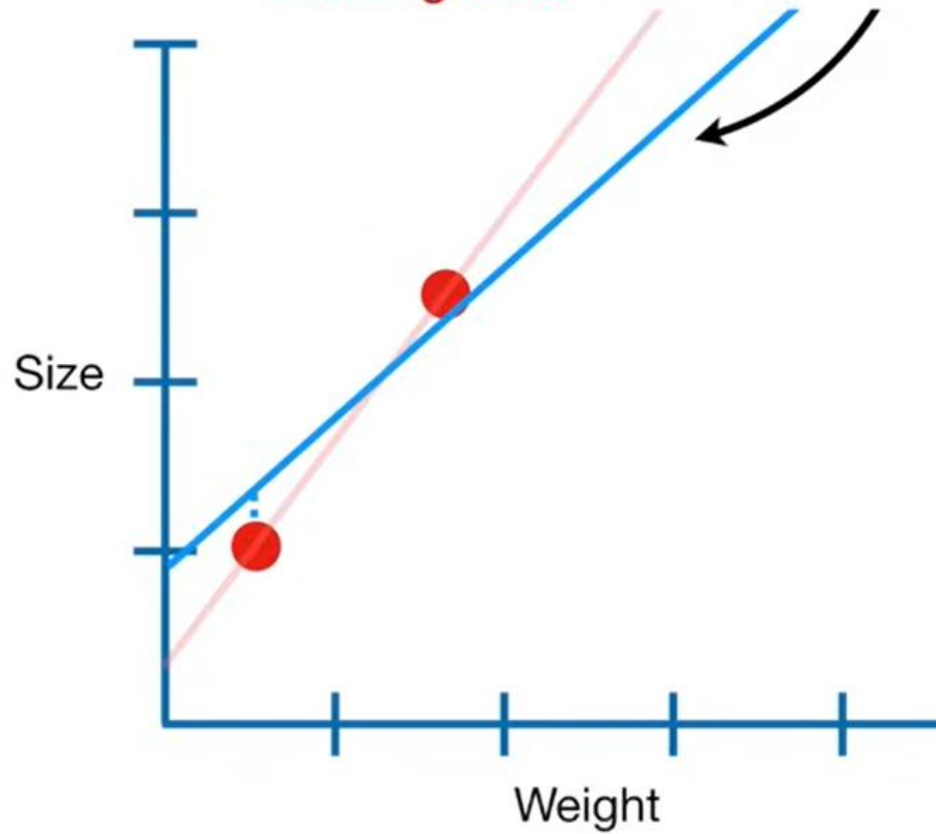
Now let's go back to just the **Training Data**...



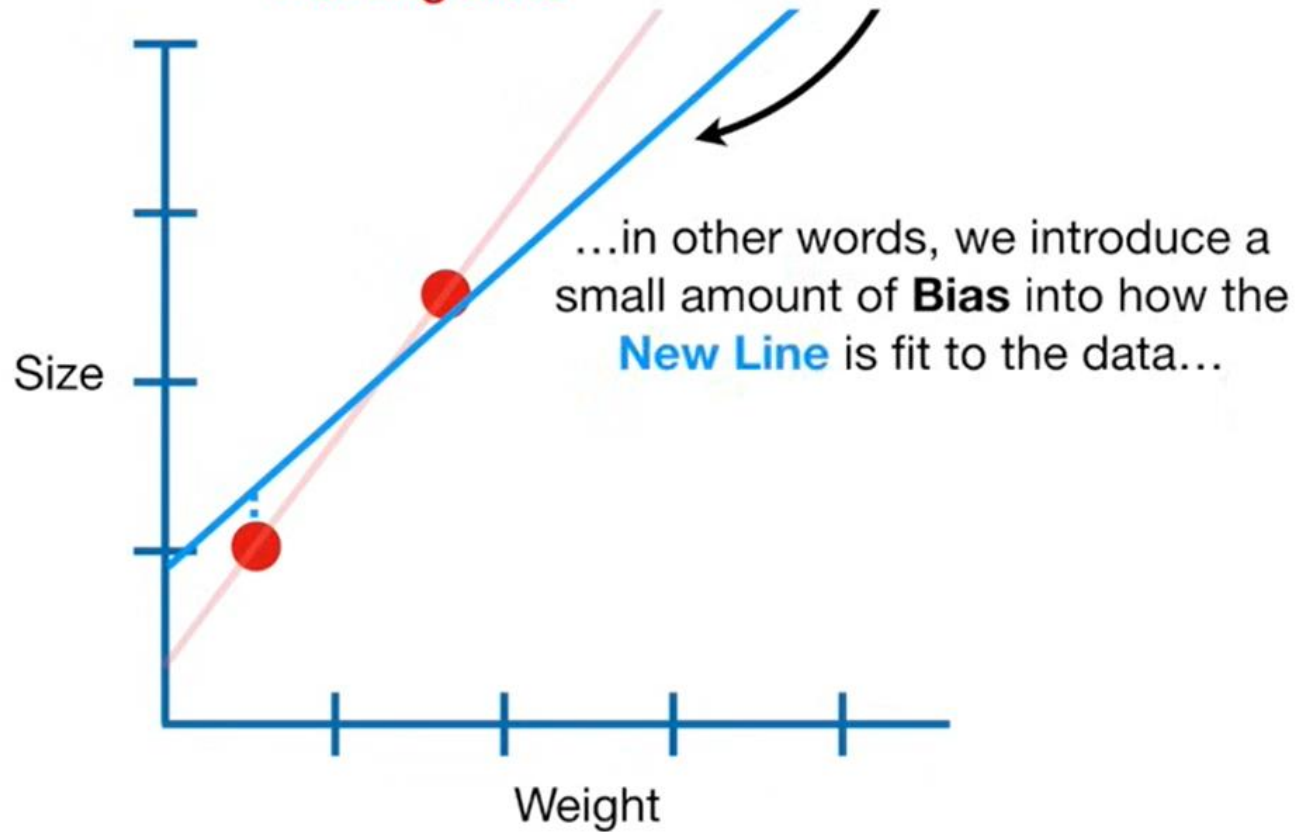
We just saw that **Least Squares** results in a **Line** that is **Over Fit** and has **High Variance**...



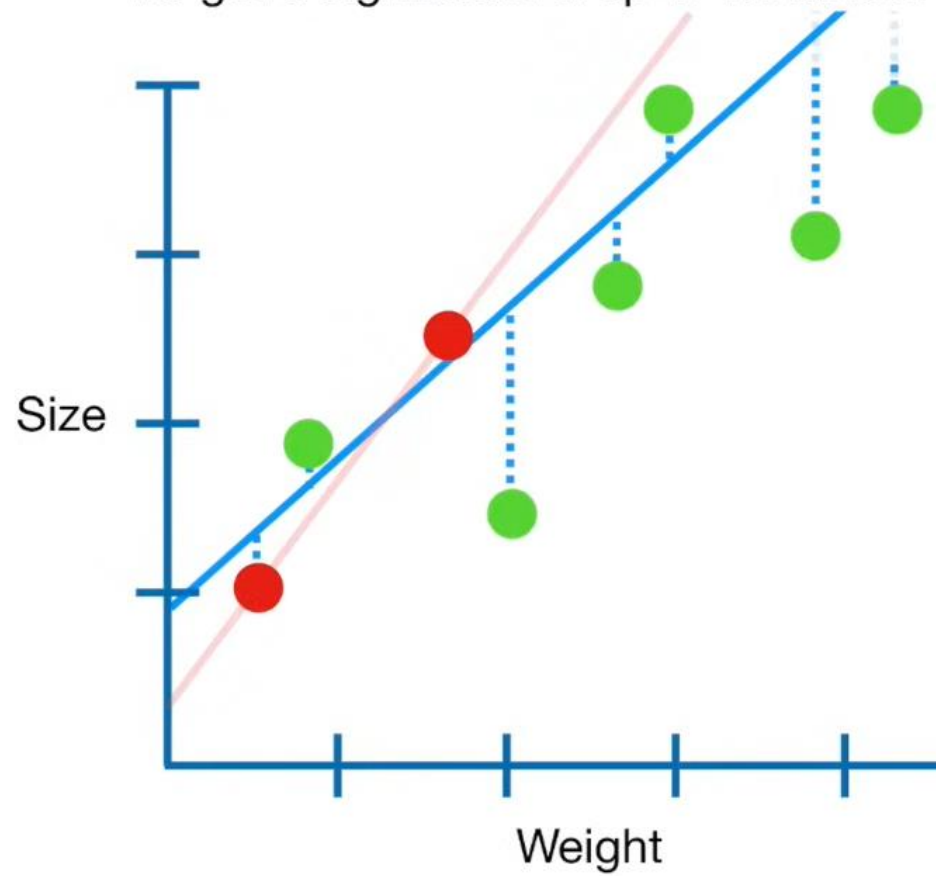
The main idea behind **Ridge Regression** is to find a **New Line** that doesn't fit the **Training Data** as well...



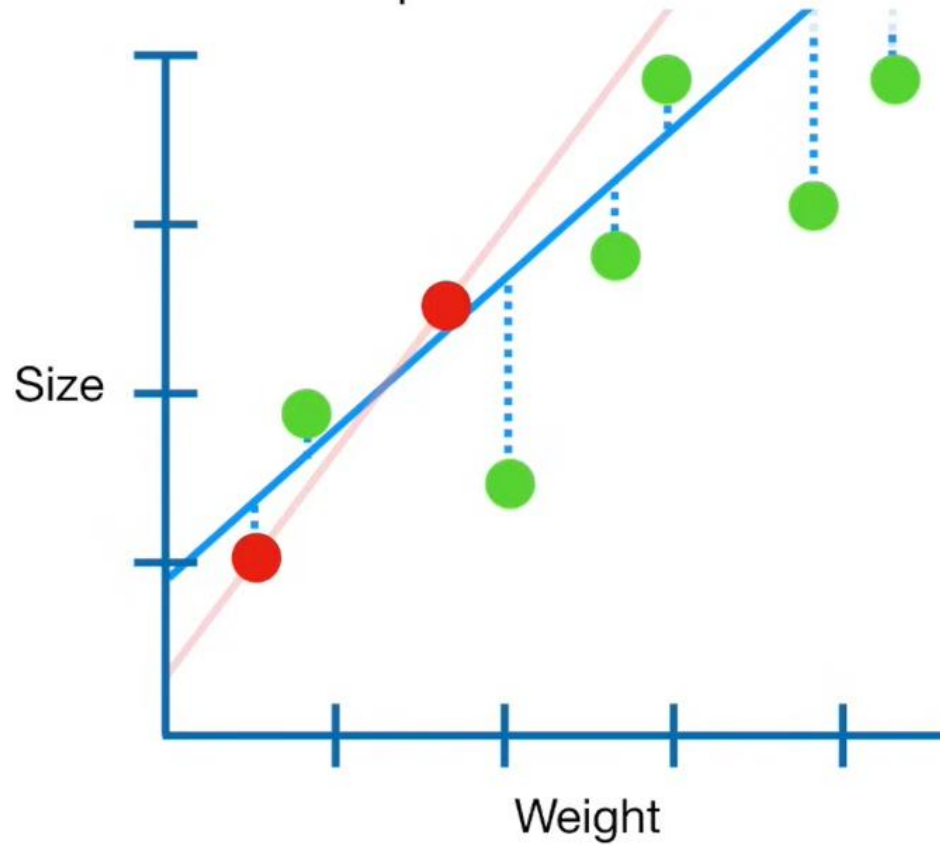
The main idea behind **Ridge Regression** is to find a **New Line** that doesn't fit the **Training Data** as well...



...but in return for that small amount of **Bias**,
we get a significant drop in **Variance**.

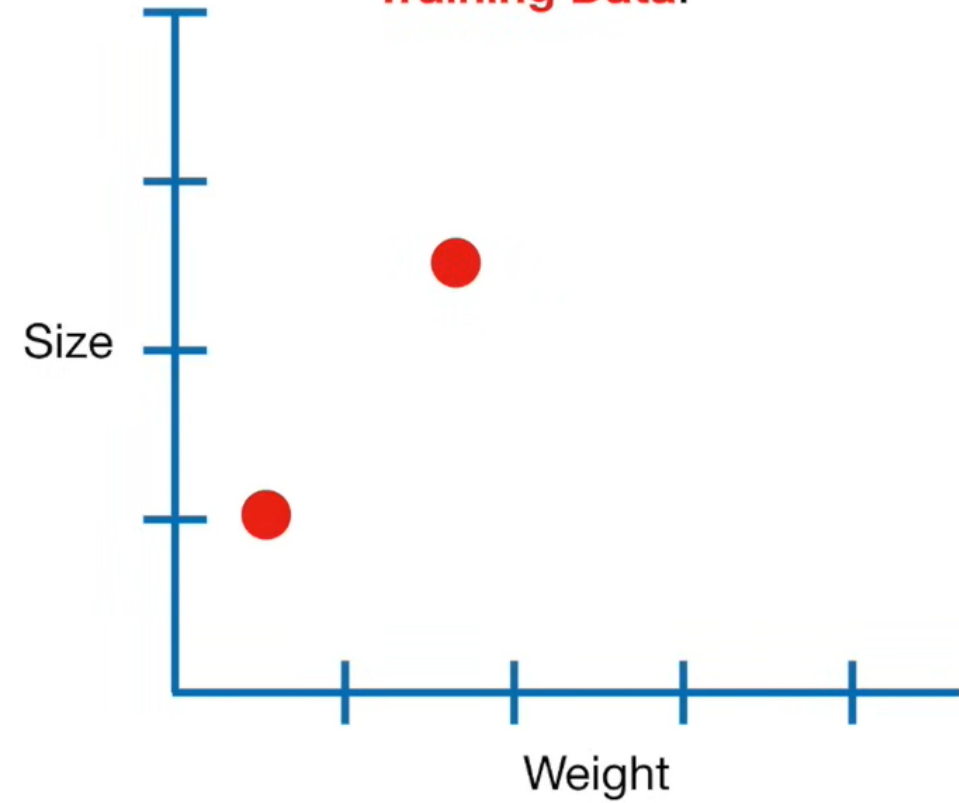


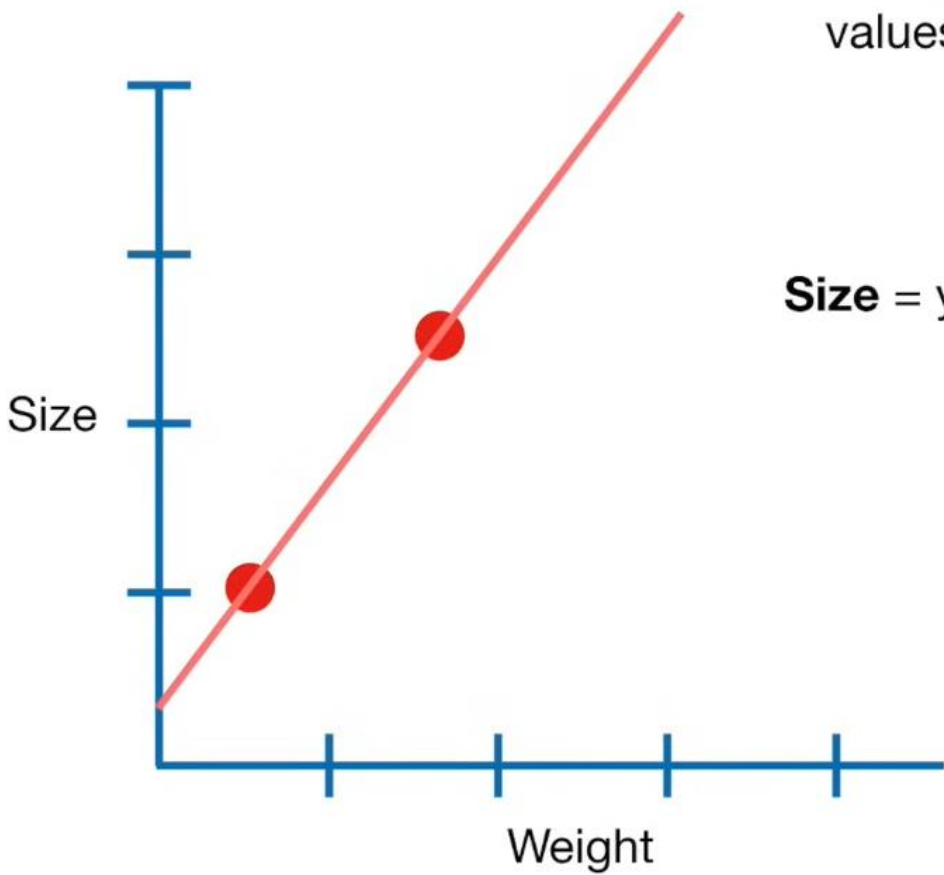
In other words, by starting with a slightly worse fit, **Ridge Regression** can provide better long term predictions.





Let's go back to just the
Training Data.

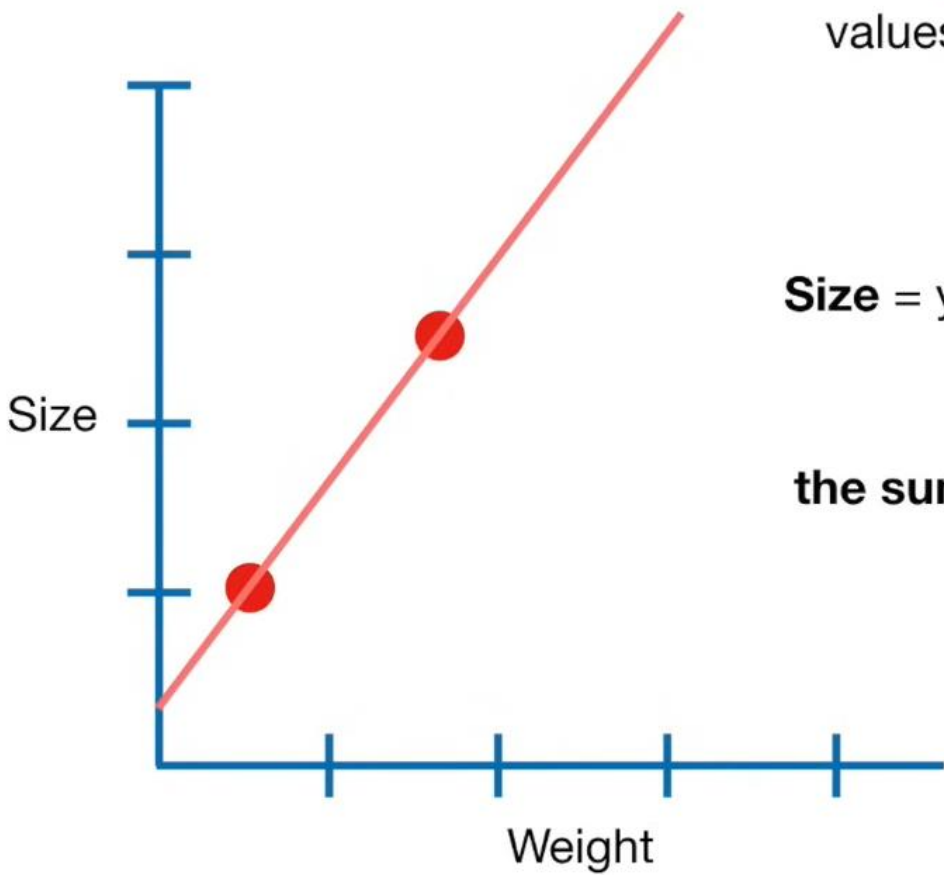




When **Least Squares** determines values for the parameters in this equation...



$$\mathbf{Size} = \text{y-axis intercept} + \text{slope} \times \mathbf{Weight}$$



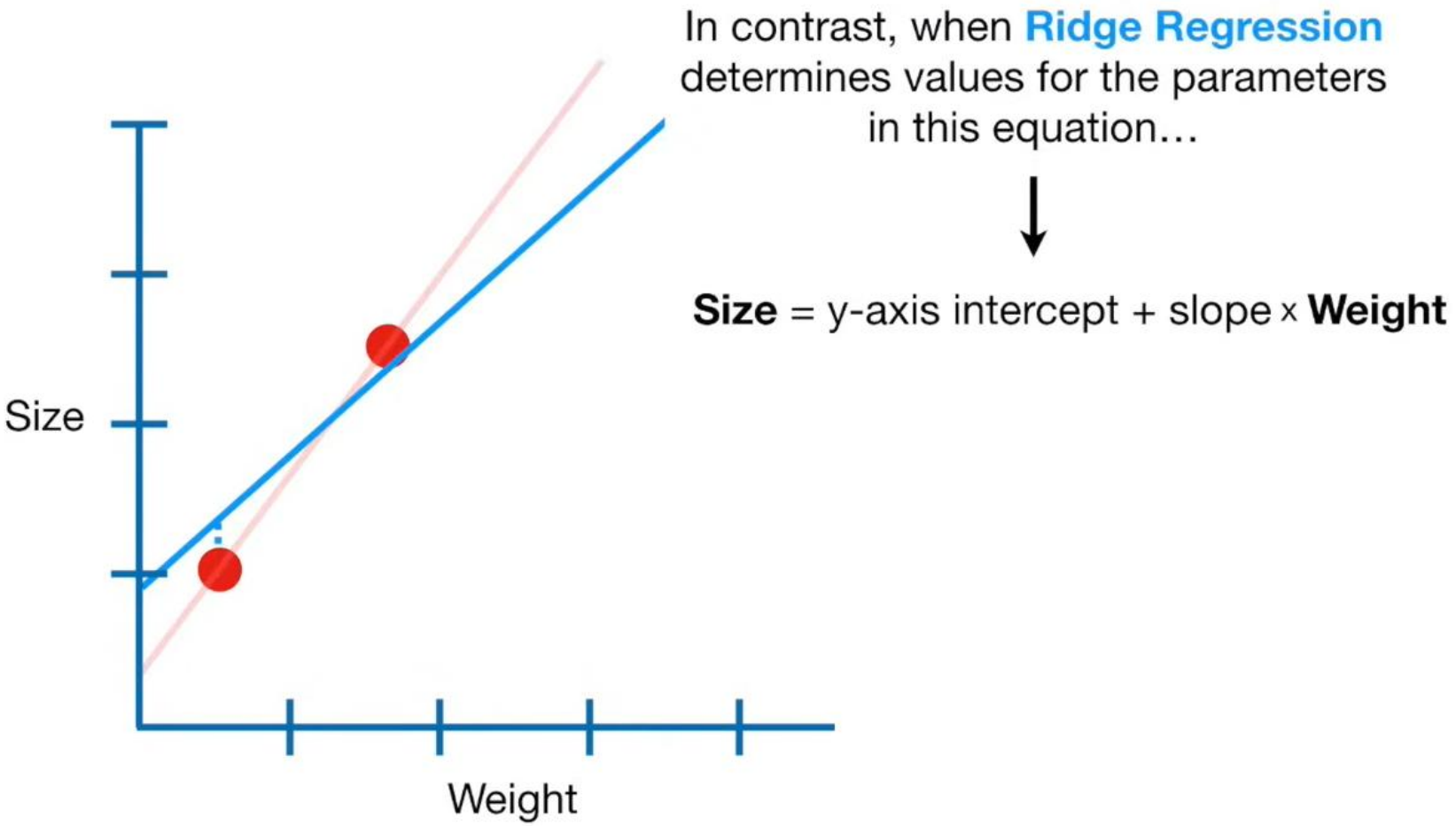
When **Least Squares** determines values for the parameters in this equation...

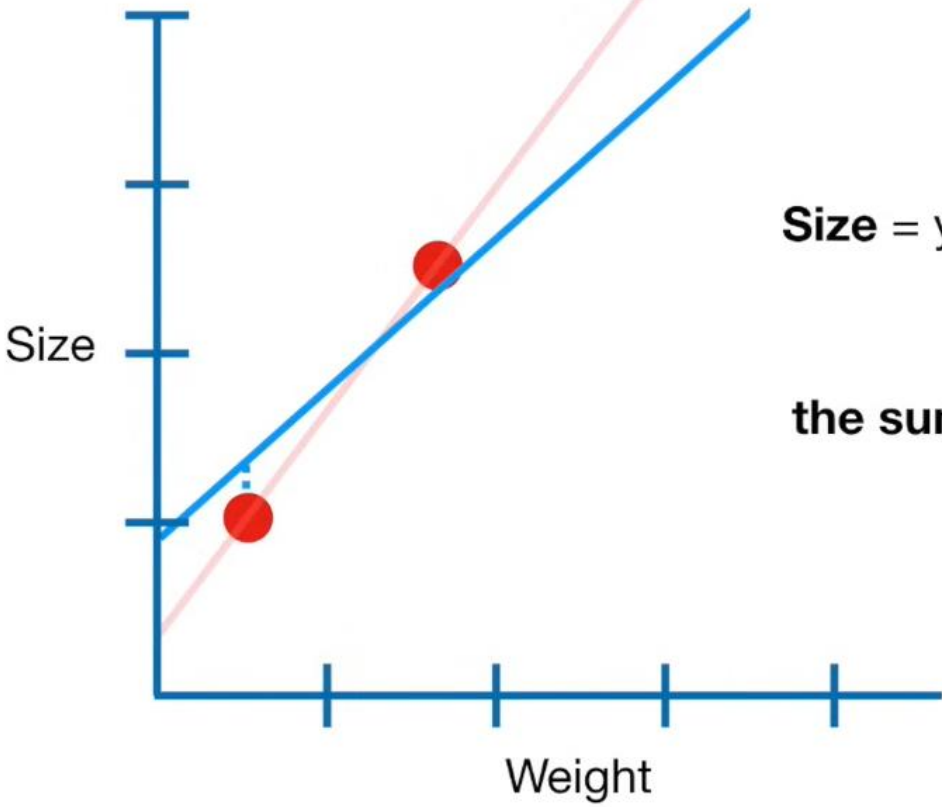


Size = y-axis intercept + slope × **Weight**

...it minimizes...

the sum of the squared residuals





In contrast, when **Ridge Regression** determines values for the parameters in this equation...



Size = y-axis intercept + slope × **Weight**

...it minimizes...

the sum of the squared residuals

+

$\lambda \times \text{the slope}^2$

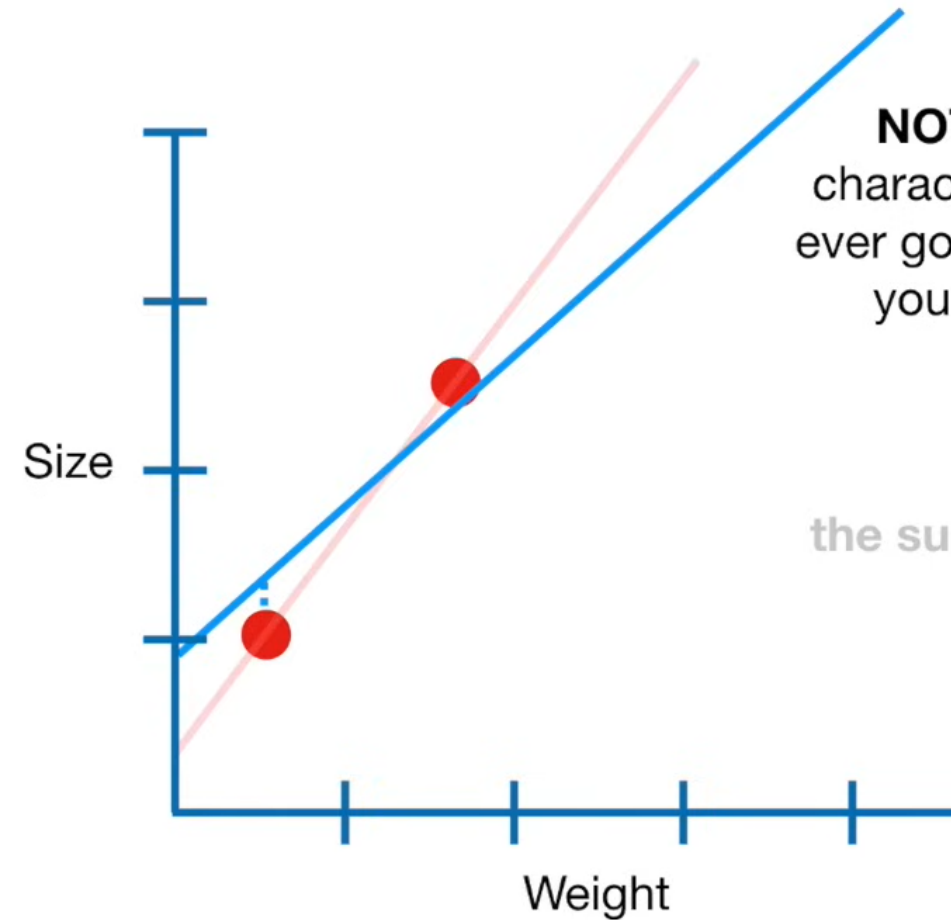
NOTE: I usually try to avoid using Greek characters as much as possible, but if you are ever going to do **Ridge Regression** in practice, you have to know that this term is called “lambda”.

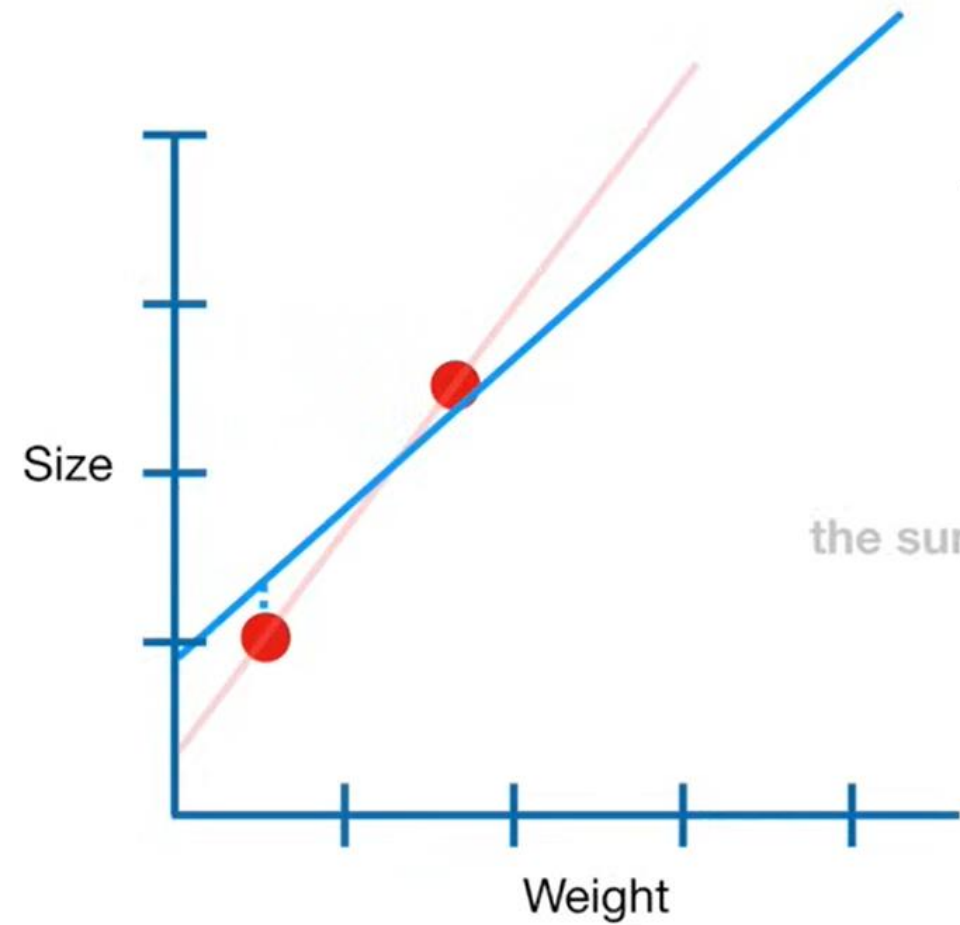
the sum of the squared residuals

+

λ

the slope²



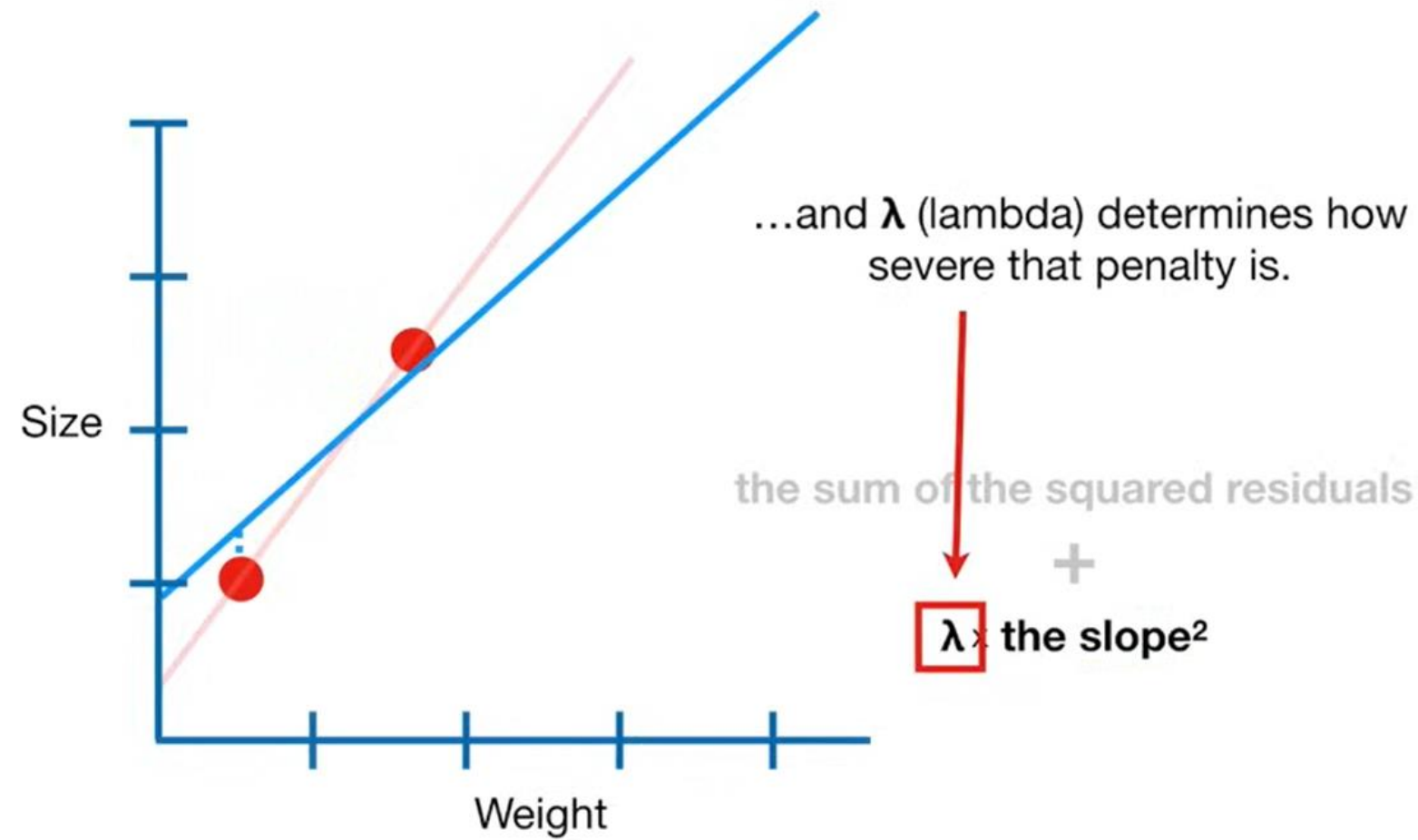


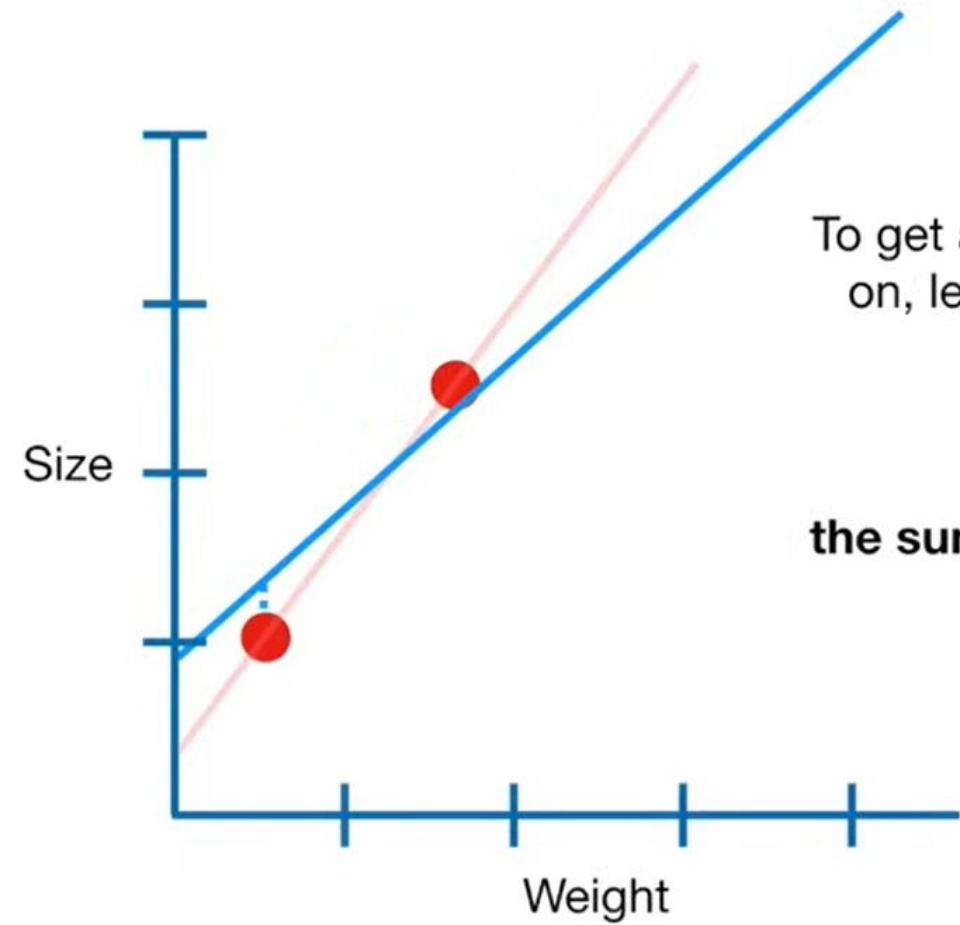
This part of the equation adds a penalty to the traditional **Least Squares** method...

the sum of the squared residuals

+

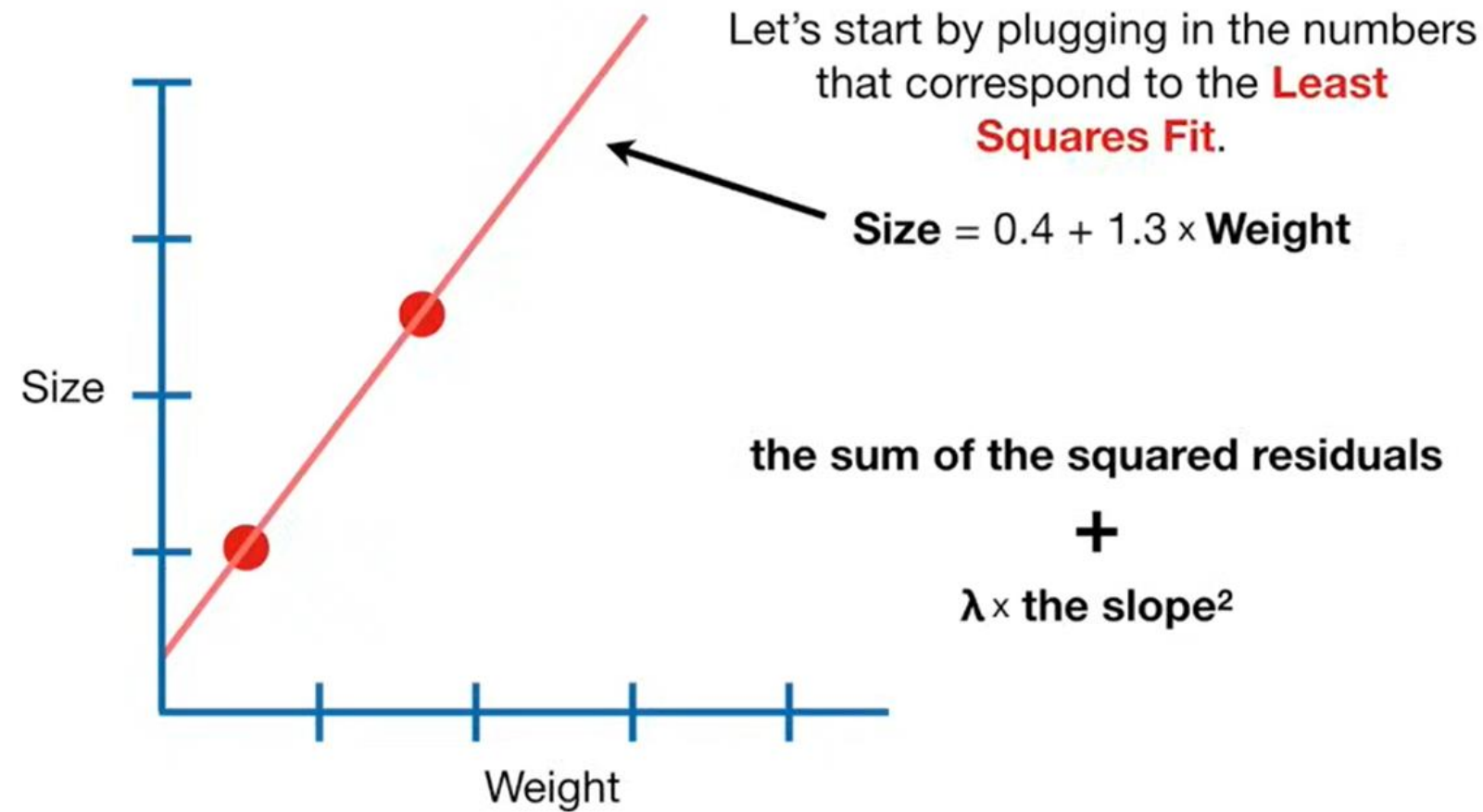
$\lambda \times$ the slope²

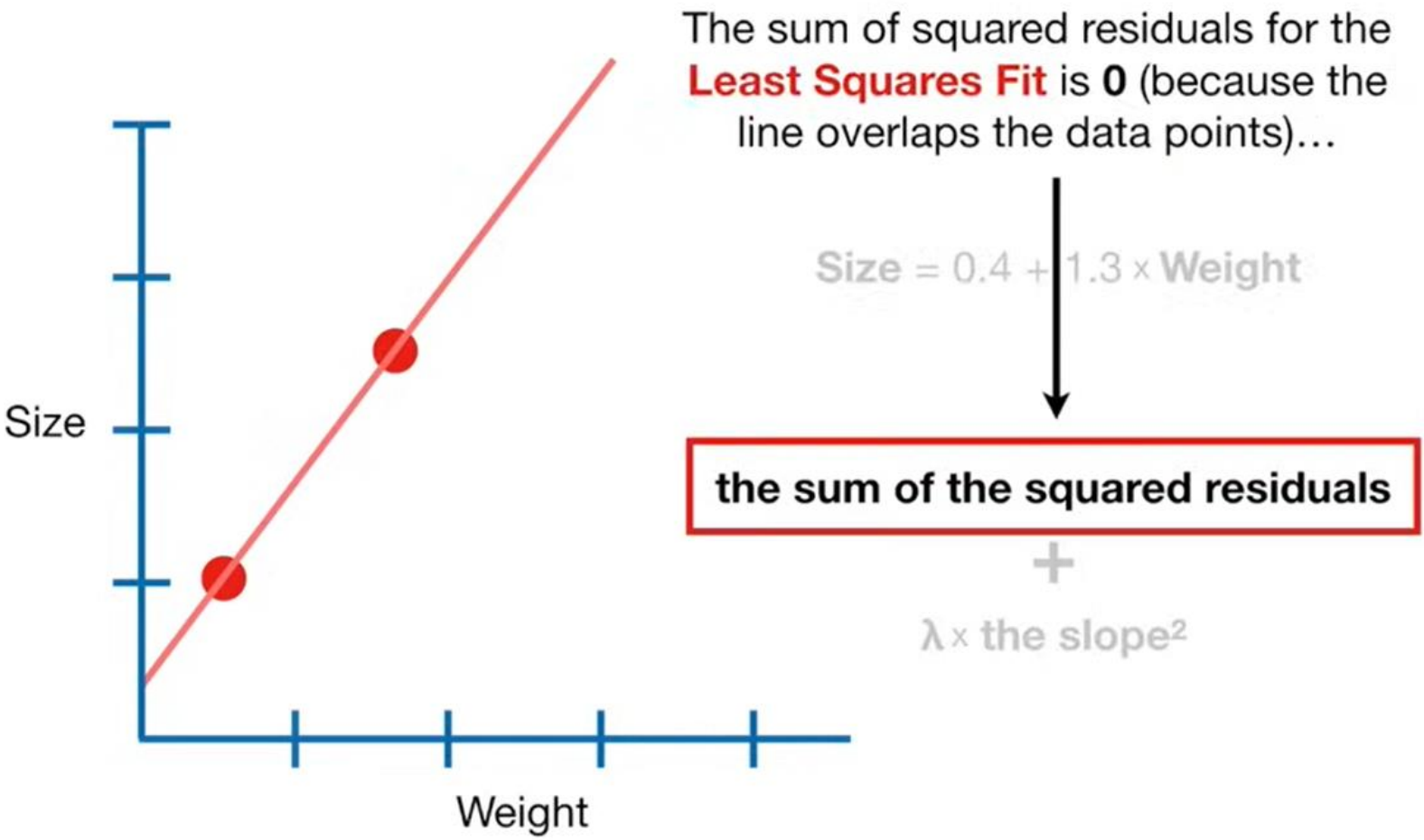


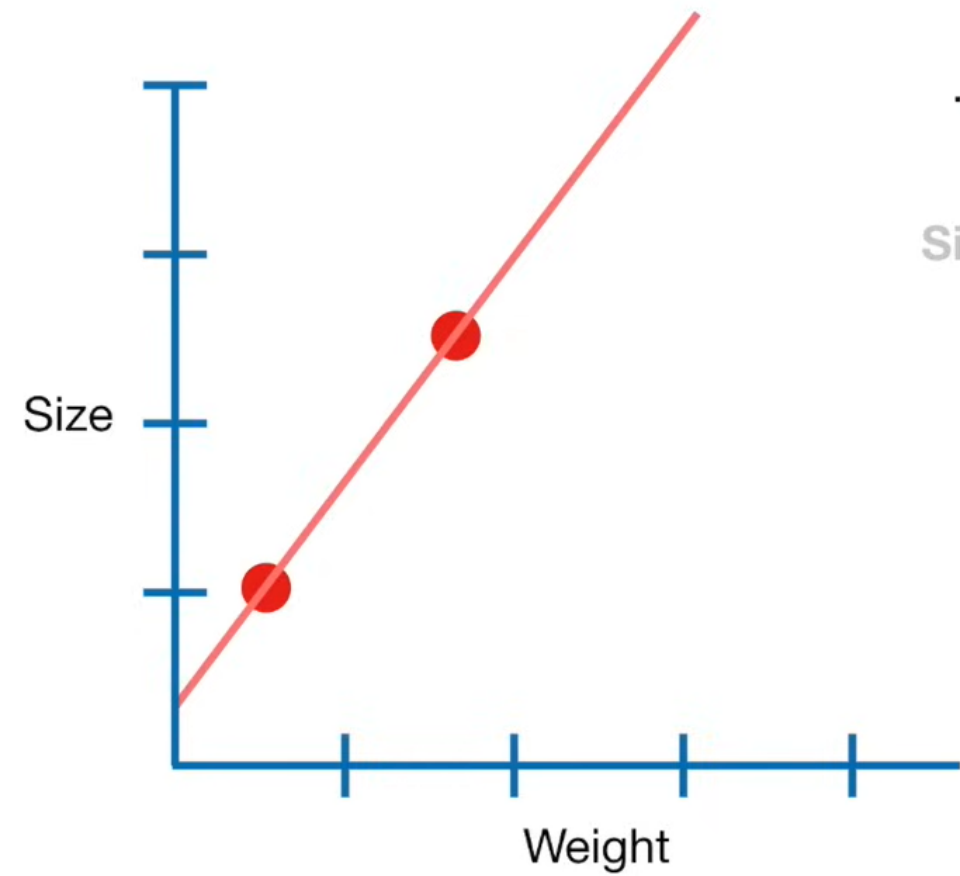


To get a better idea of what's going on, let's plug in some numbers!!!

the sum of the squared residuals
+
 $\lambda \times \text{slope}^2$



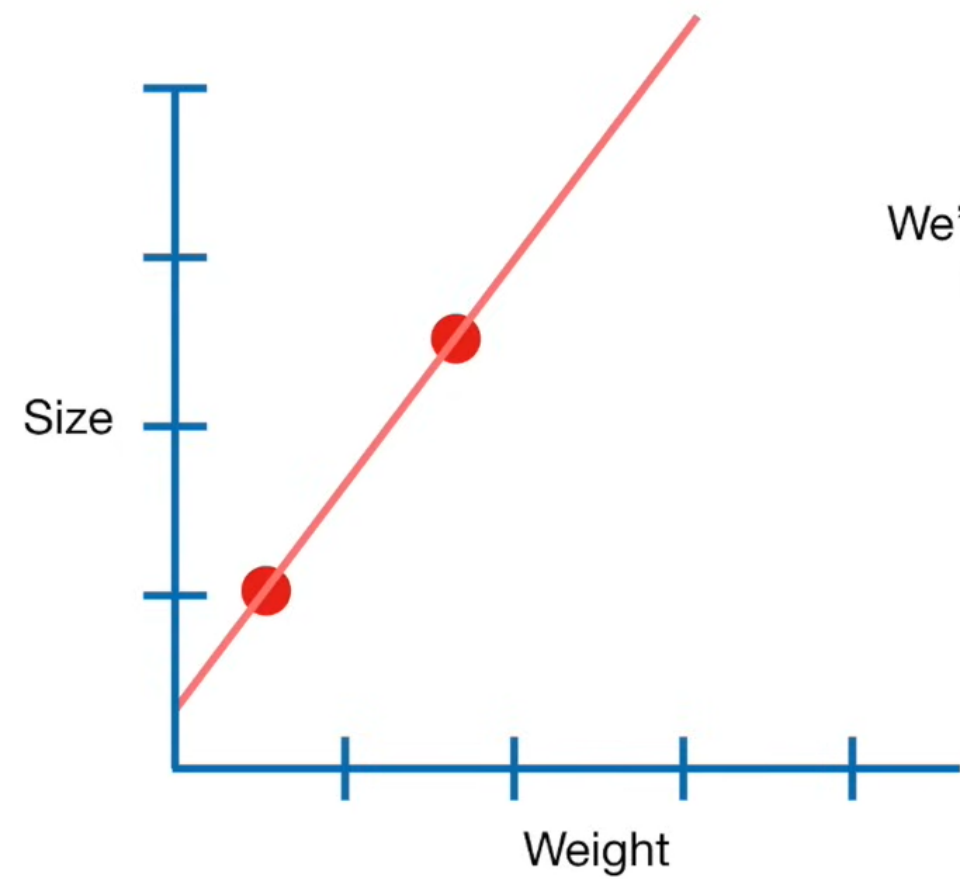




...and the slope is **1.3**.

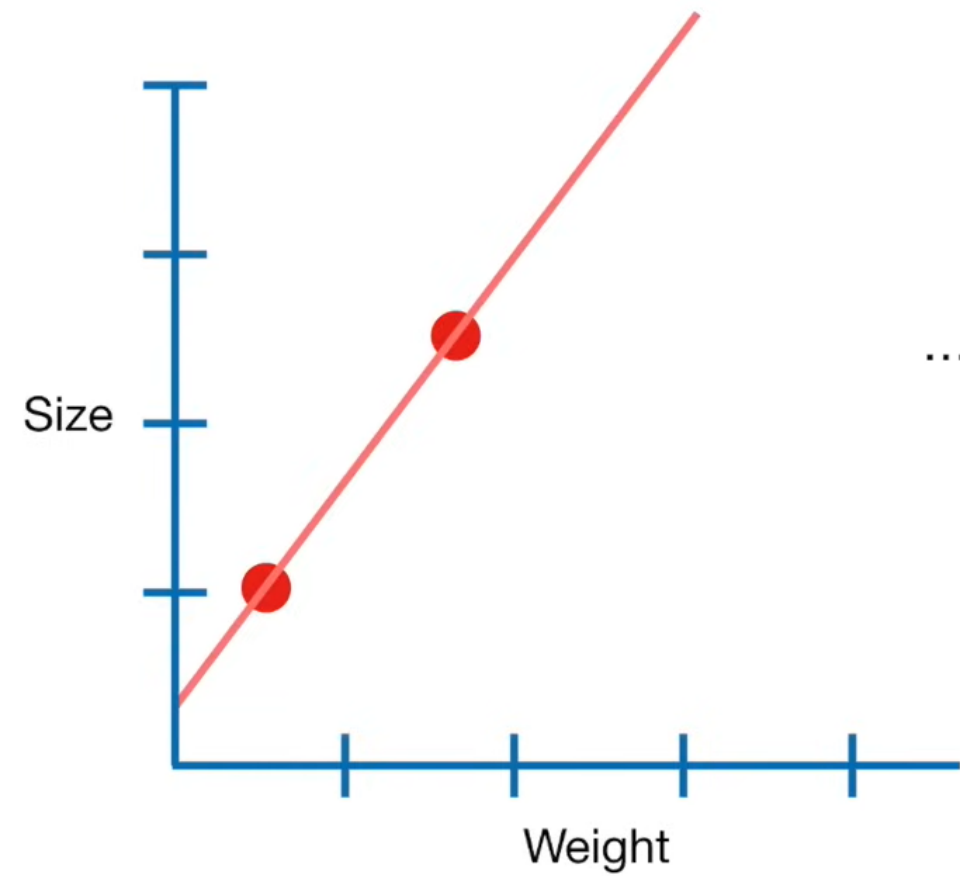
$$\text{Size} = 0.4 + \boxed{1.3} \times \text{Weight}$$

$$\lambda \times \begin{matrix} 0 \\ + \end{matrix} \downarrow \boxed{\text{the slope}^2}$$



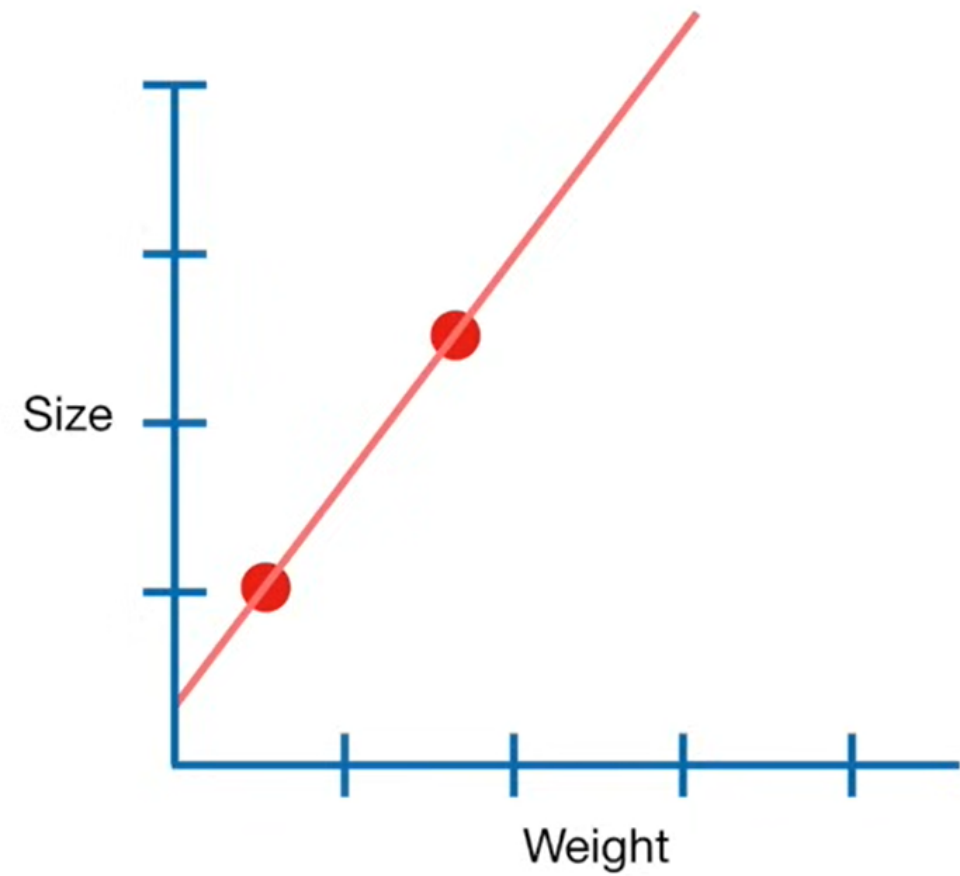
We'll talk more about λ later,
but for now, let $\lambda = 1$.

$$0 + \boxed{\lambda \times 1.3^2}$$



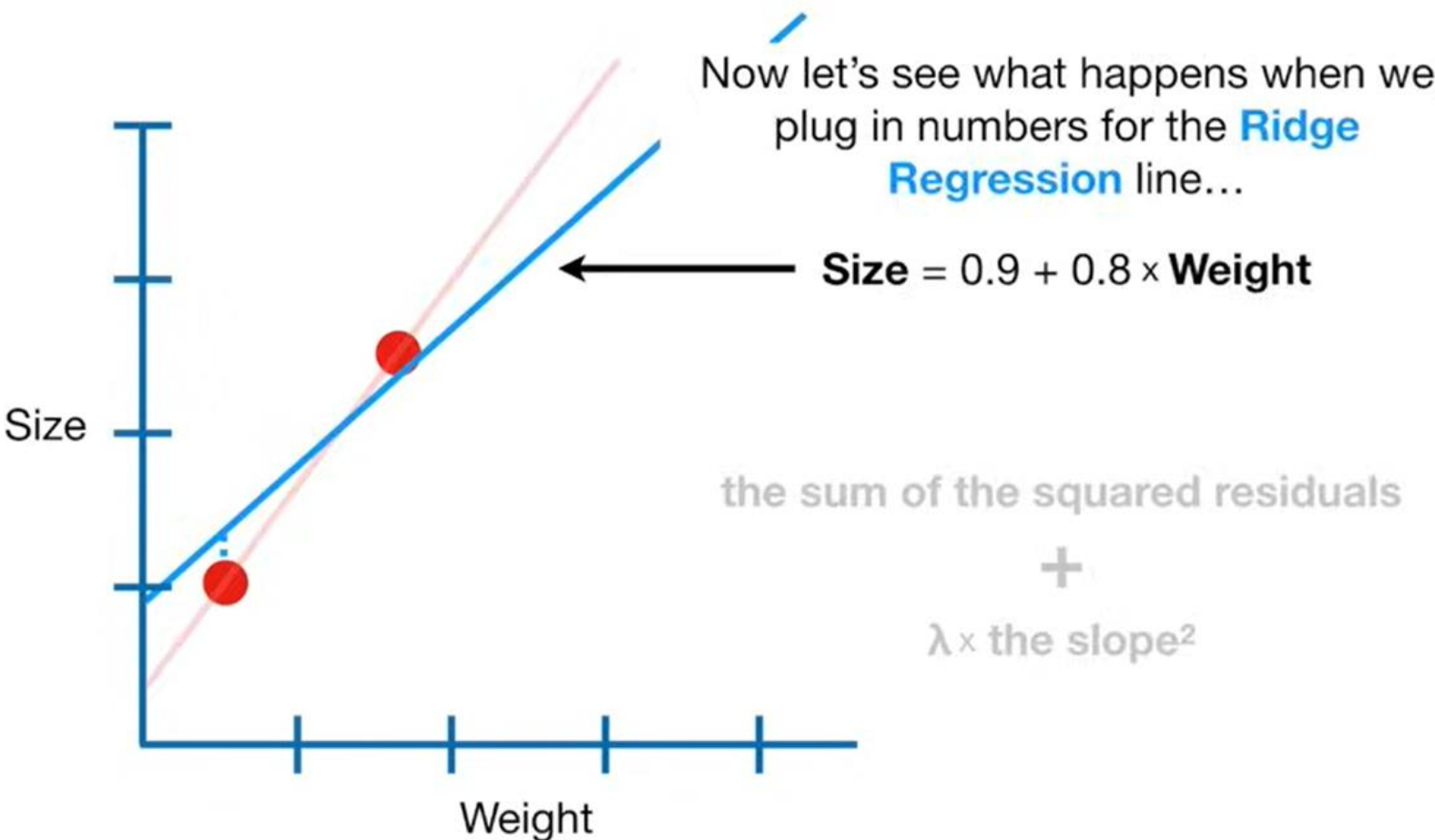
...all together, we have...

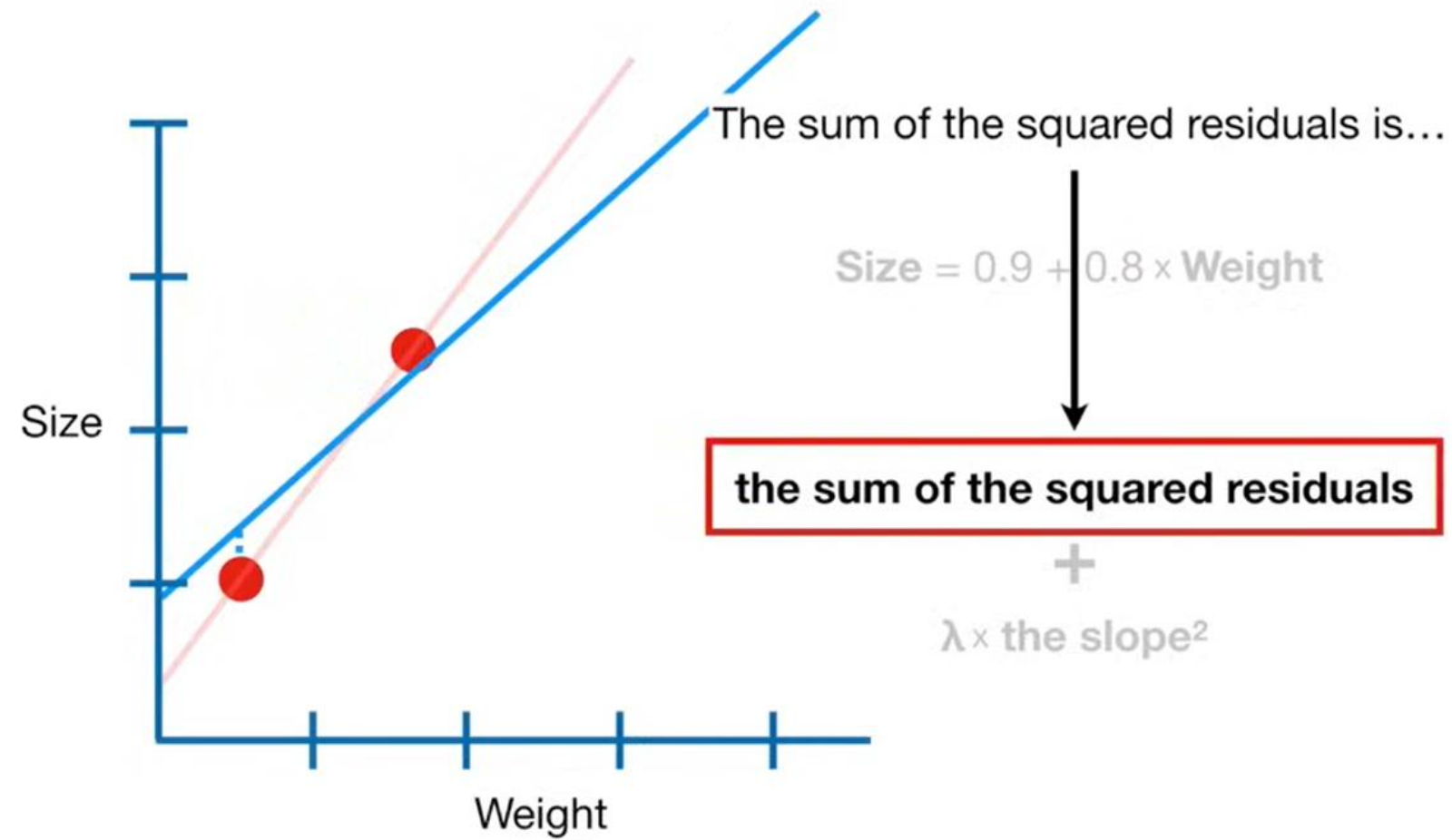
$$\begin{array}{c} 0 \\ + \\ 1 \times 1.3^2 \end{array}$$

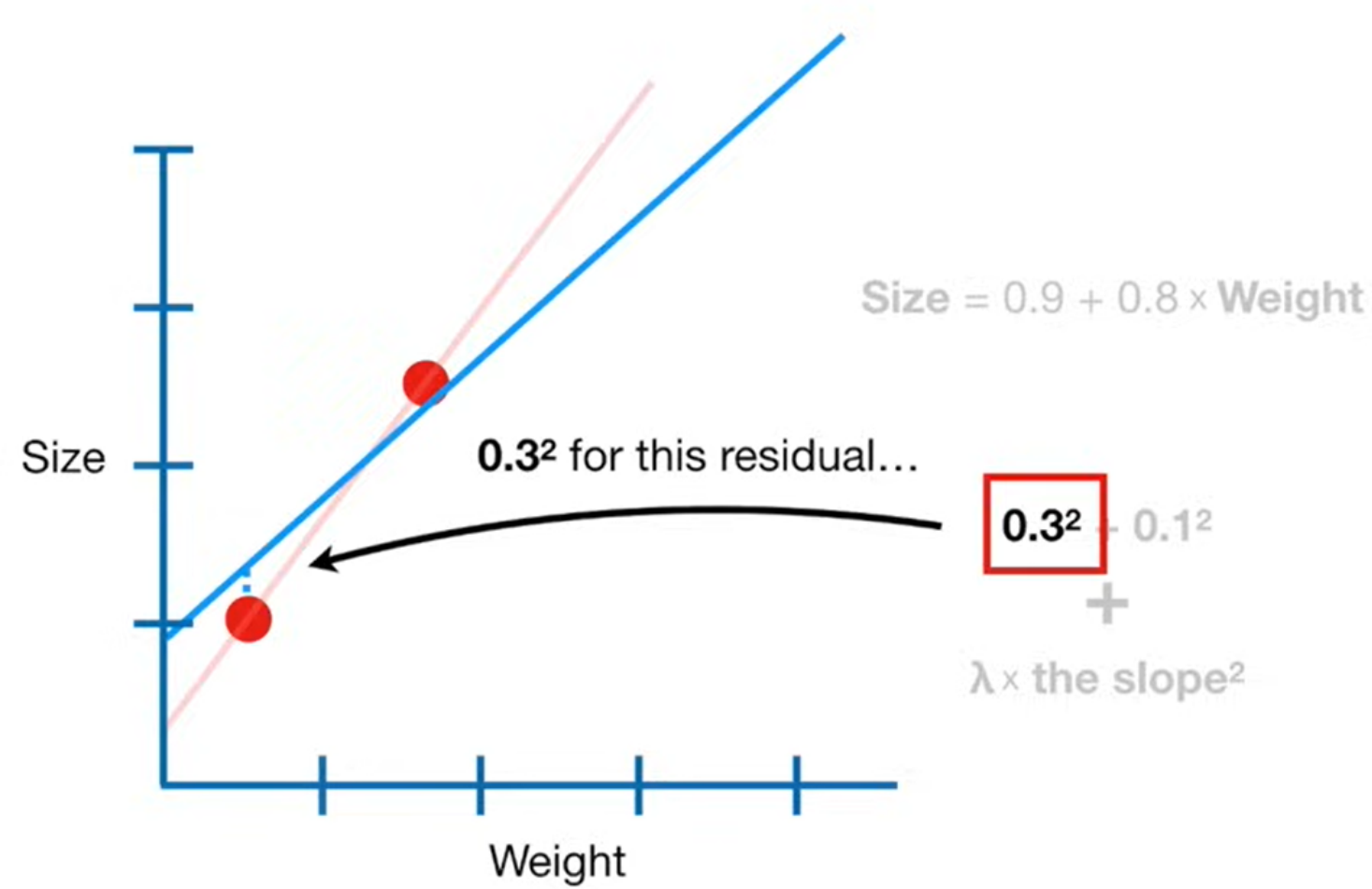


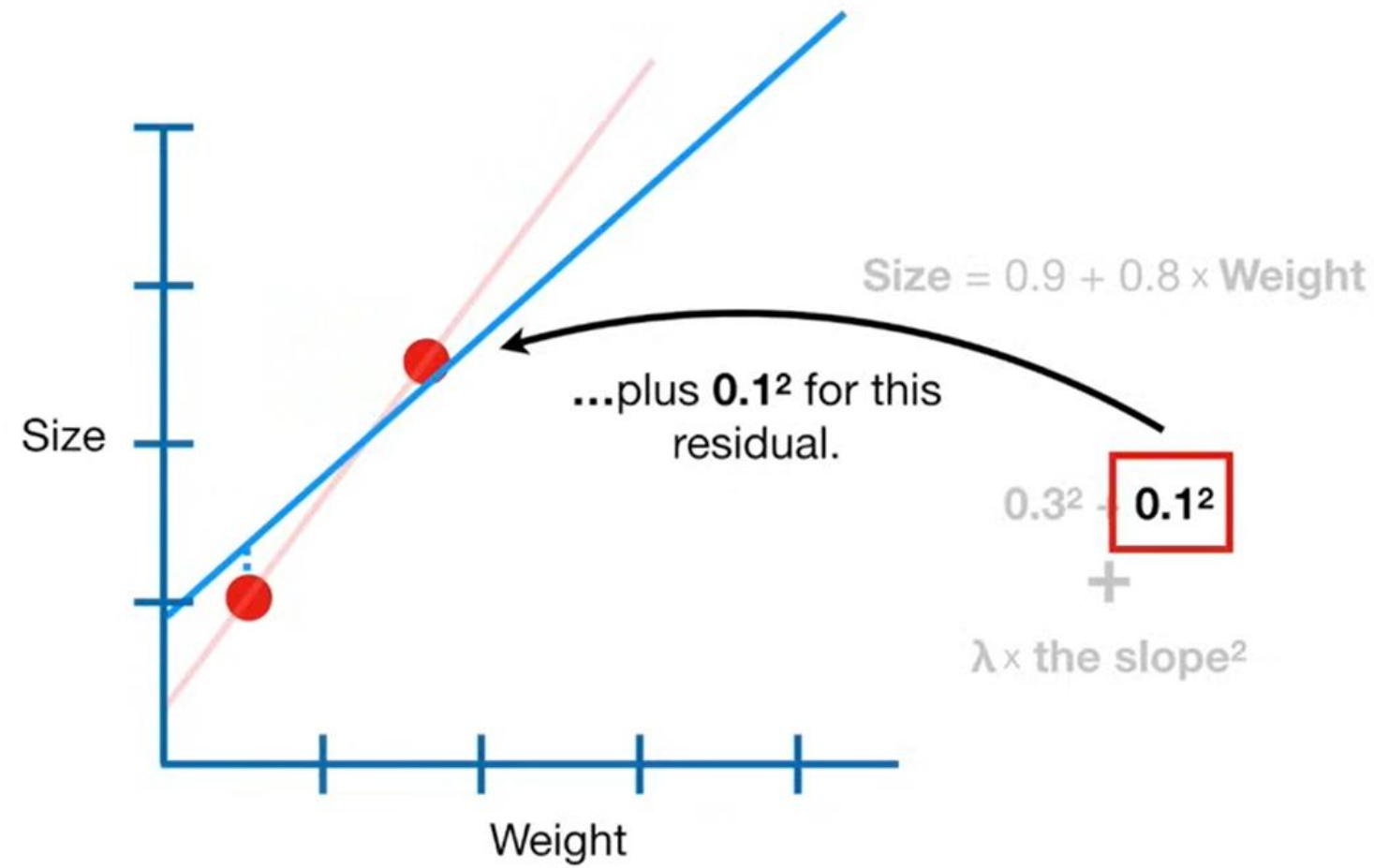
...and when we do
the math we get...

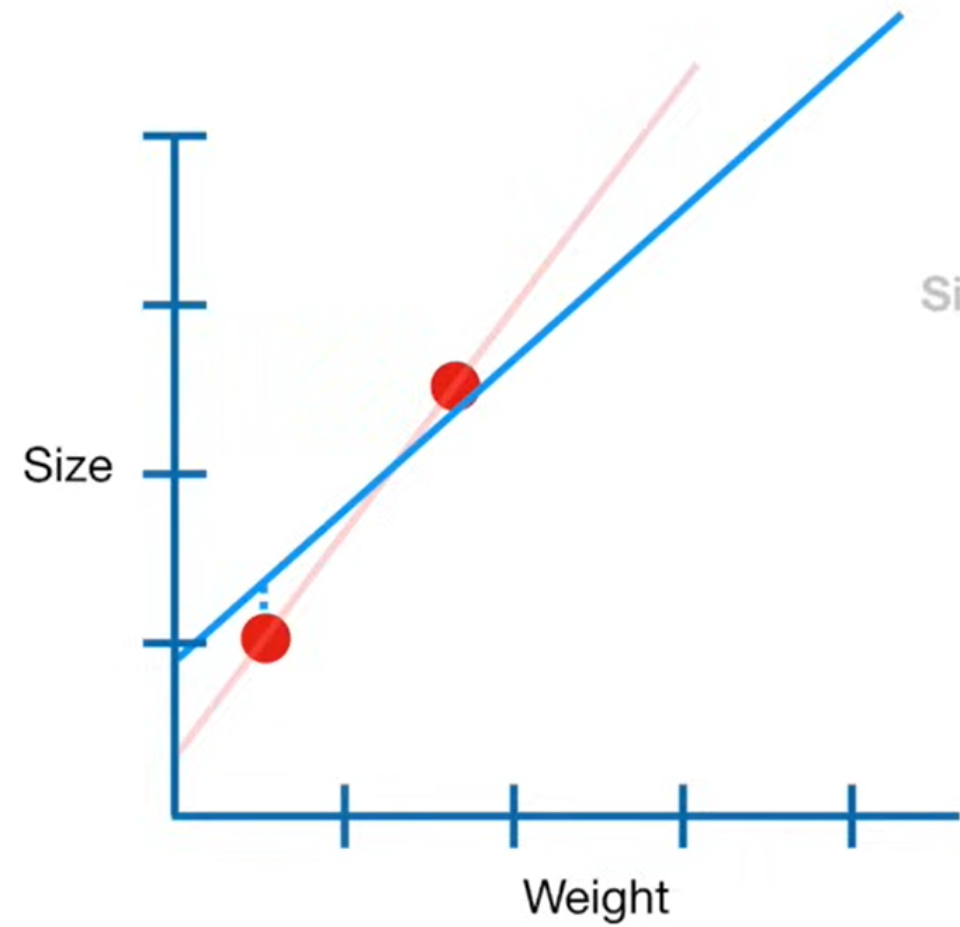
$$\begin{array}{r} 0 \\ + \\ 1 \times 1.3^2 \end{array} = 0 + 1.69$$











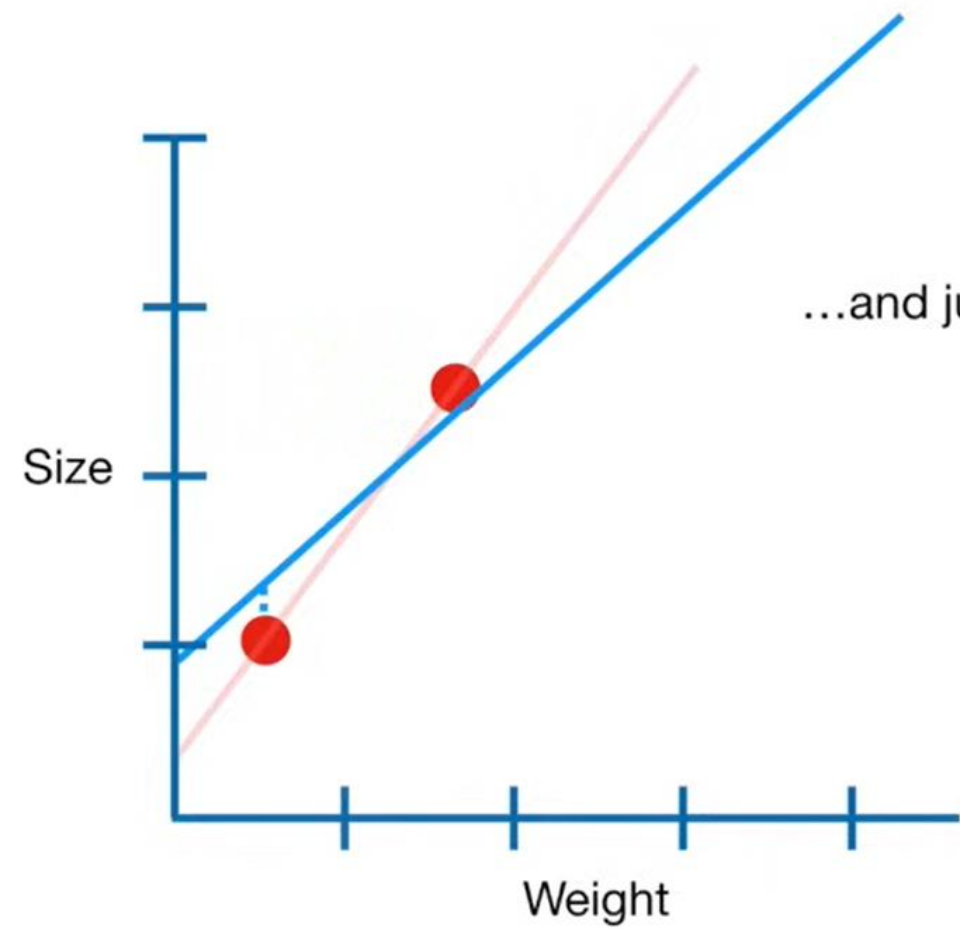
The slope is **0.8...**

$$\text{Size} = 0.9 + \boxed{0.8} \times \text{Weight}$$

$$0.3^2 + 0.1^2$$

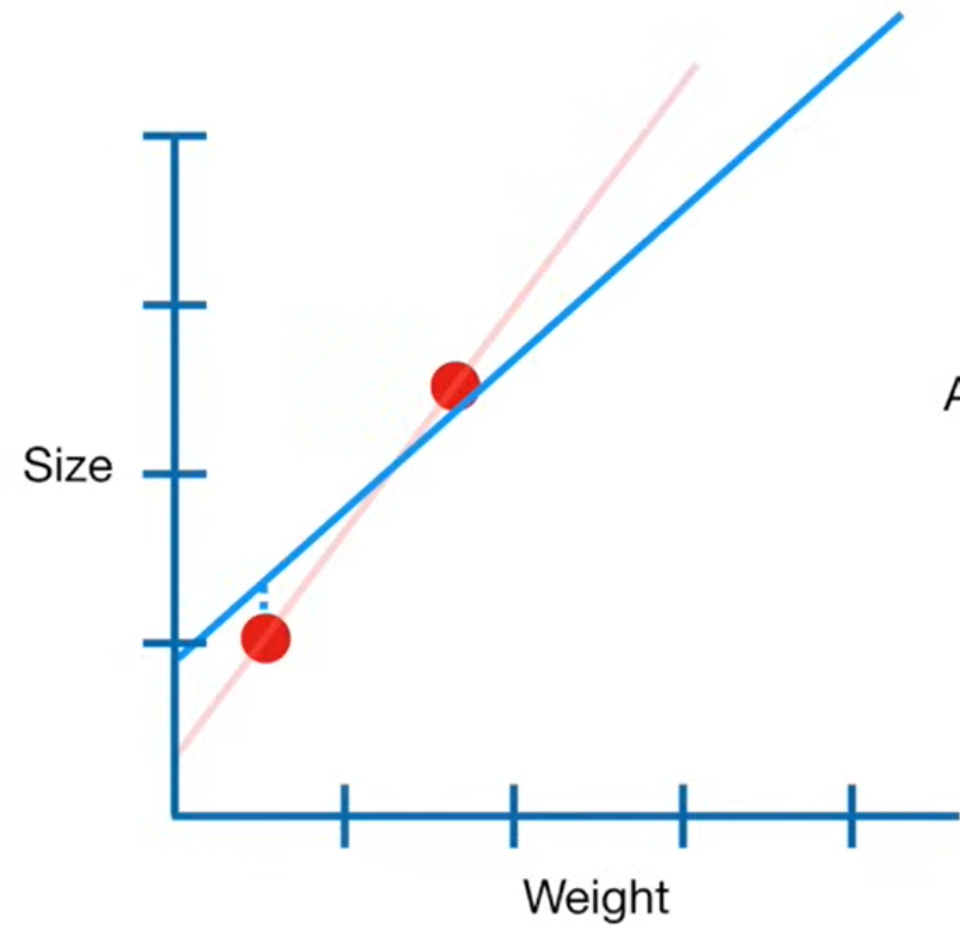
+

$$\lambda \times \boxed{\text{the slope}^2}$$



...and just like before, we'll let $\lambda = 1$.

$$0.3^2 + 0.1^2 + \boxed{\lambda} \times 0.8^2$$

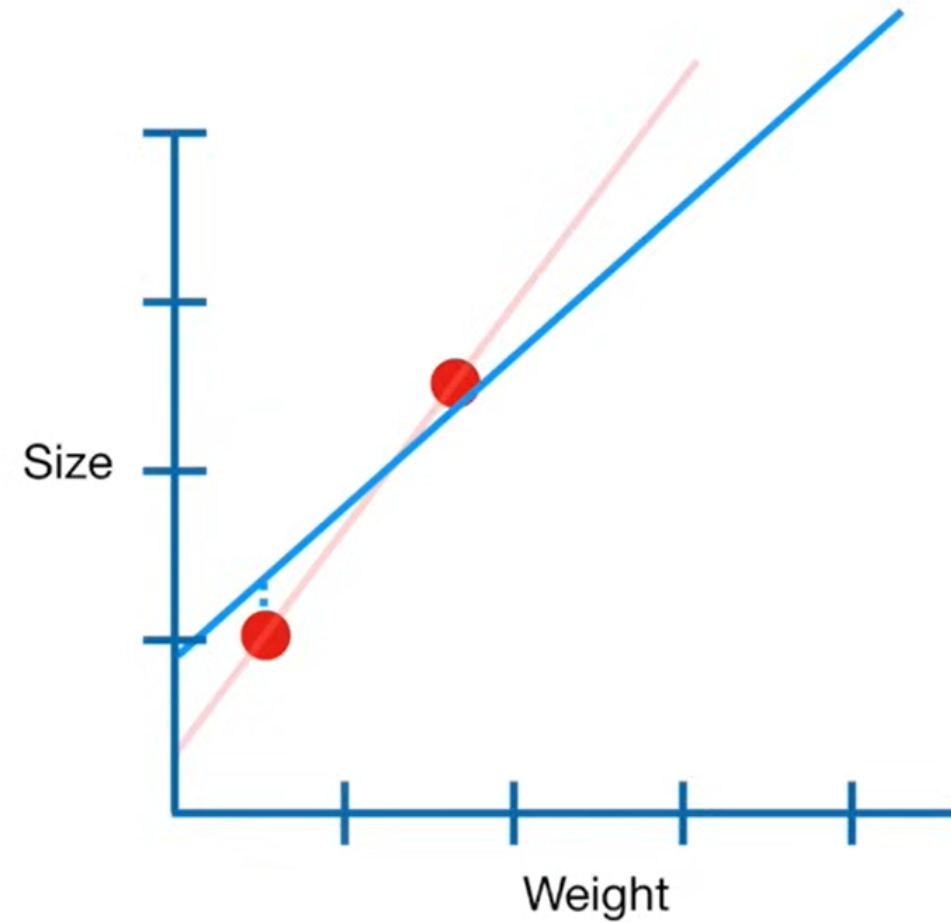


All together, we have...

$$0.3^2 + 0.1^2$$

+

$$1 \times 0.8^2$$



...and when we do the math we get...

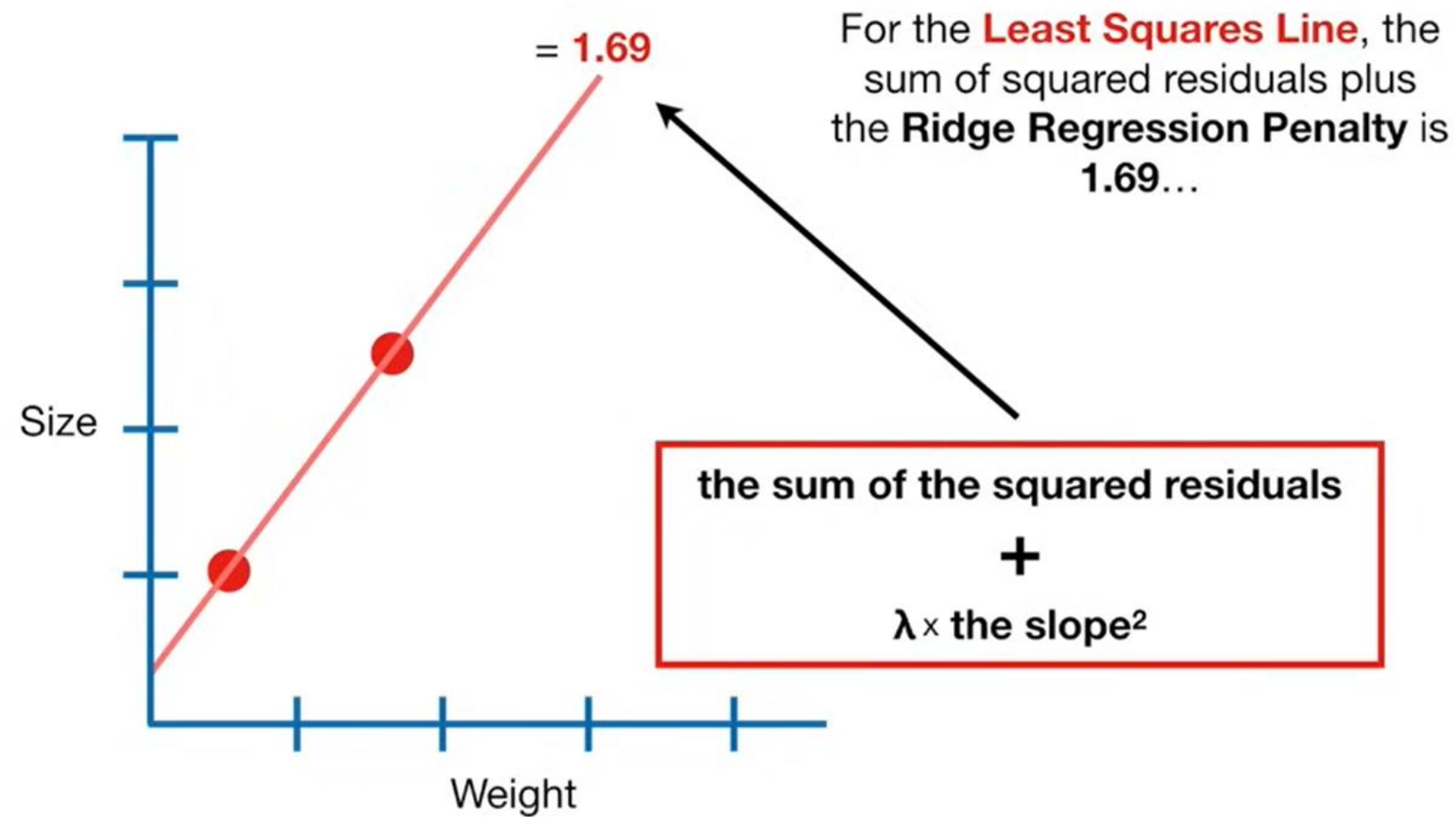
$$0.3^2 + 0.1^2$$

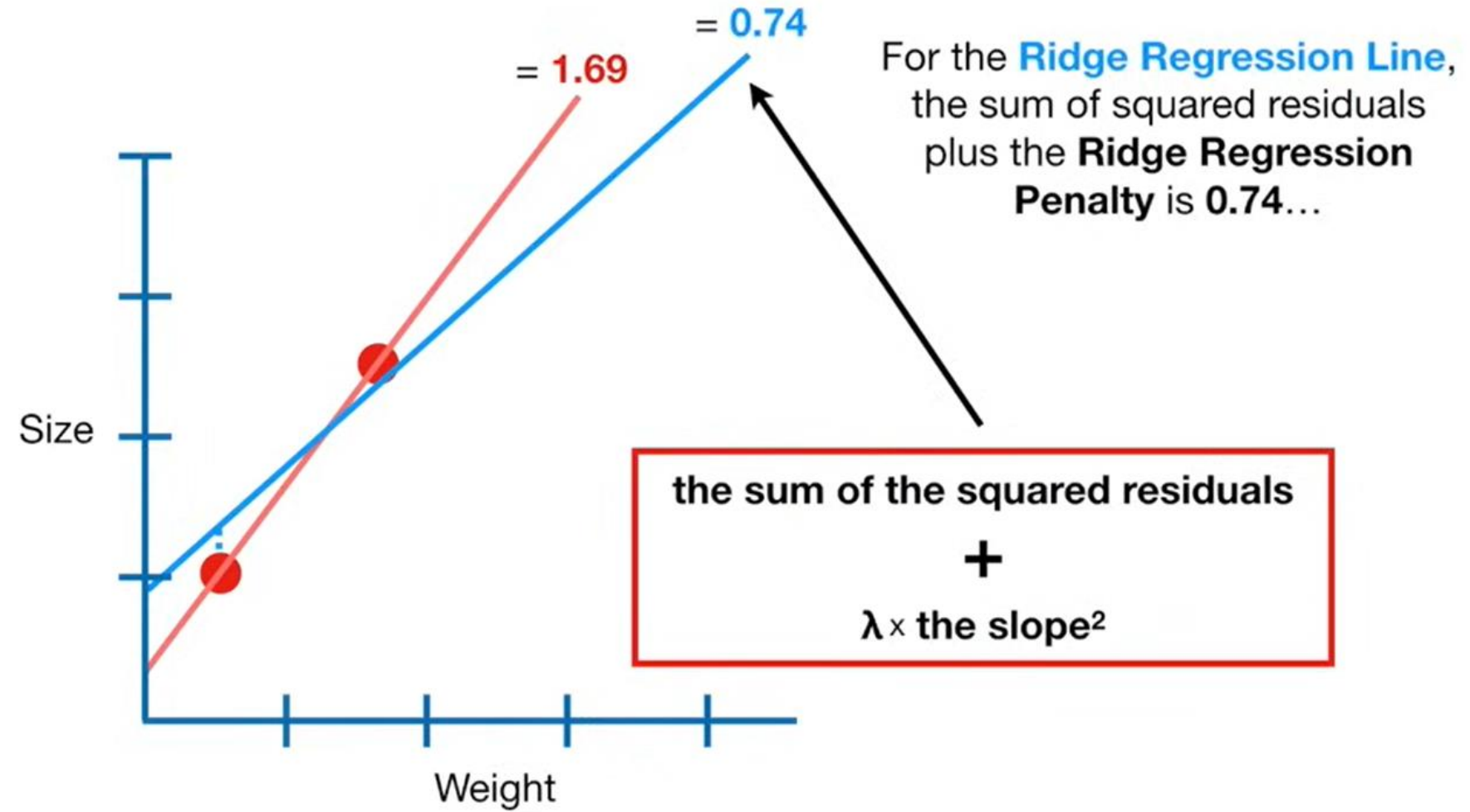
+

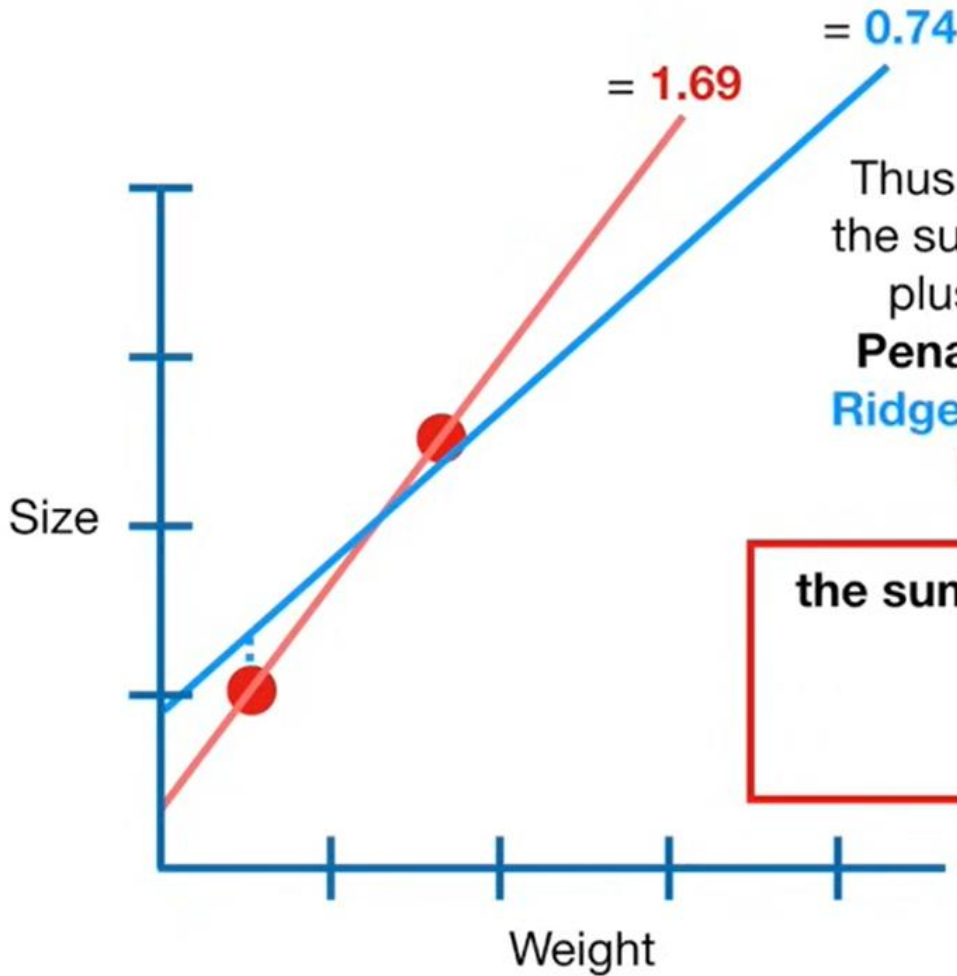
$$1 \times 0.8^2$$

$$= 0.09 + 0.01 + 0.64$$

$$= 0.74$$

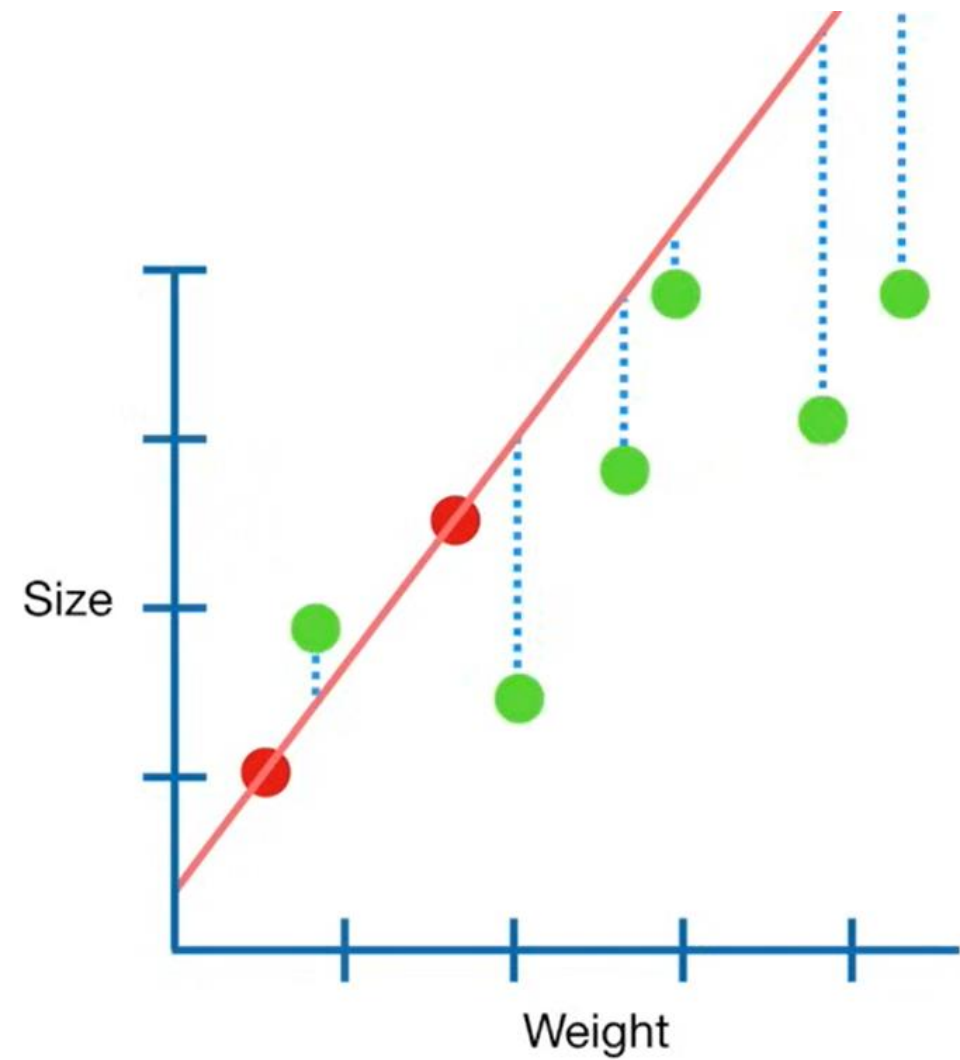




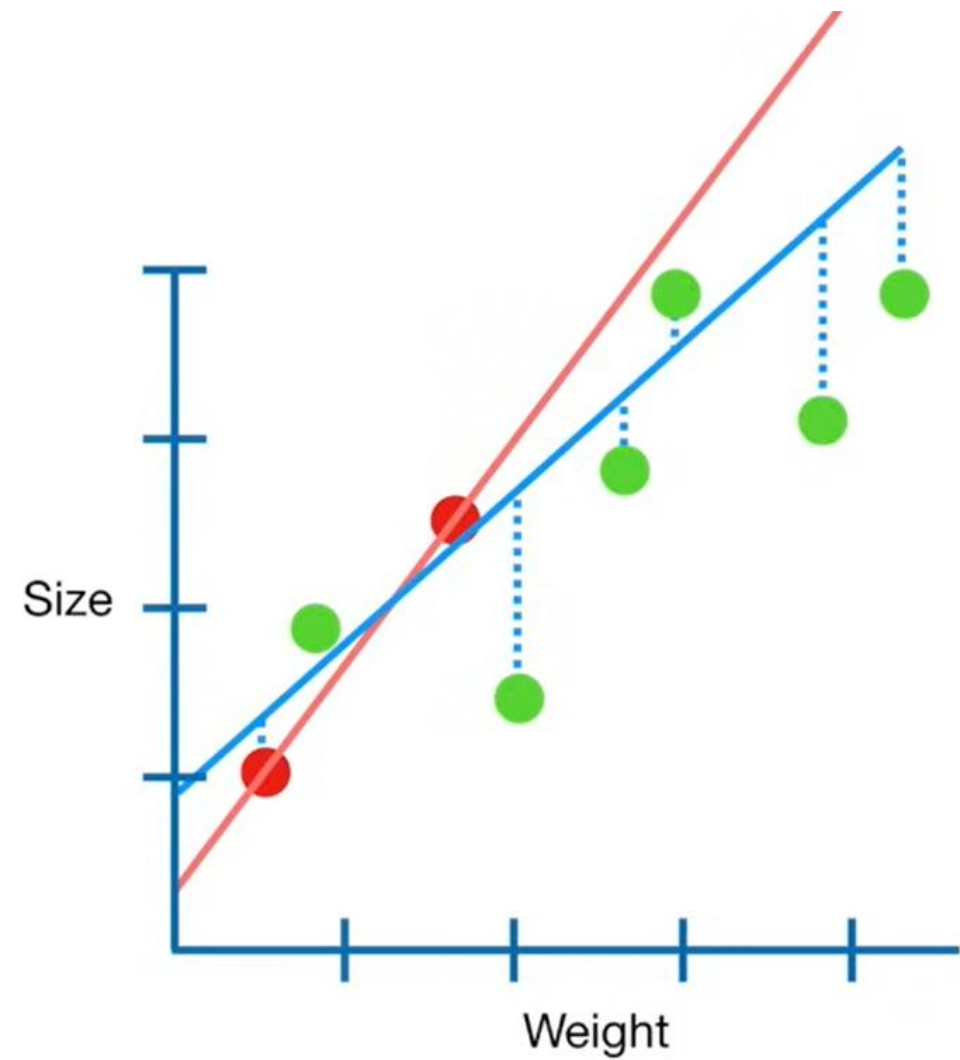


Thus, if we wanted to minimize the sum of the squared residuals plus the **Ridge Regression Penalty**, we would choose the **Ridge Regression Line** over the **Least Squares Line**.

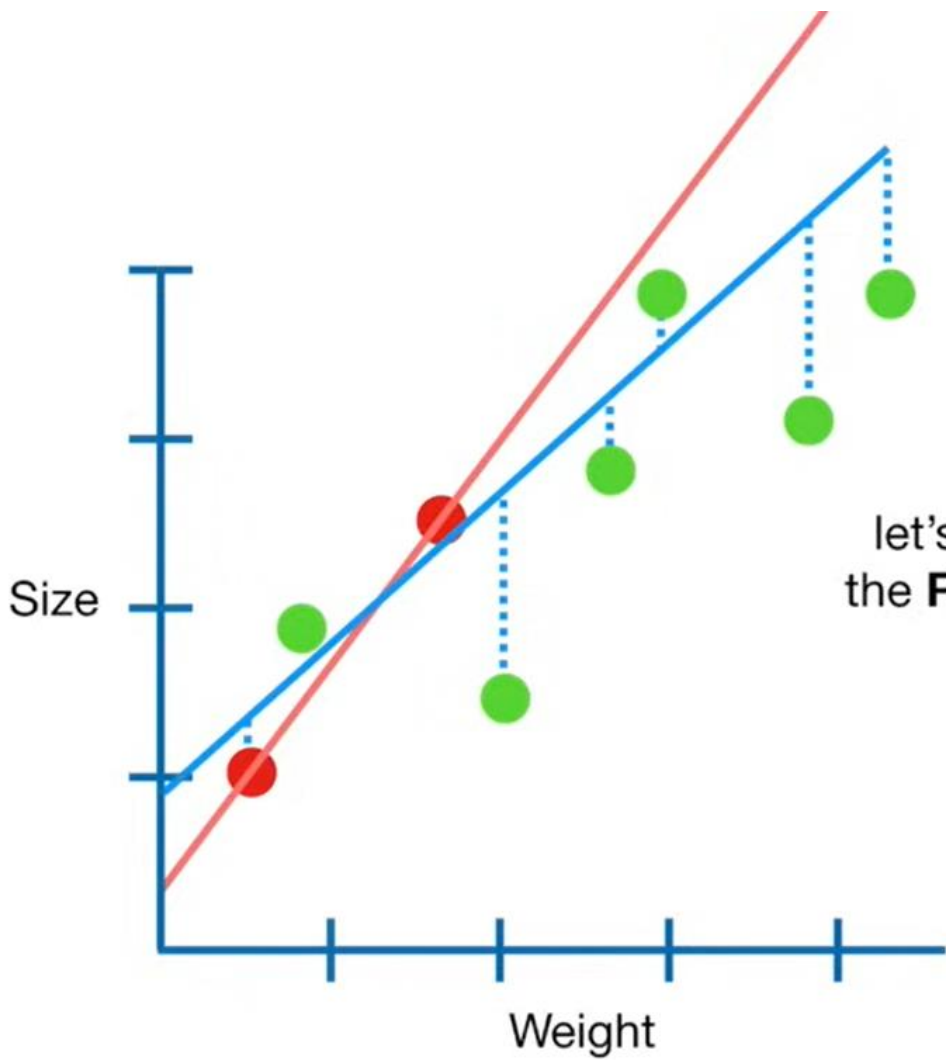
the sum of the squared residuals
+
 $\lambda \times \text{the slope}^2$



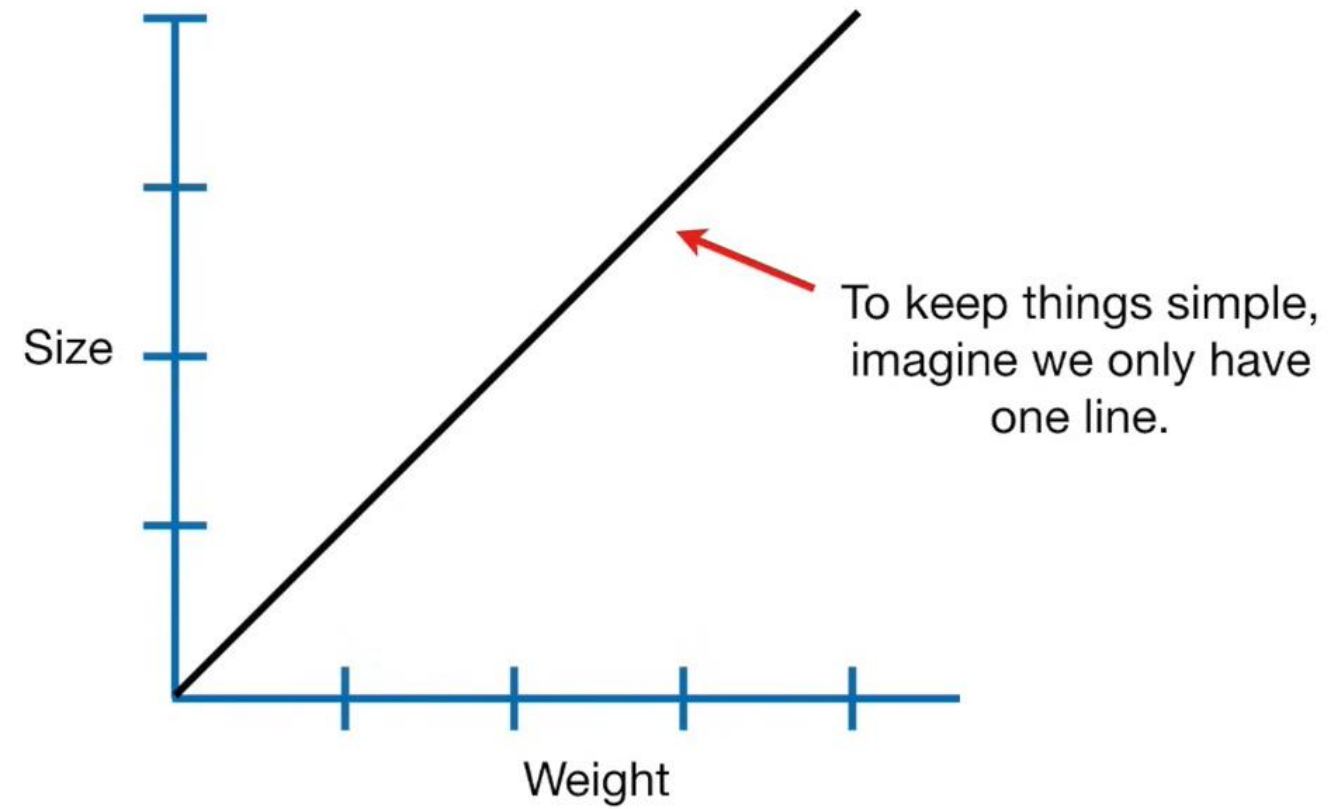
Without the small amount of **Bias** that the penalty creates, the **Least Squares Fit** has a large amount of **Variance**.

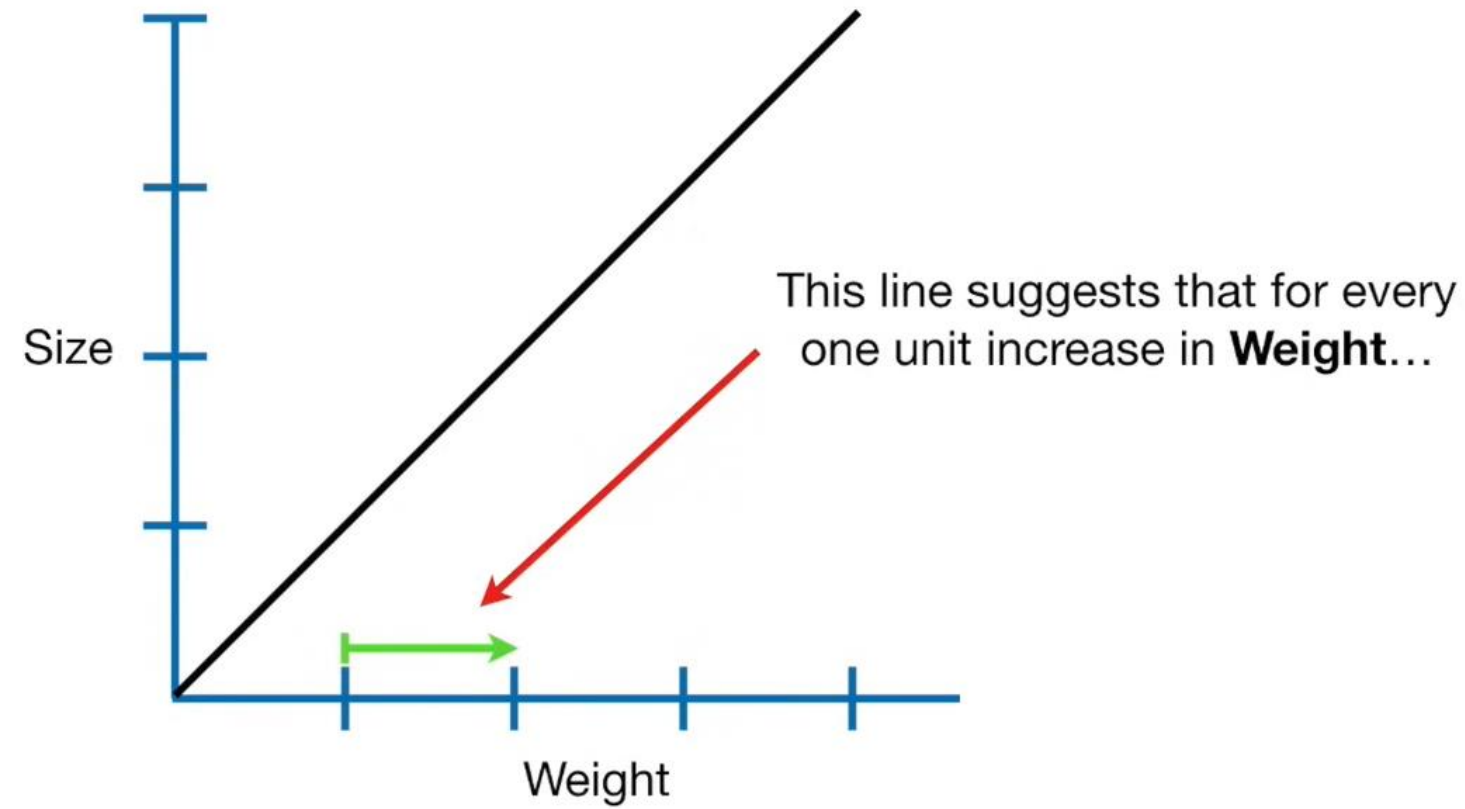


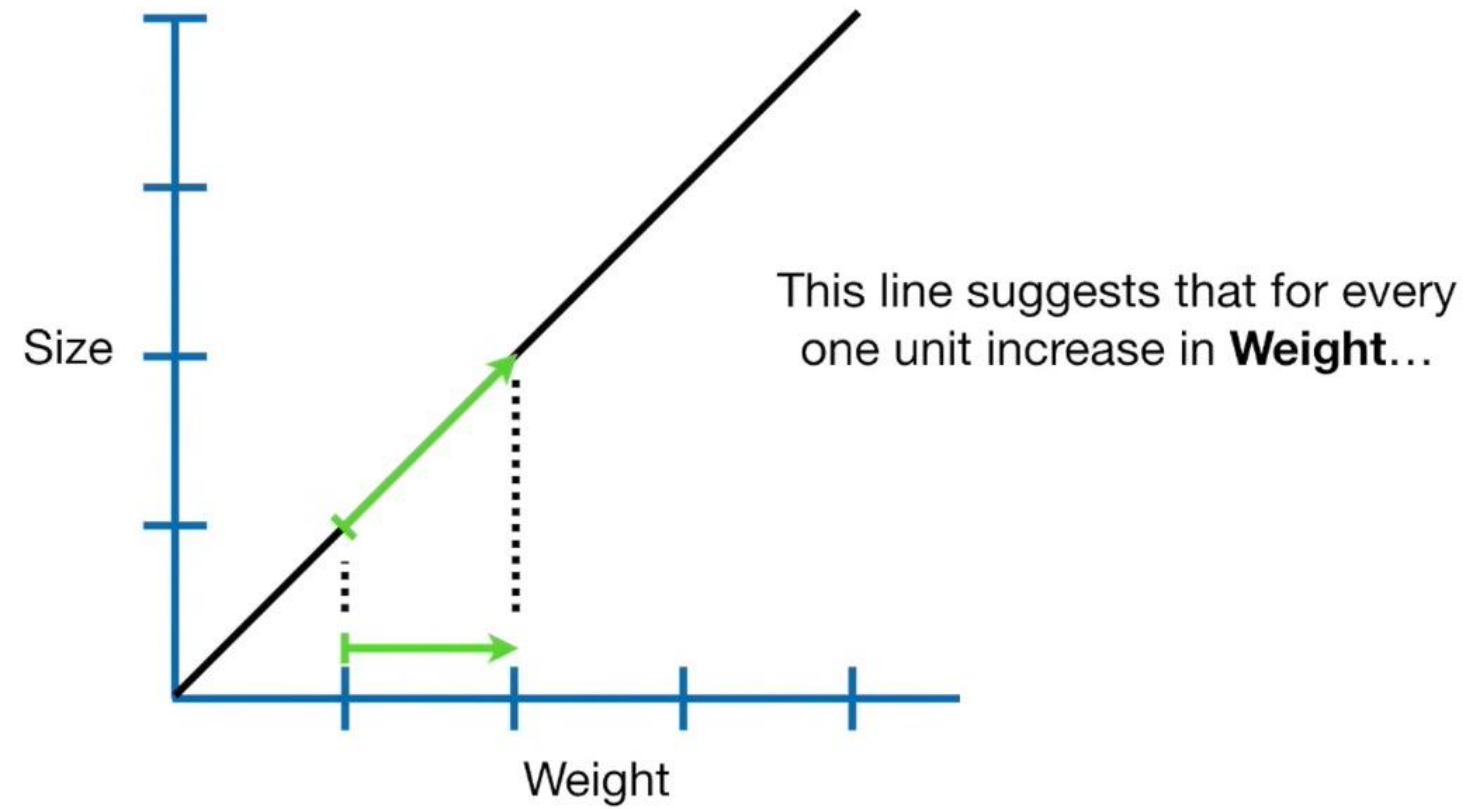
In contrast, the **Ridge Regression Line**, which has the small amount of **Bias** due to the penalty, has less **Variance**.

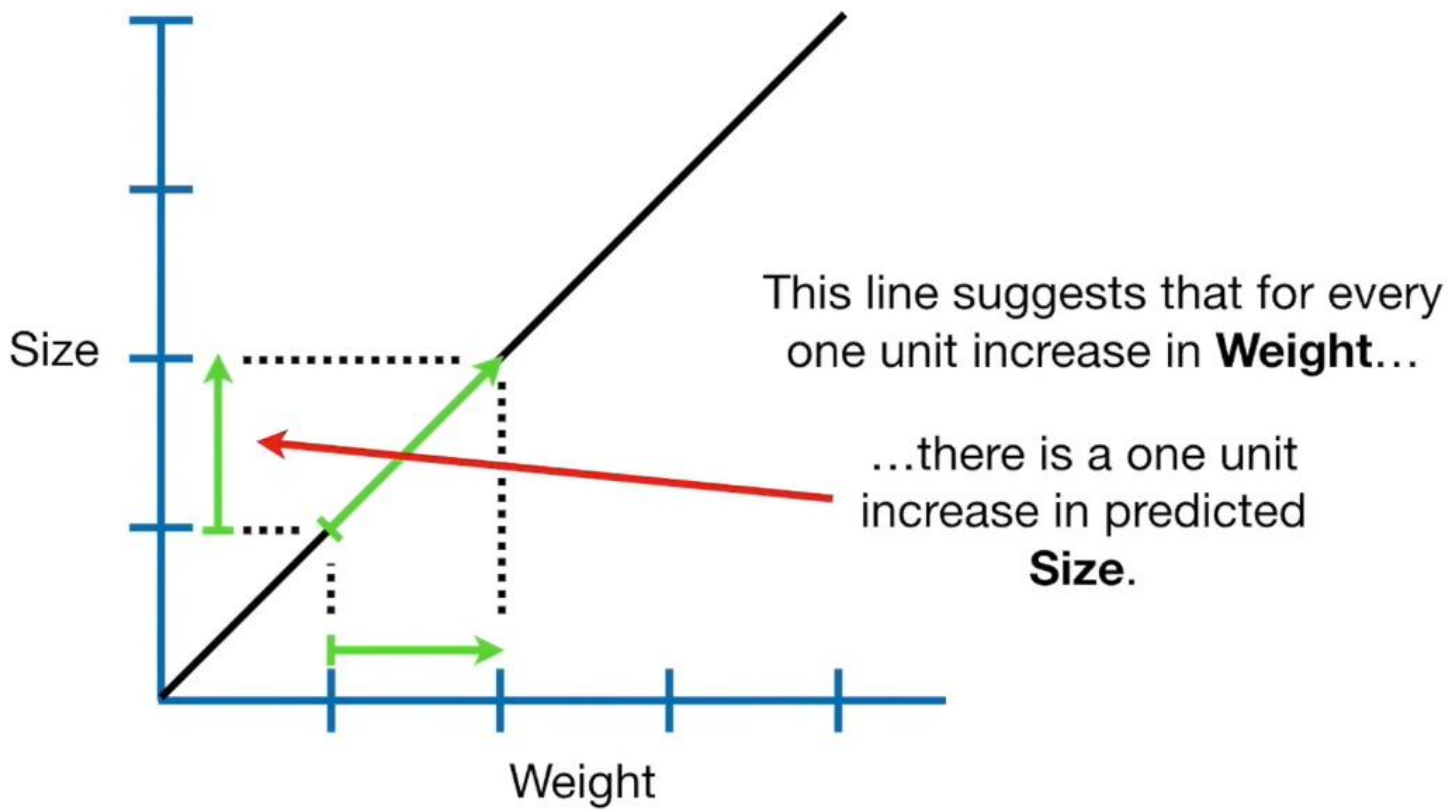


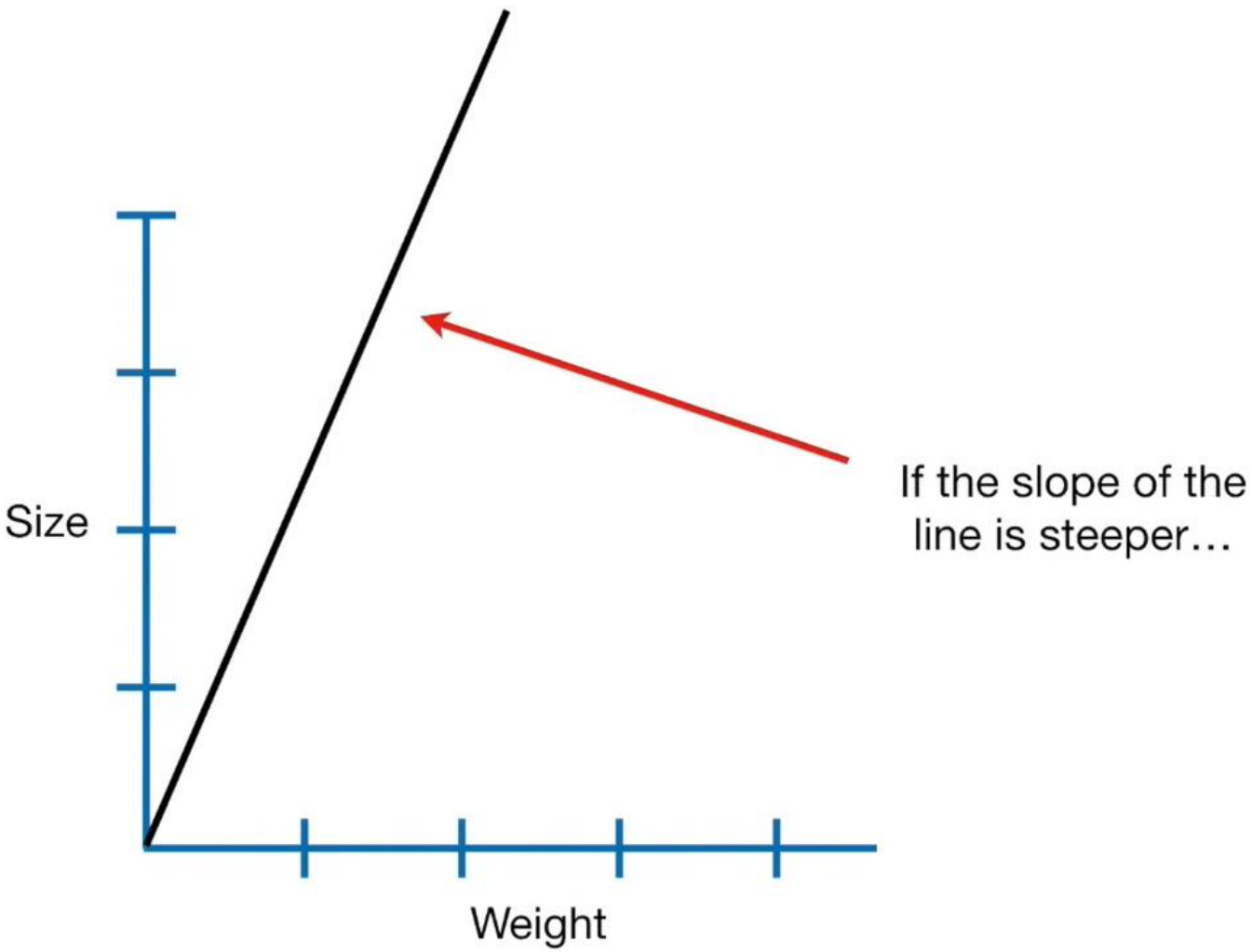
Now before we talk about λ ,
let's talk a little bit more about the effect that
the **Ridge Regression Penalty** has on how the
line is fit to the data.

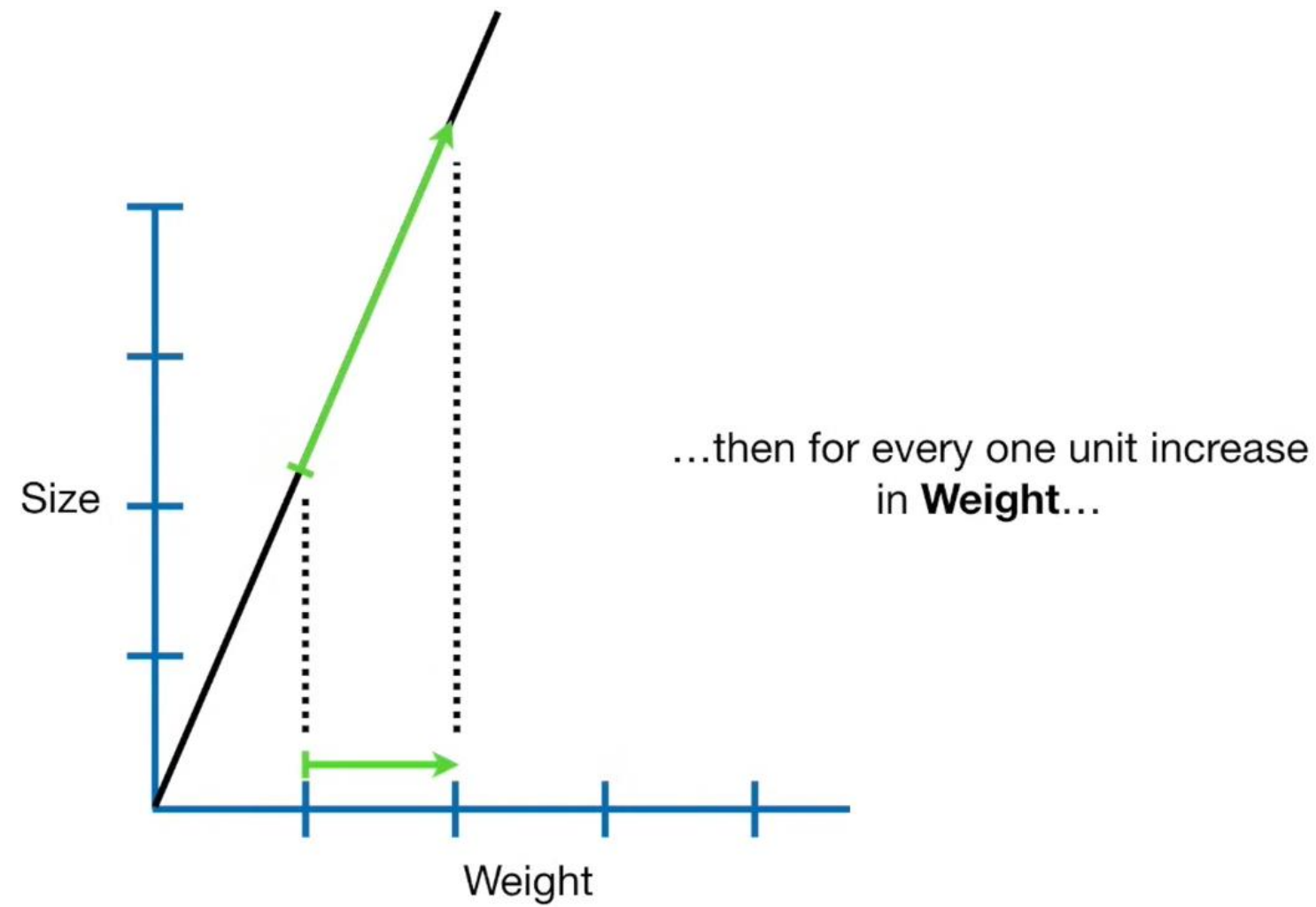


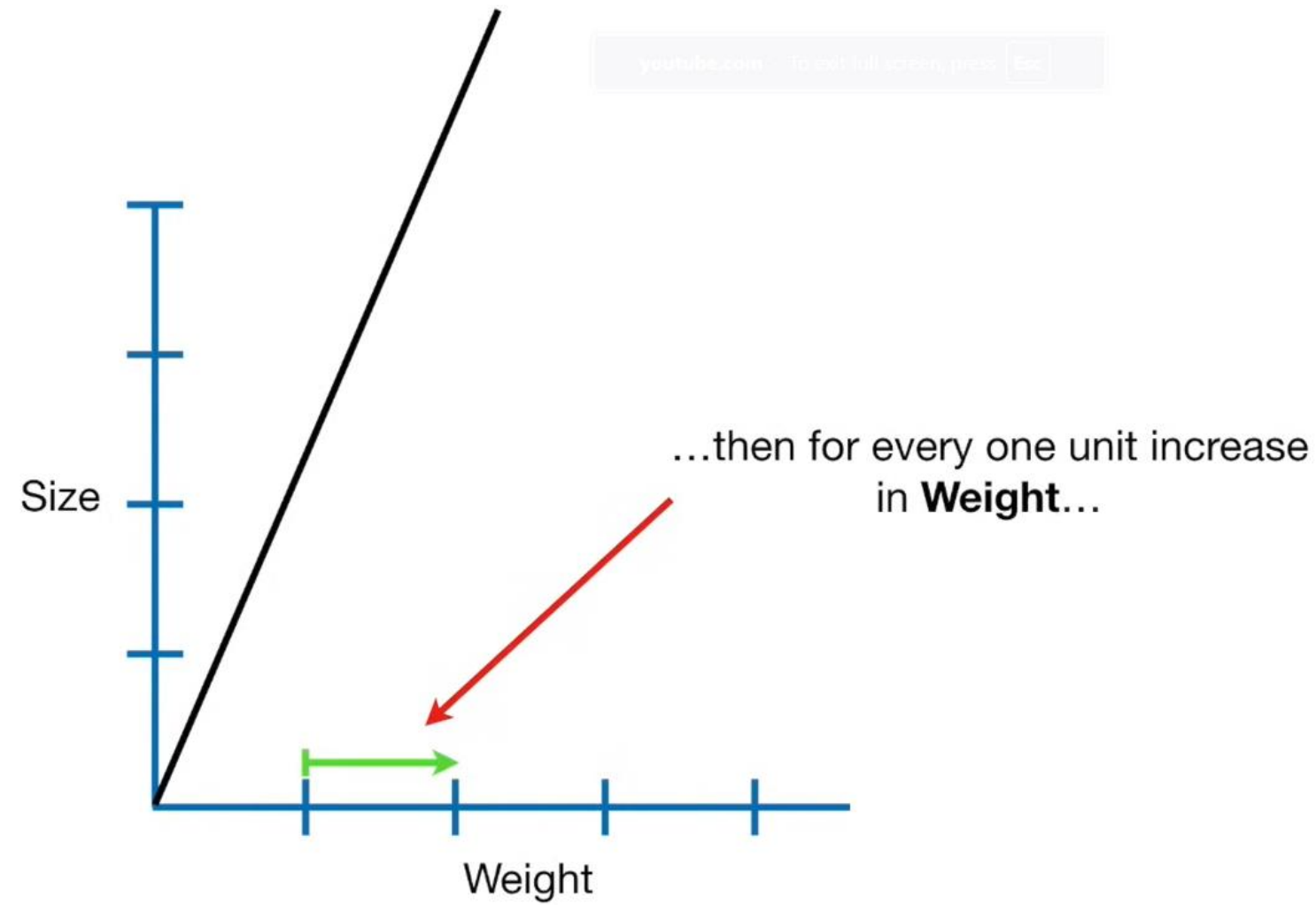


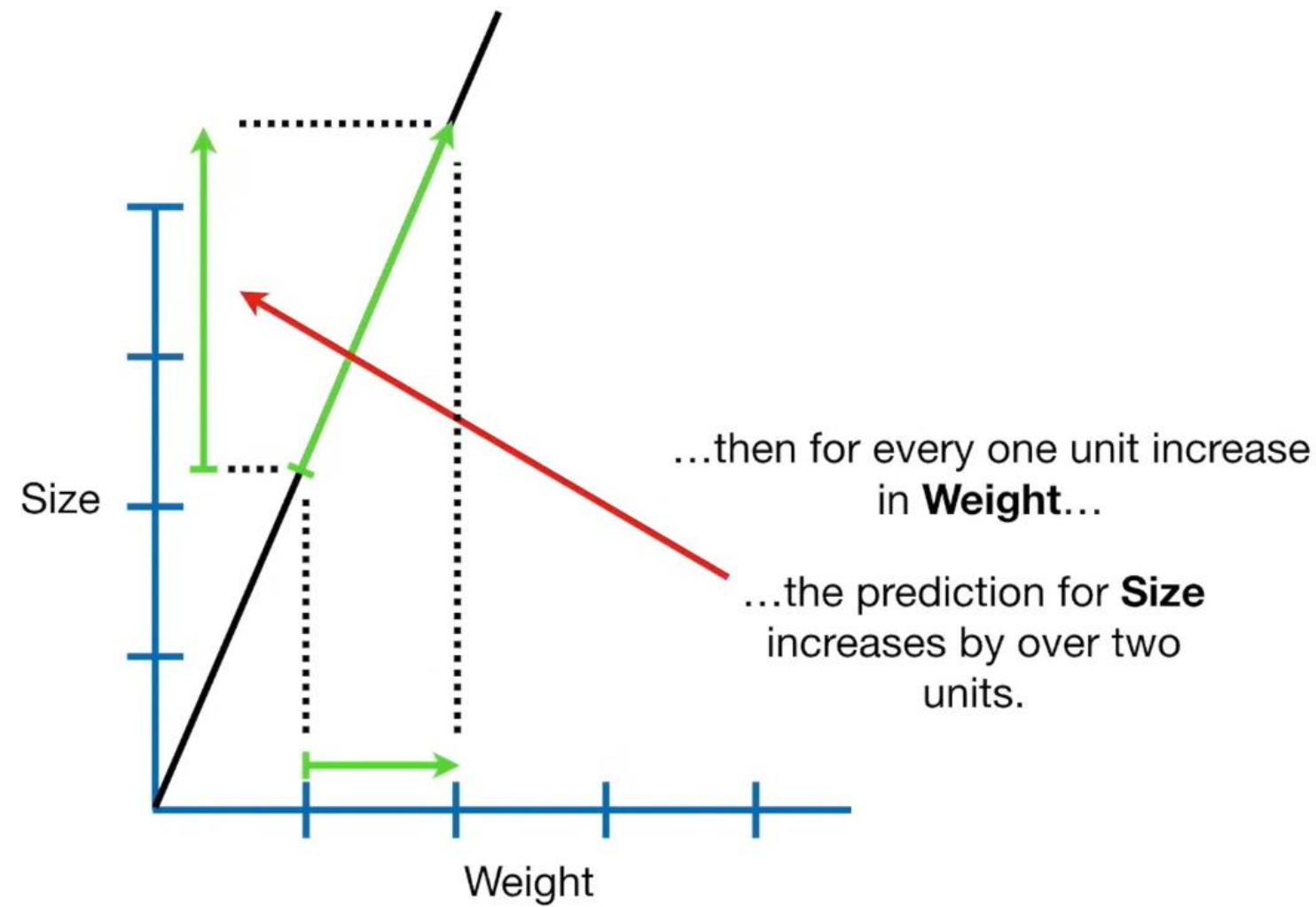


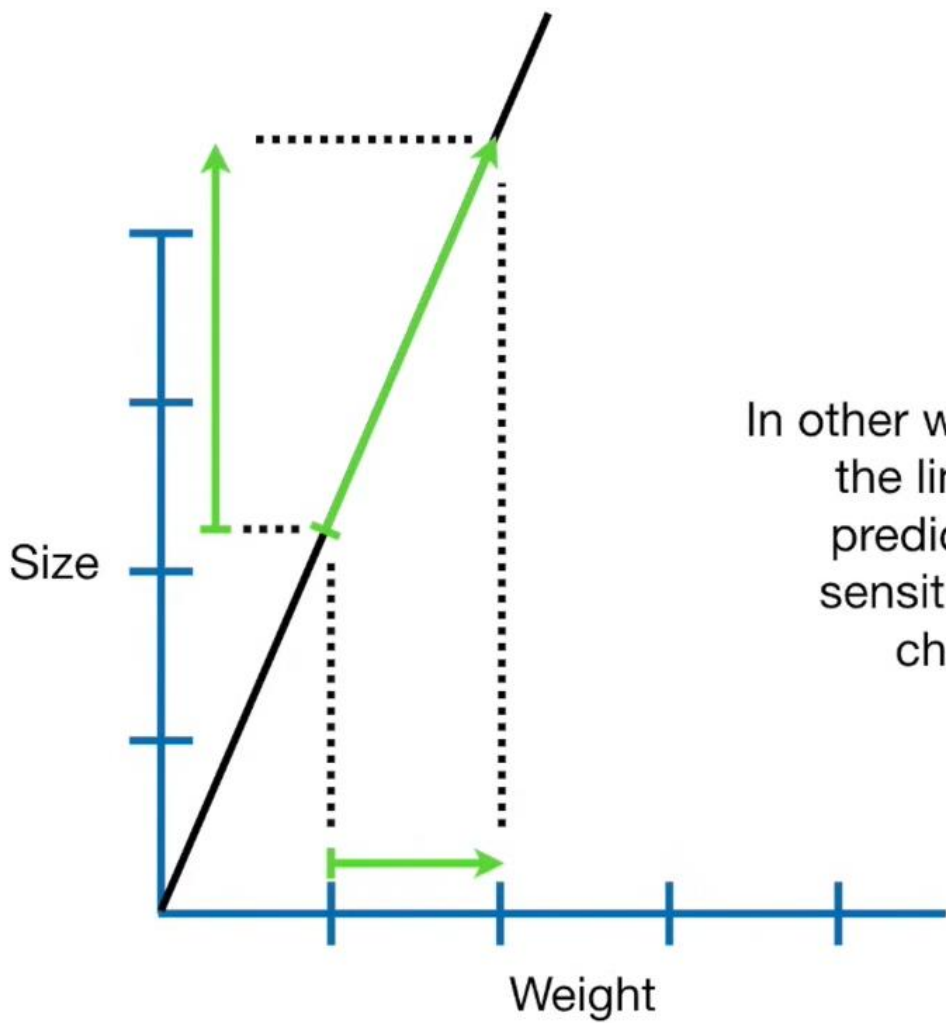




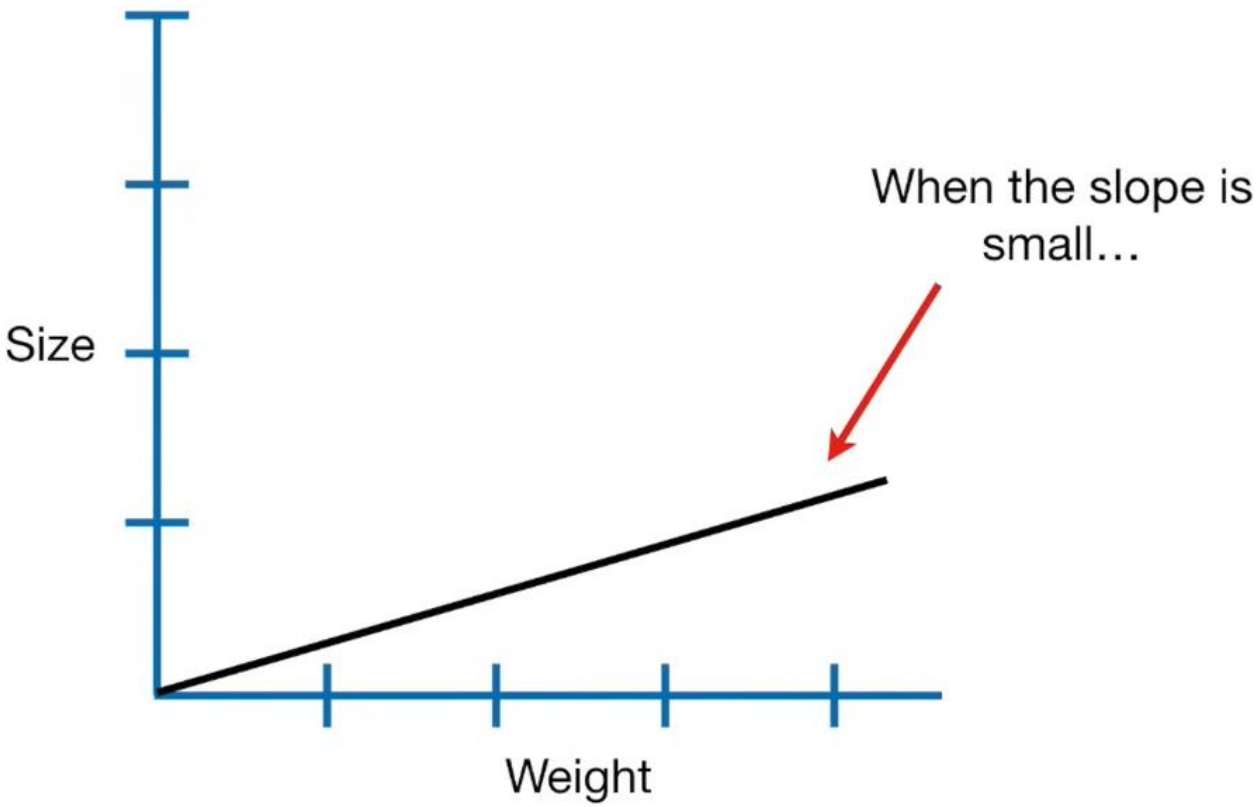


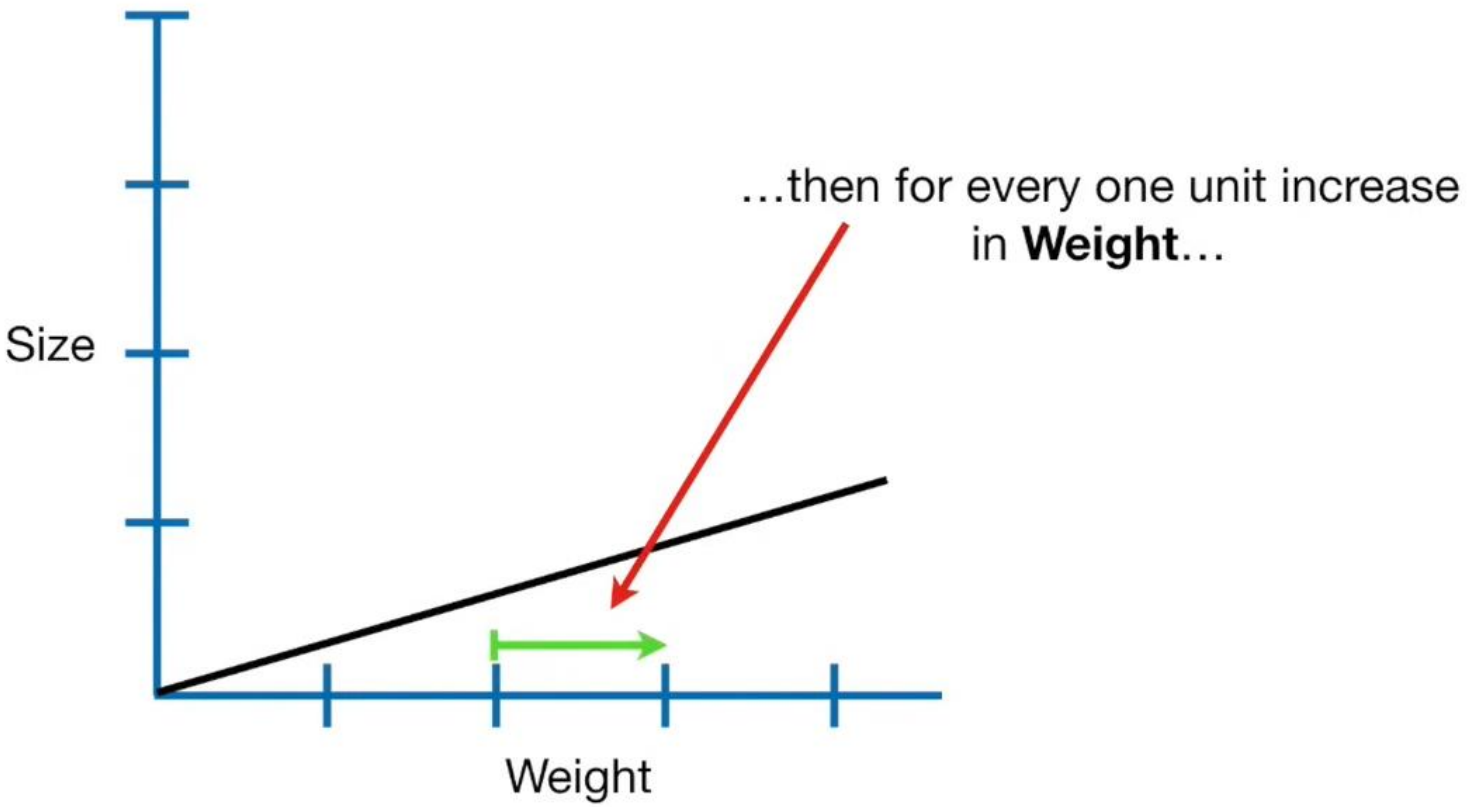




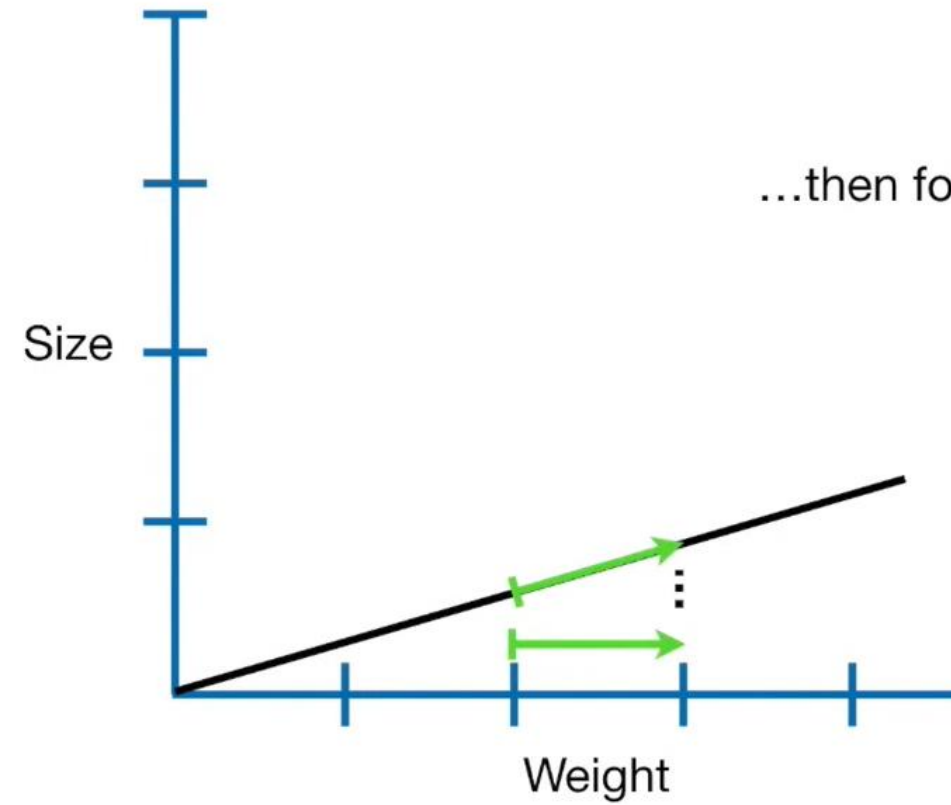


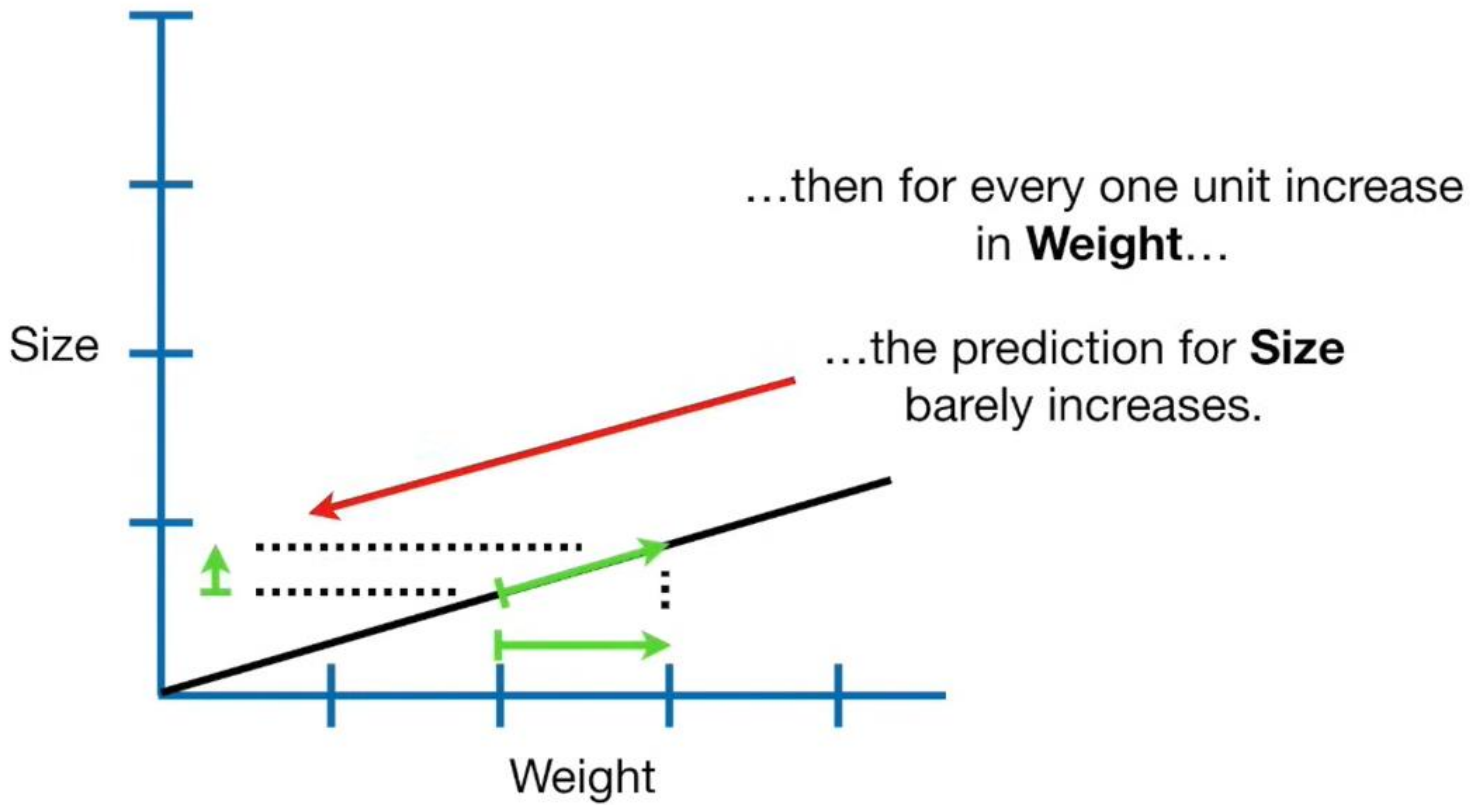
In other words, when the slope of the line is steep, then the prediction for **Size** is very sensitive to relatively small changes in **Weight**.

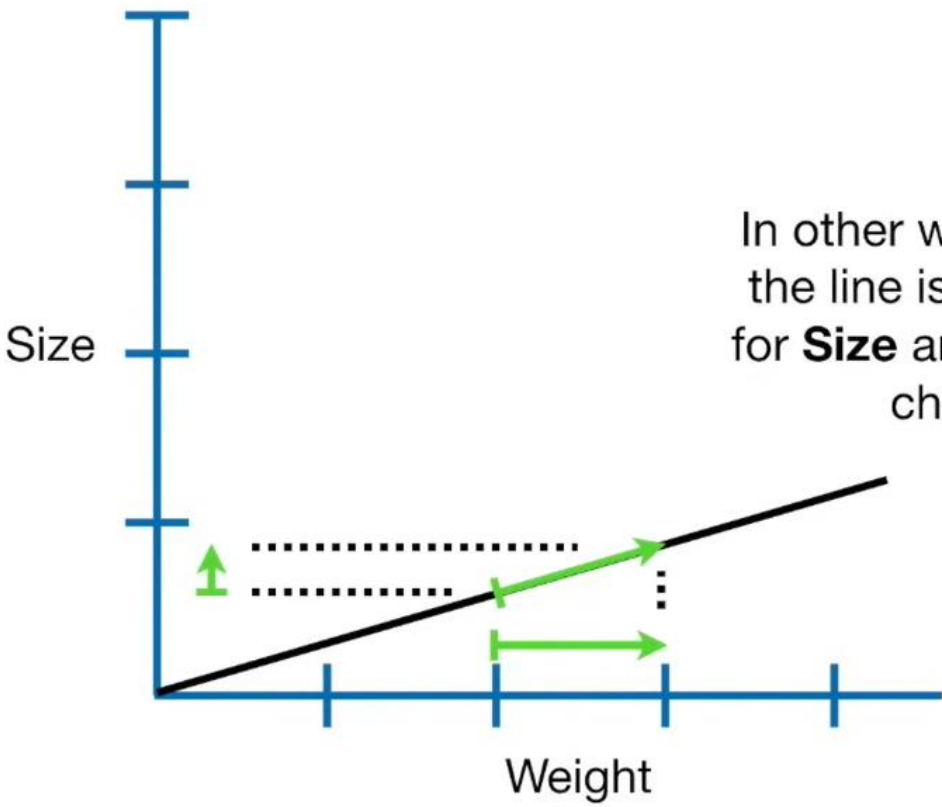




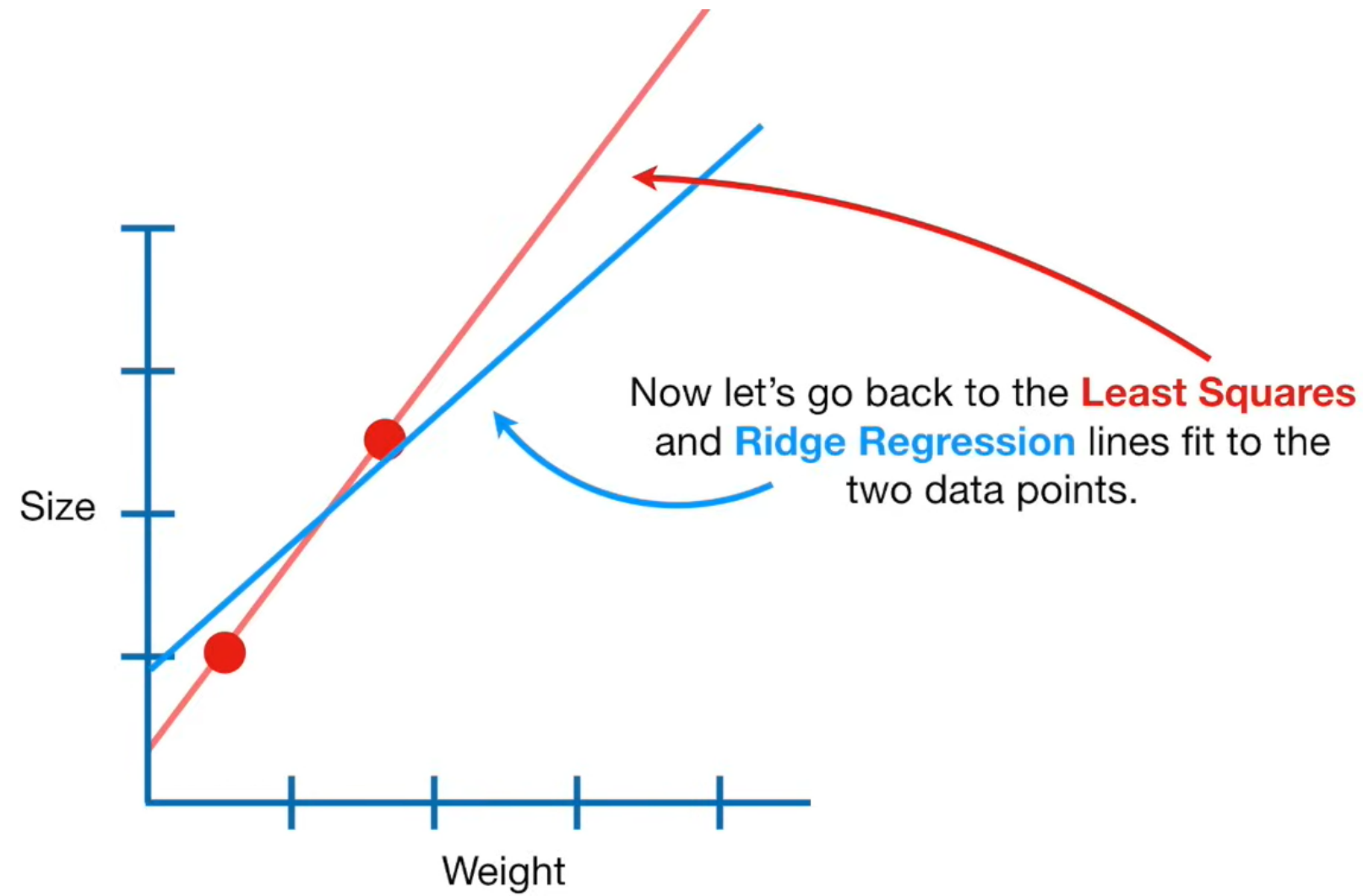
...then for every one unit increase
in **Weight**...

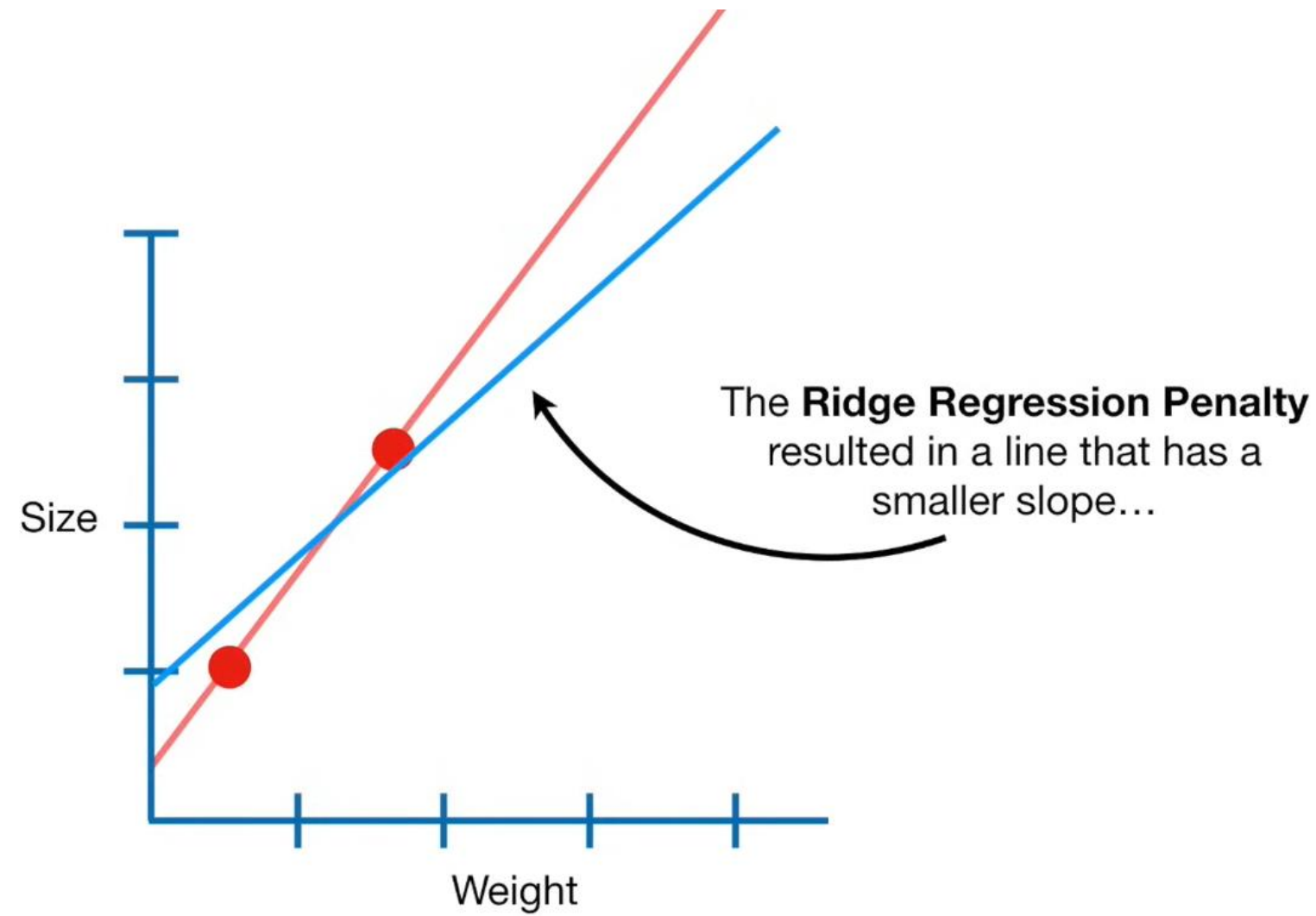


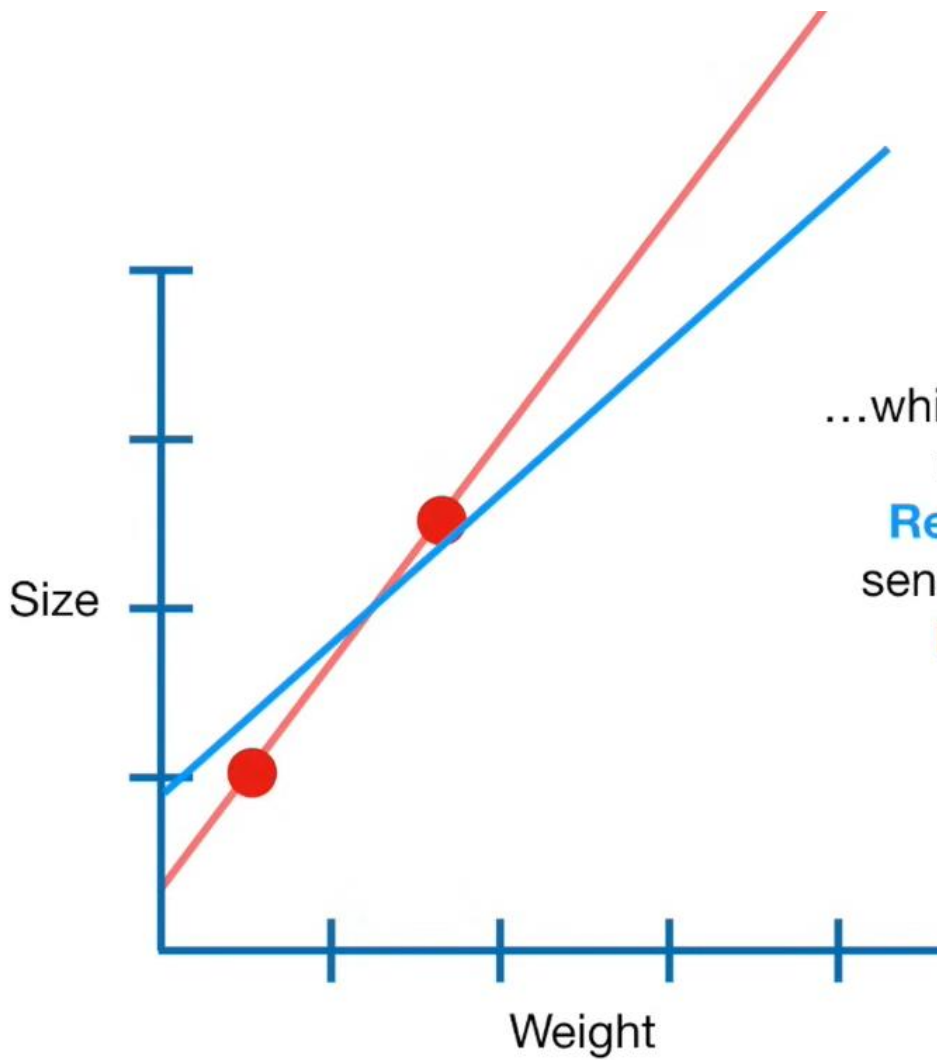




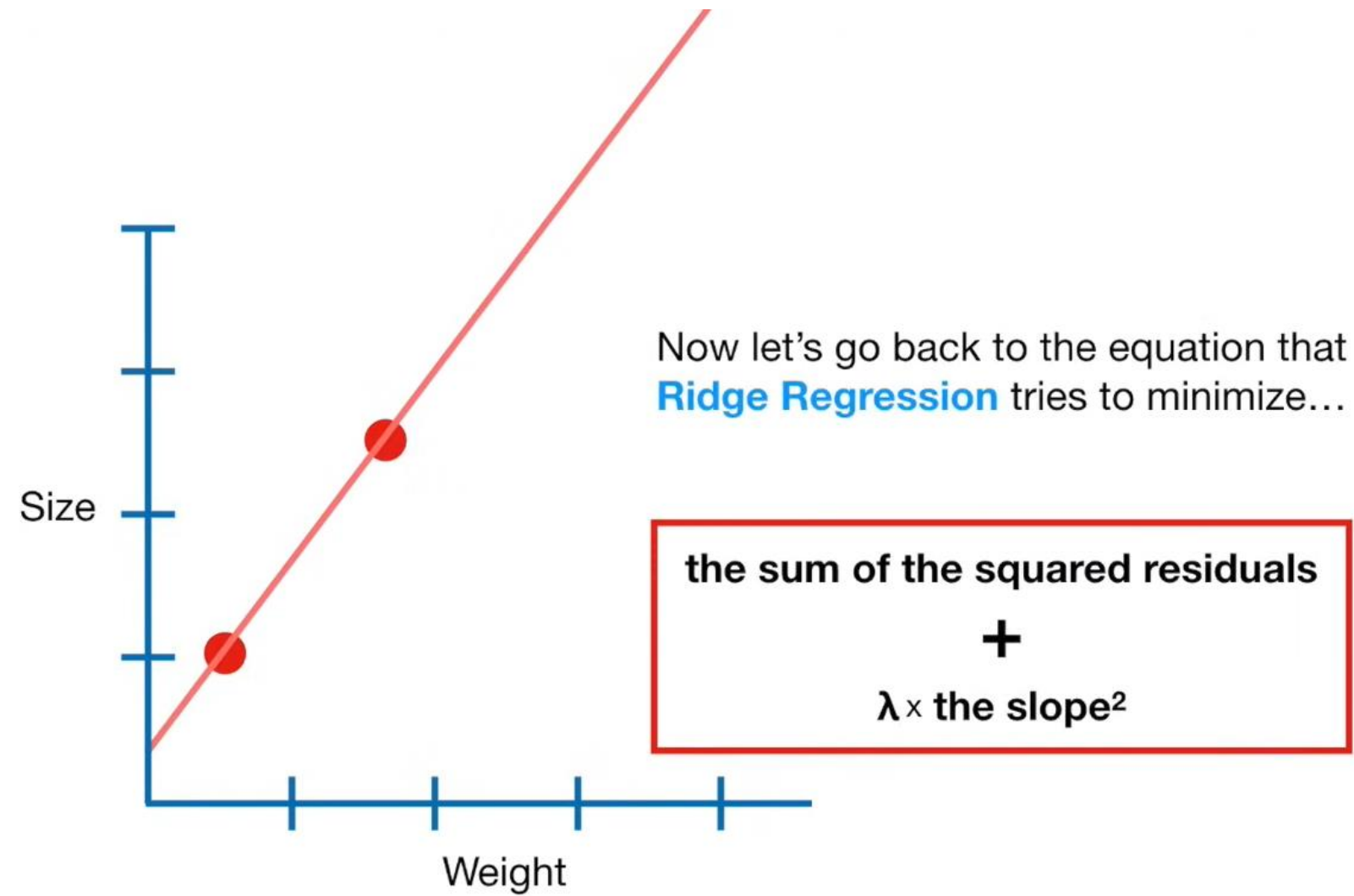
In other words, when the slope of the line is small, then predictions for **Size** are much less sensitive to changes in **Weight**.

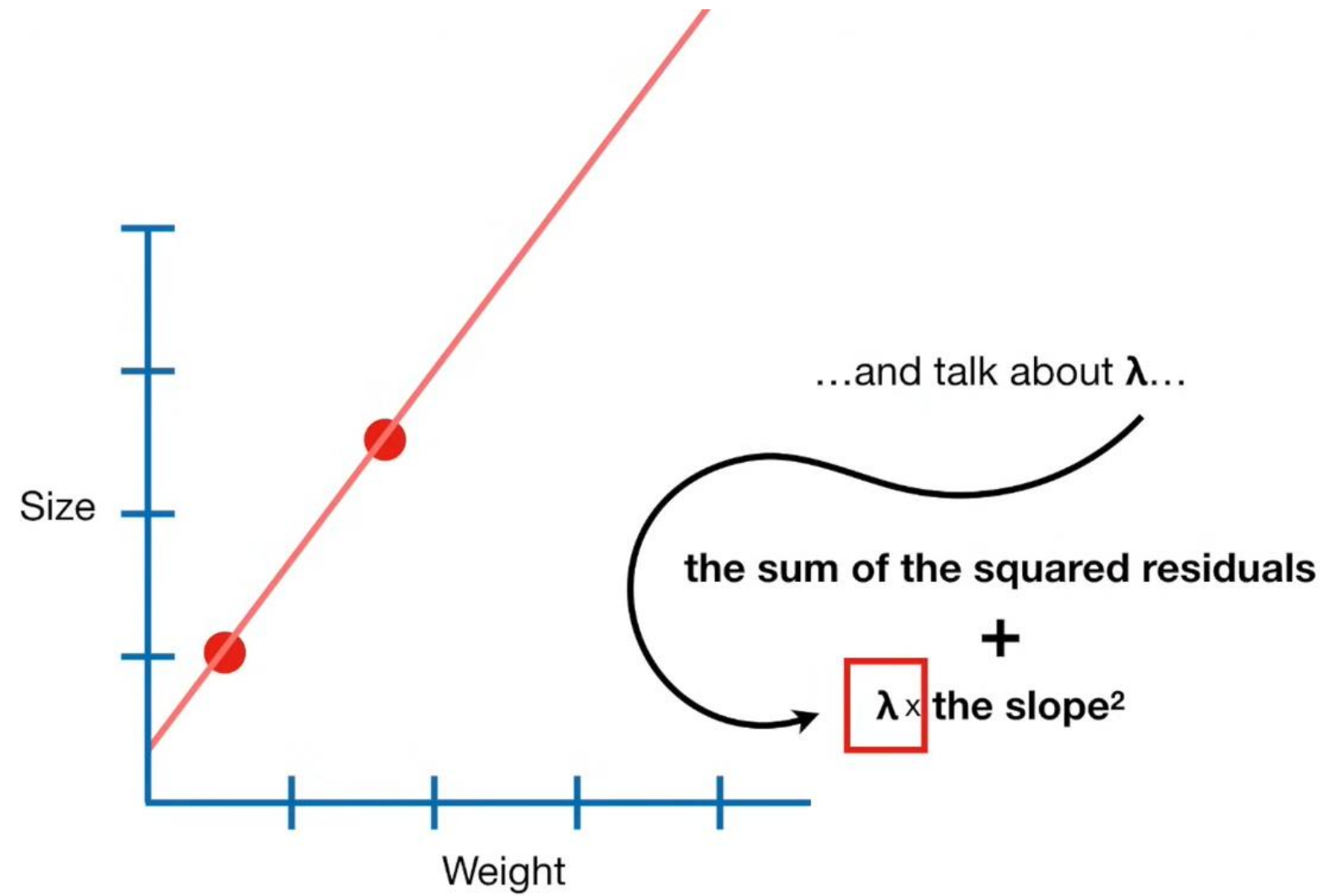


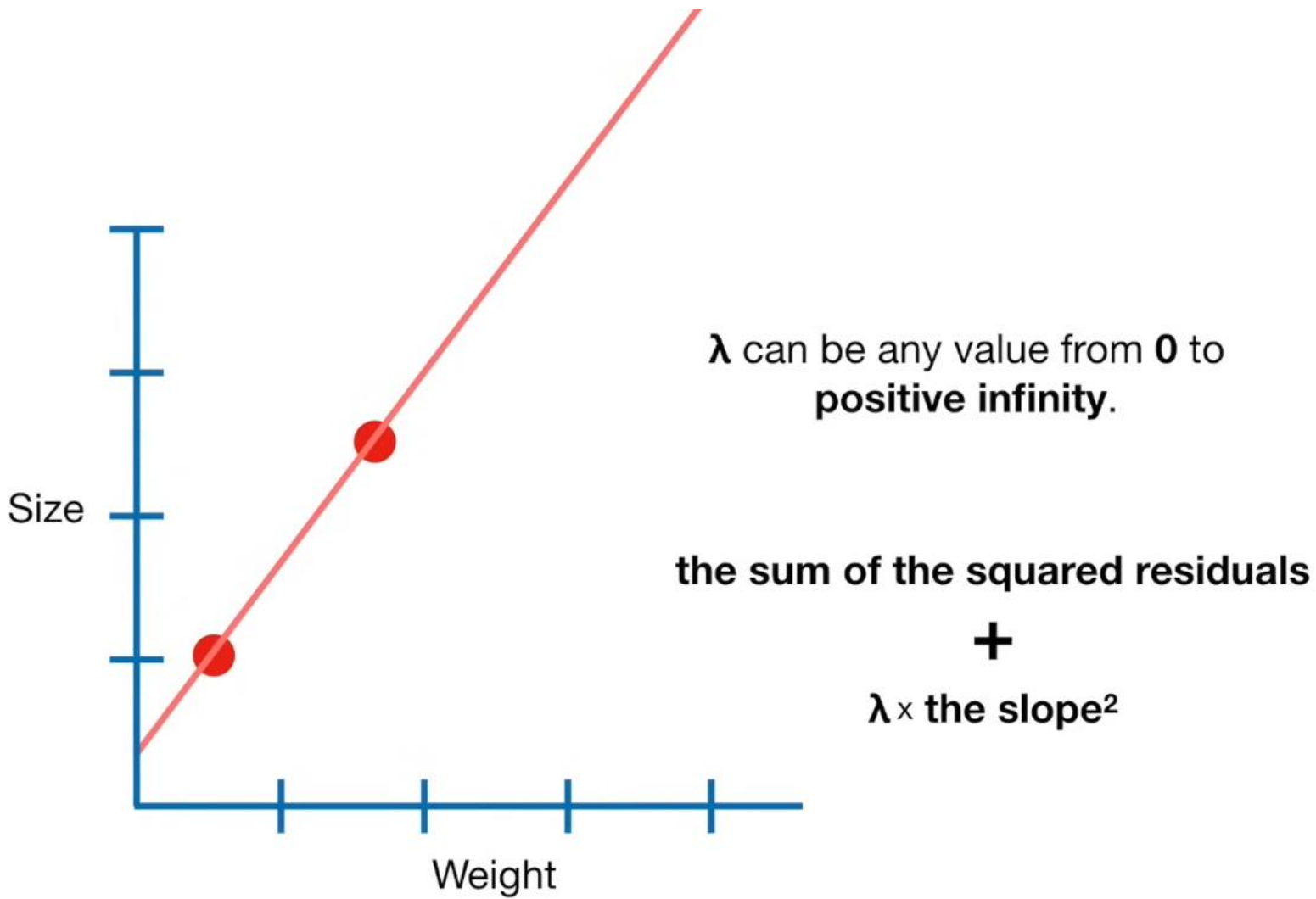


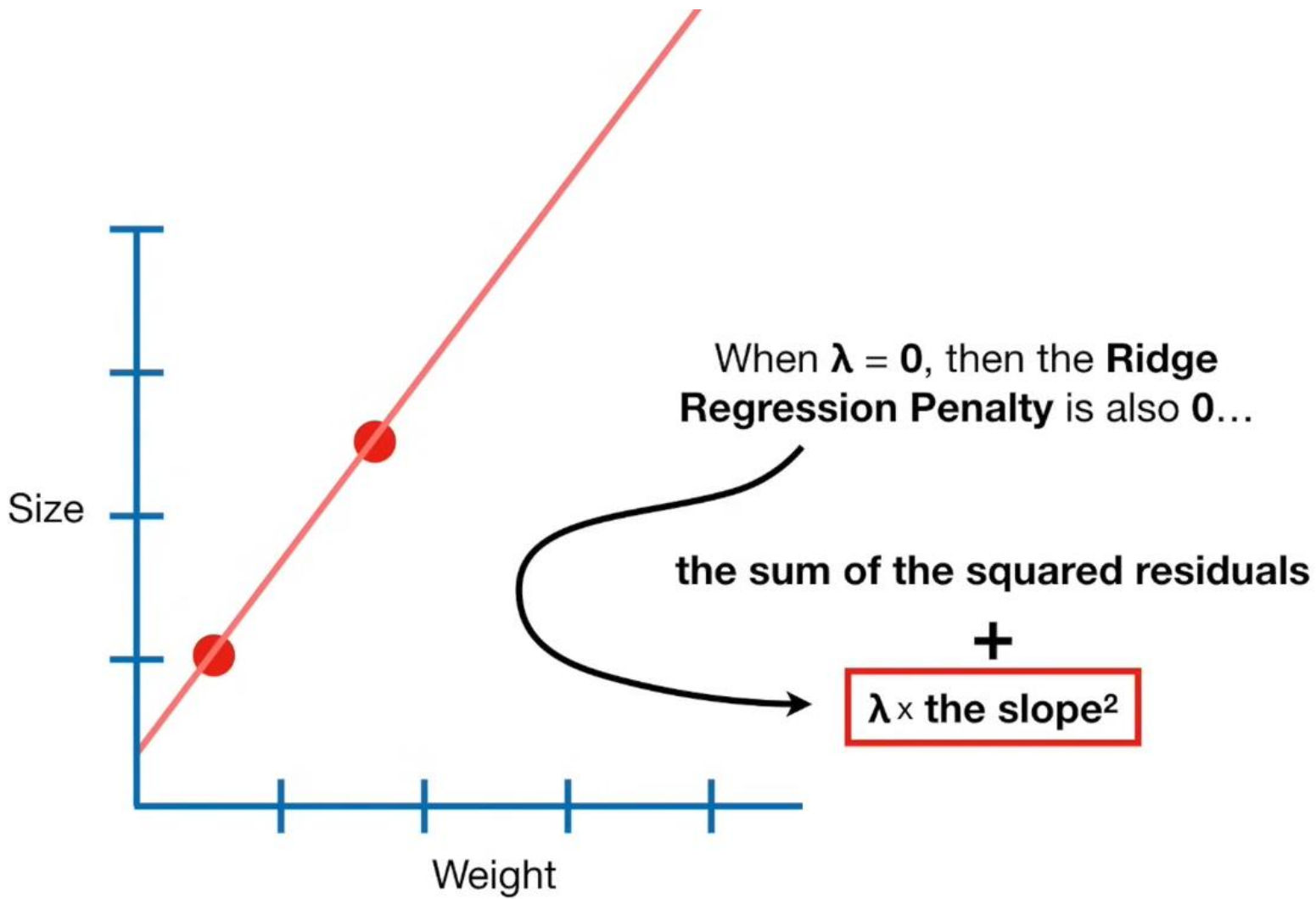


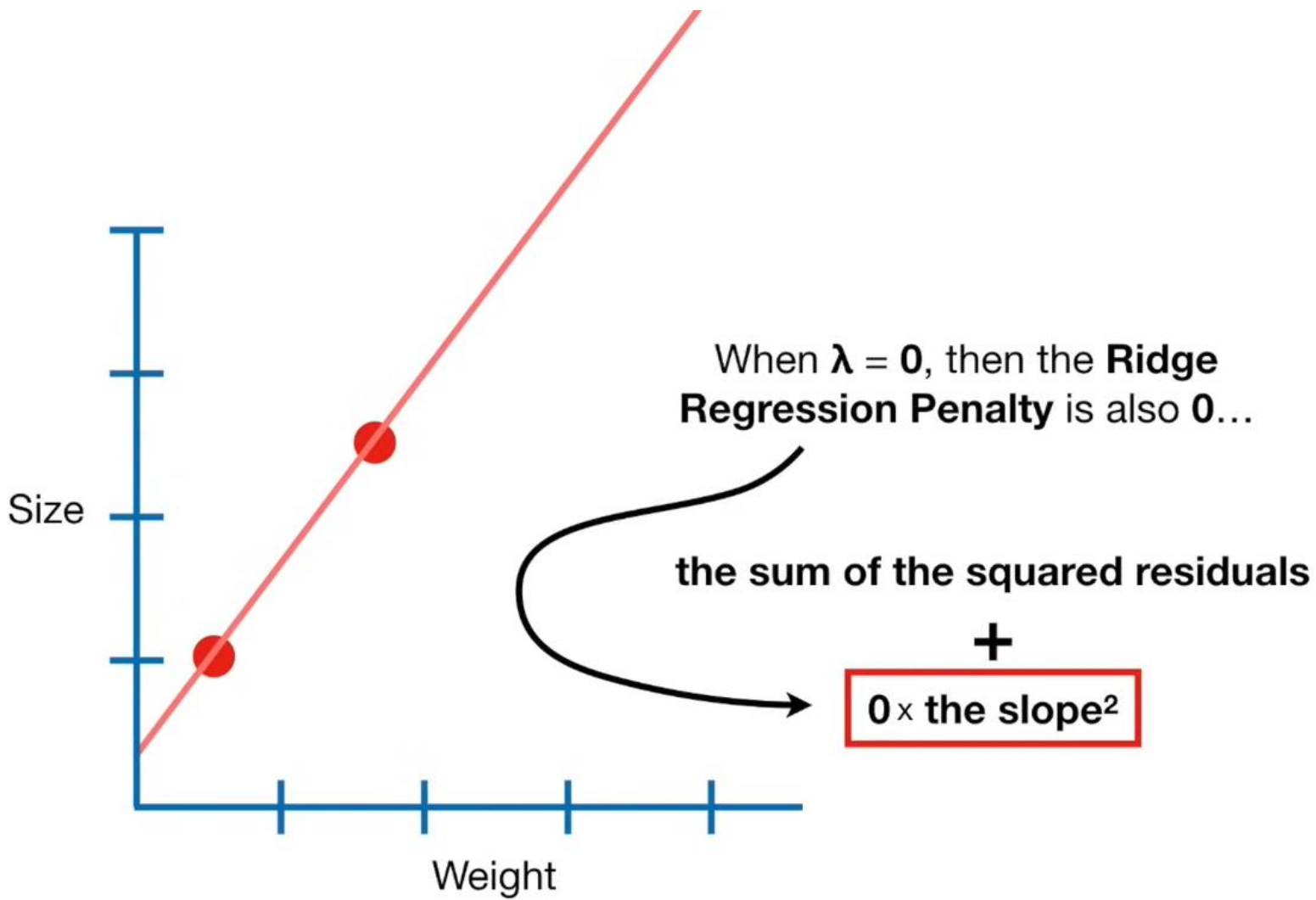
...which means that predictions made with the **Ridge Regression Line** are less sensitive to **Weight** than the **Least Squares Line**.

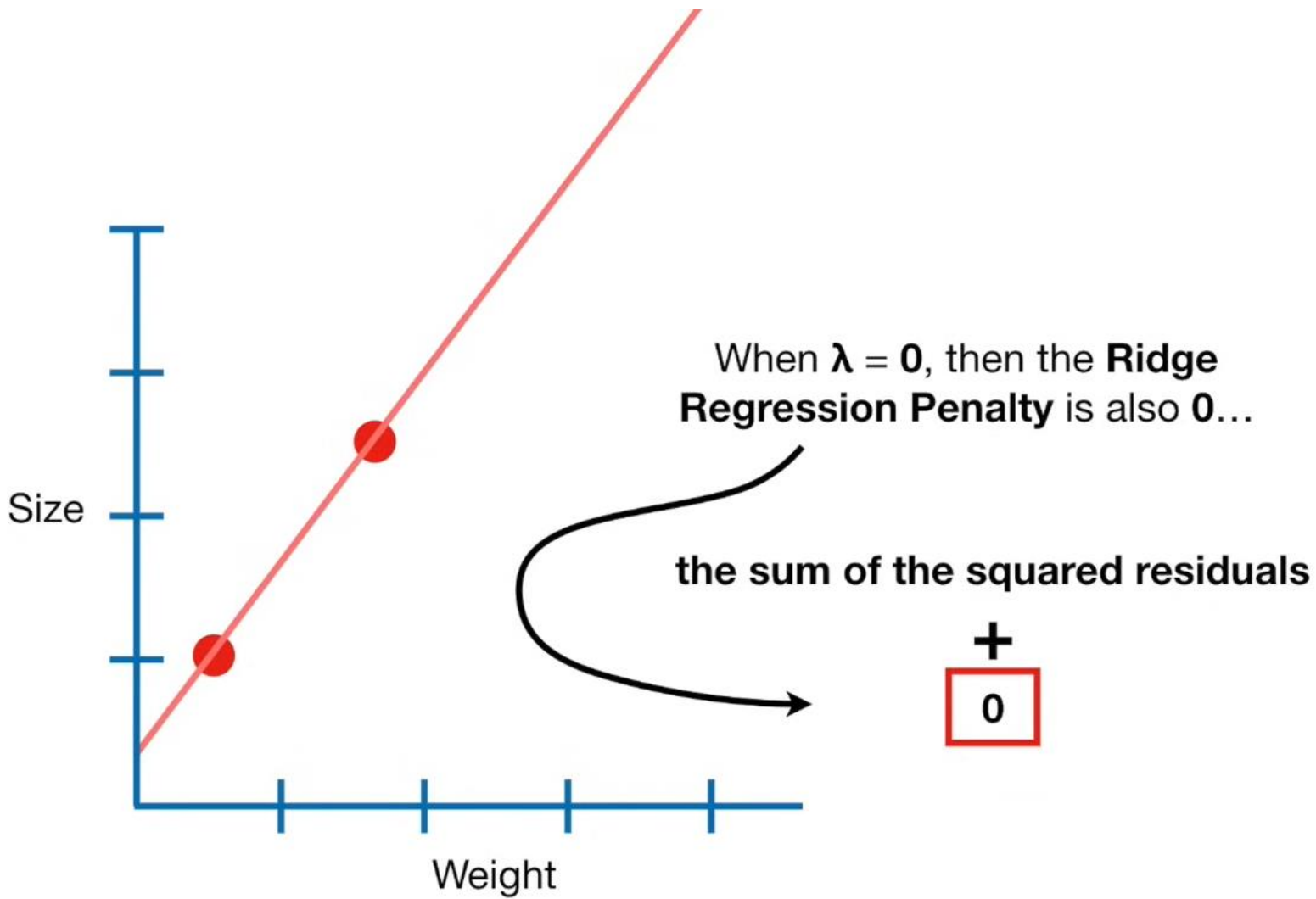


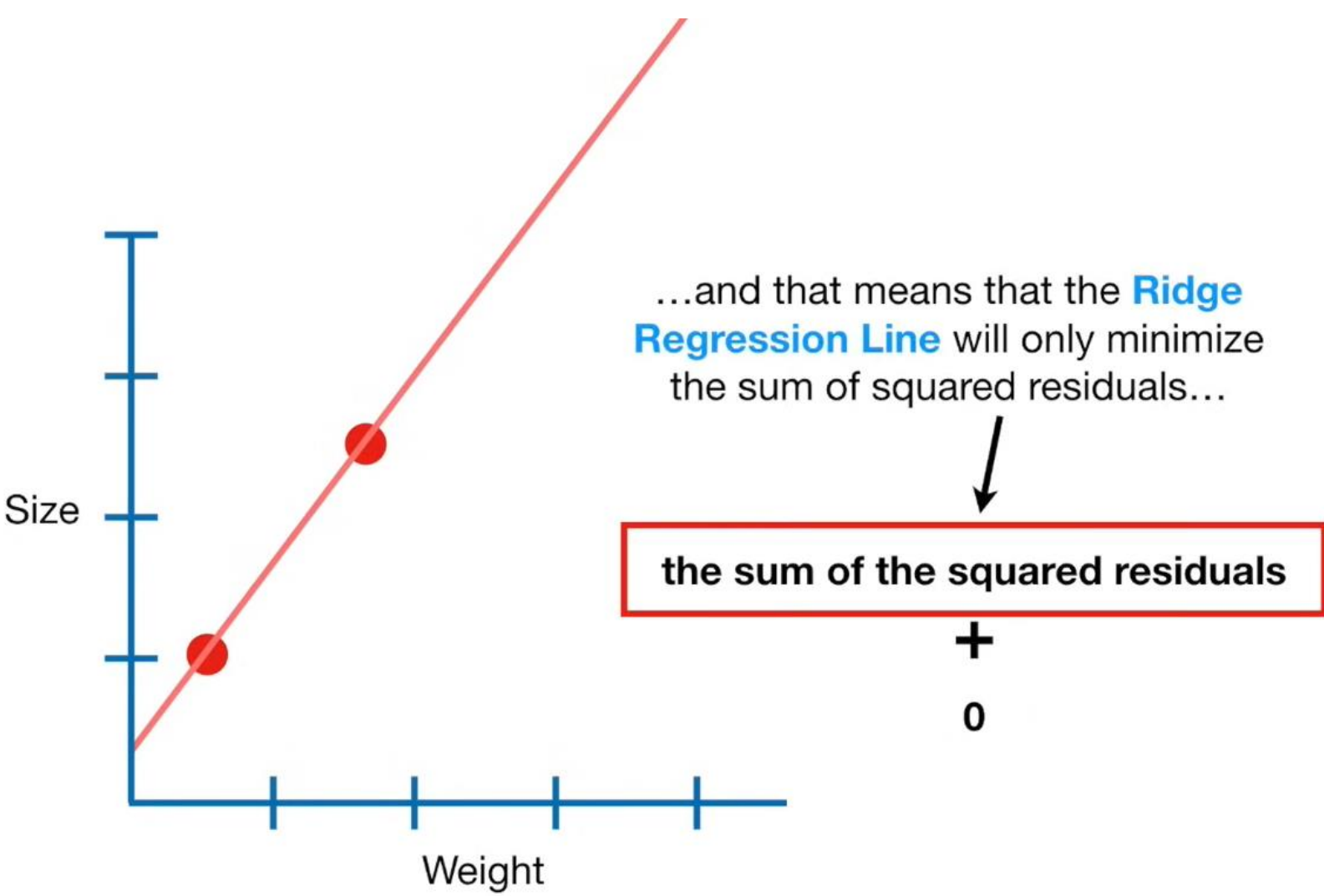


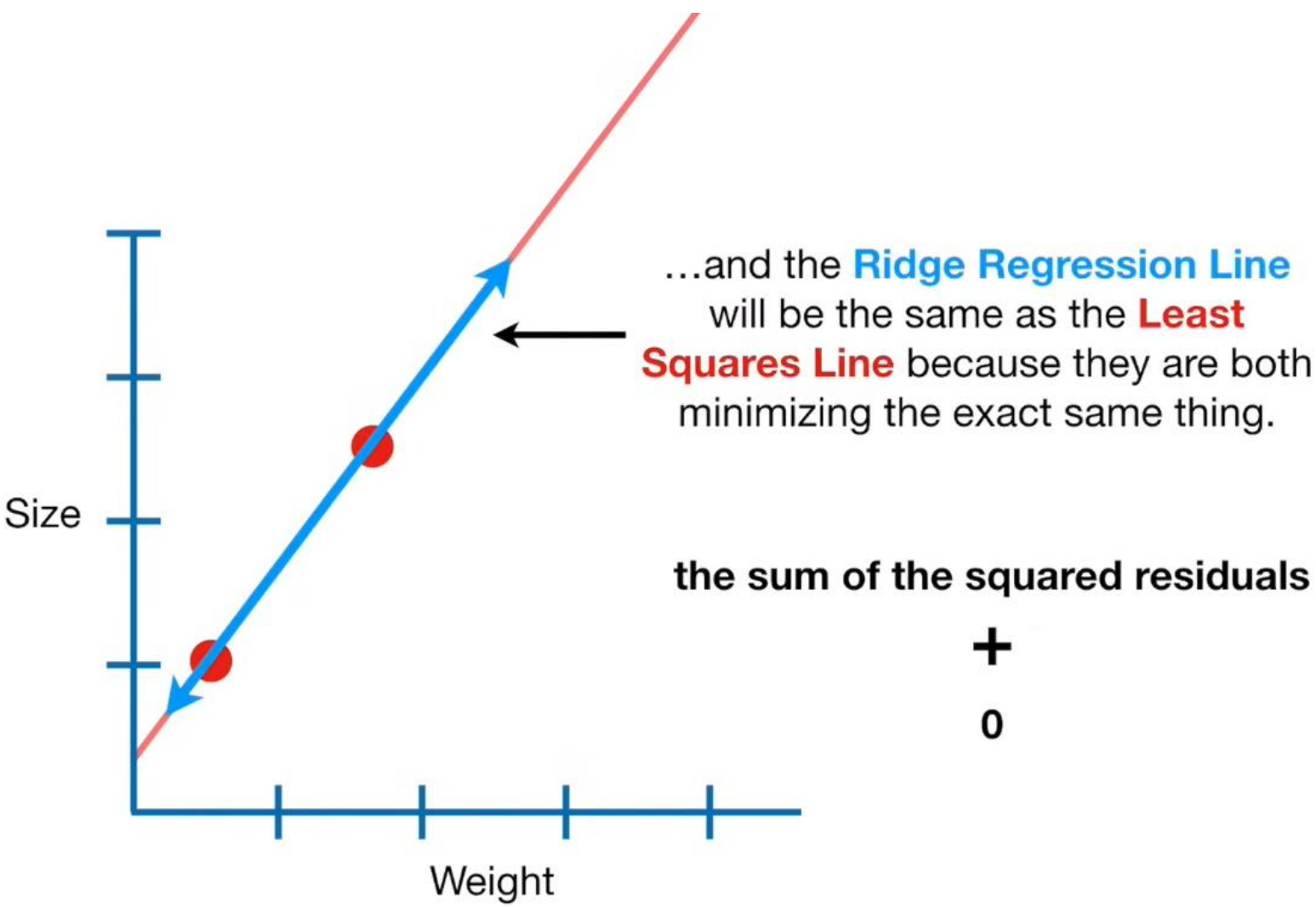


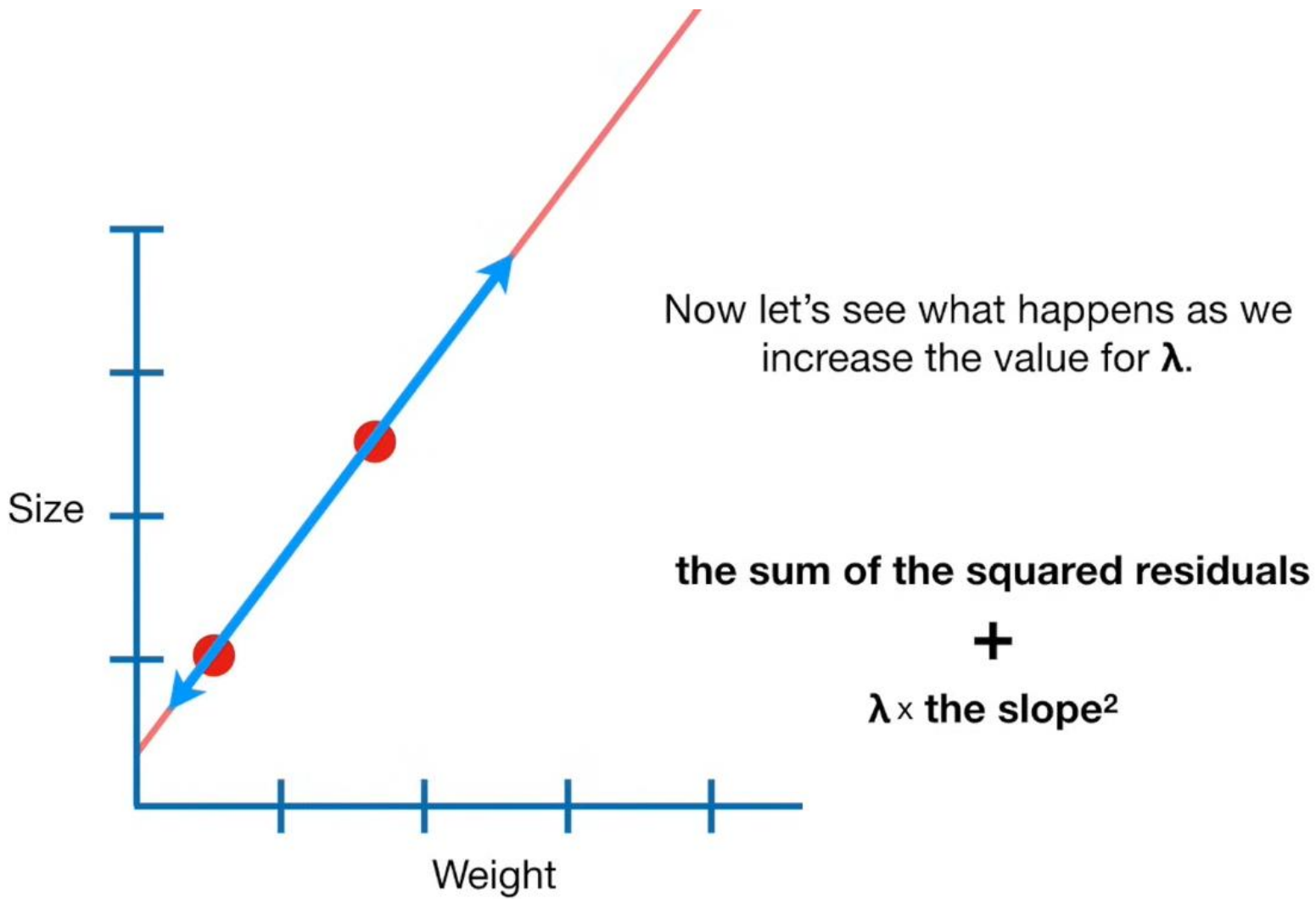


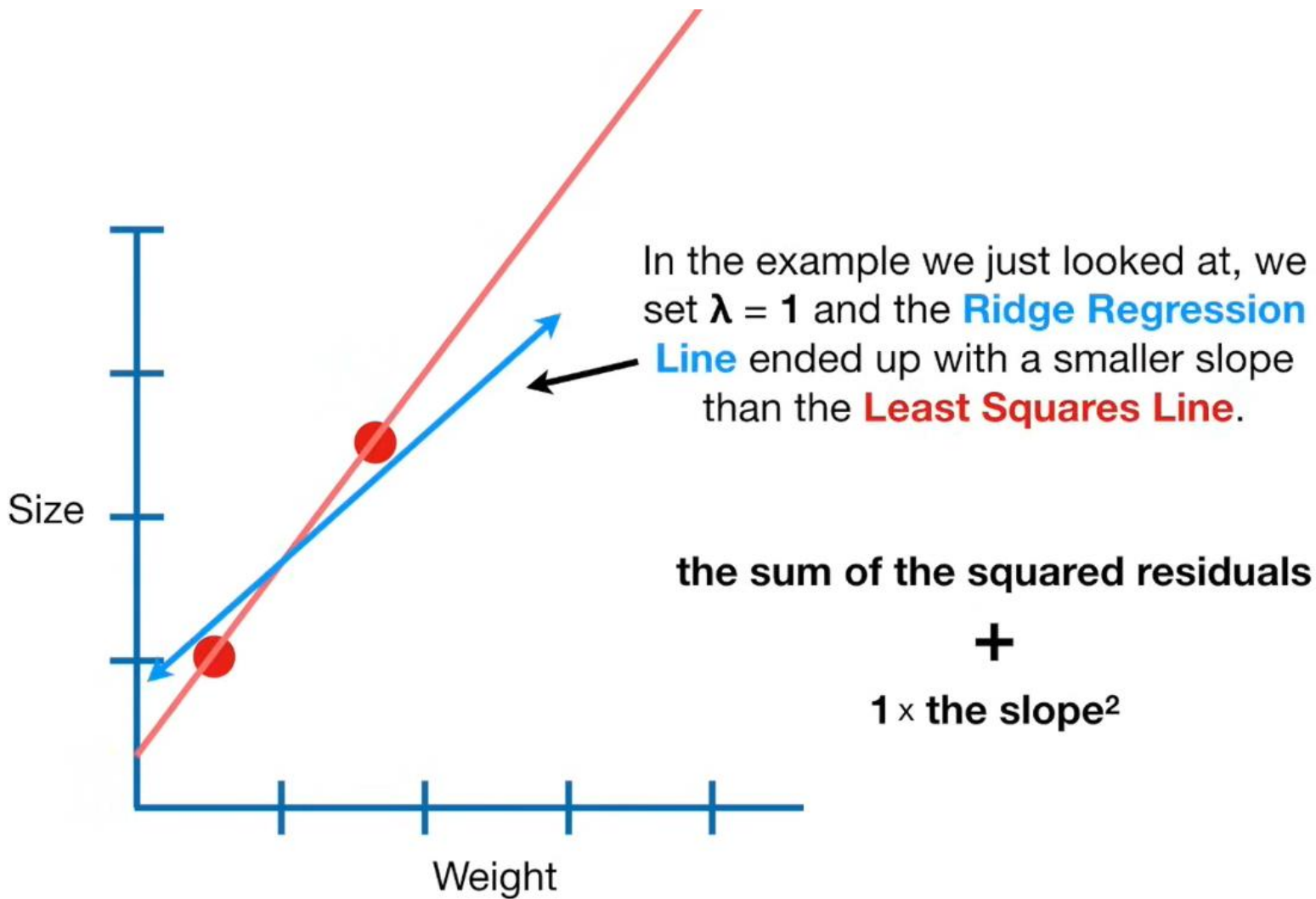


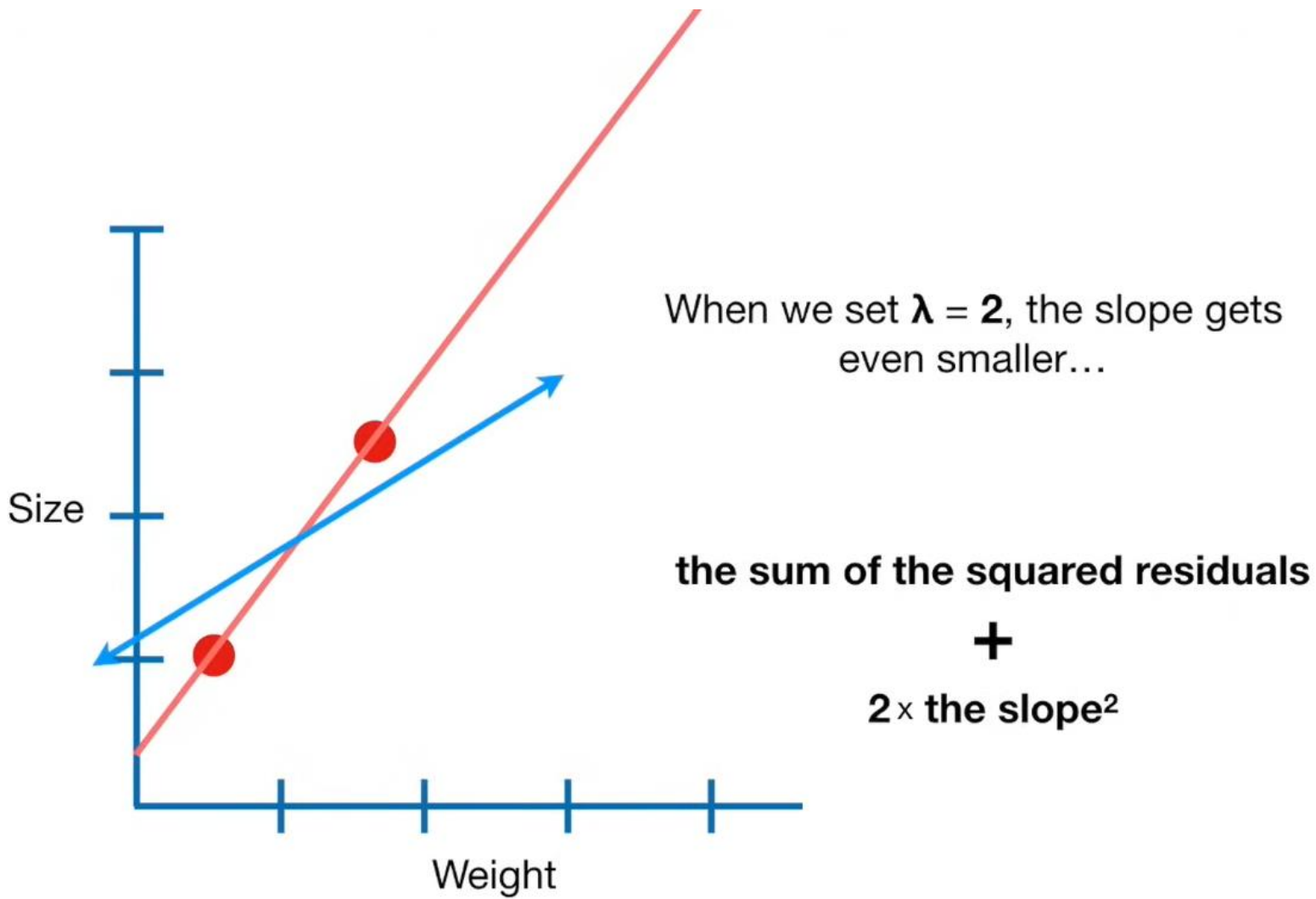


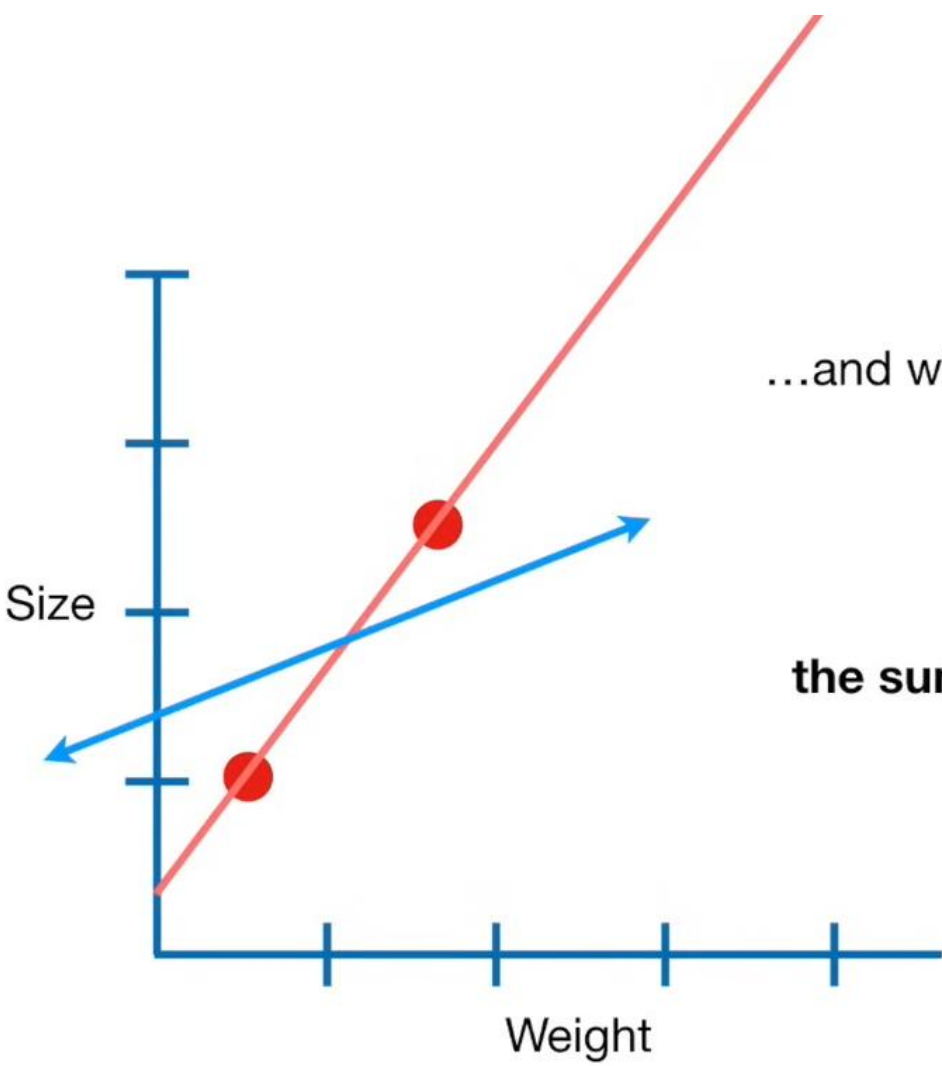






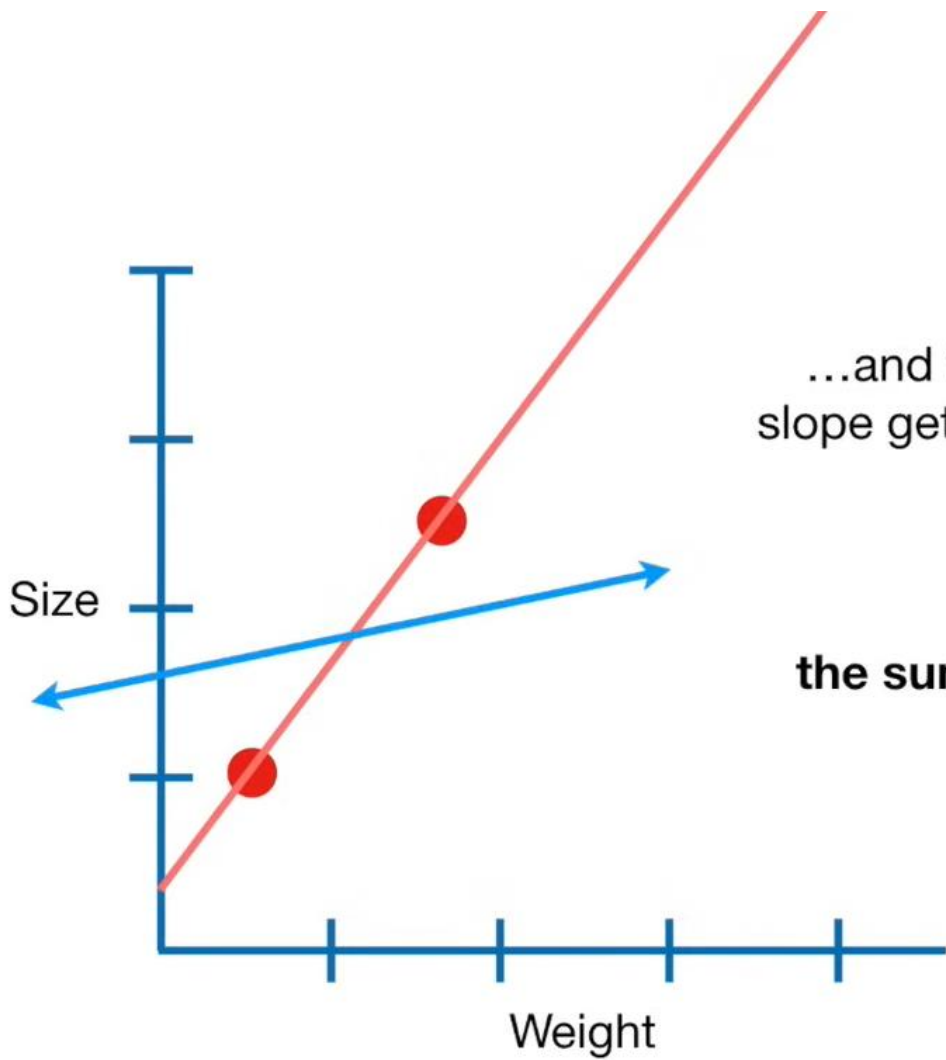






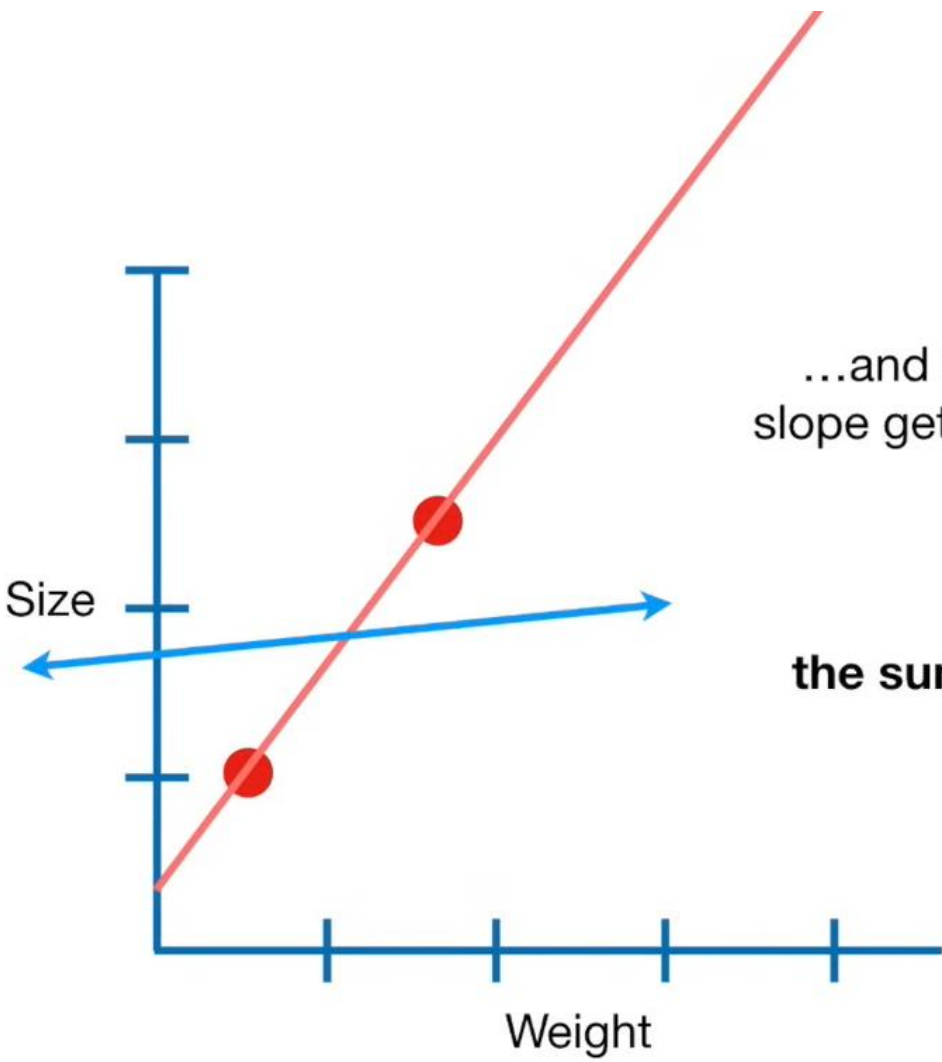
...and when we set $\lambda = 3$, the slope is even smaller...

the sum of the squared residuals
+
 $3 \times \text{the slope}^2$



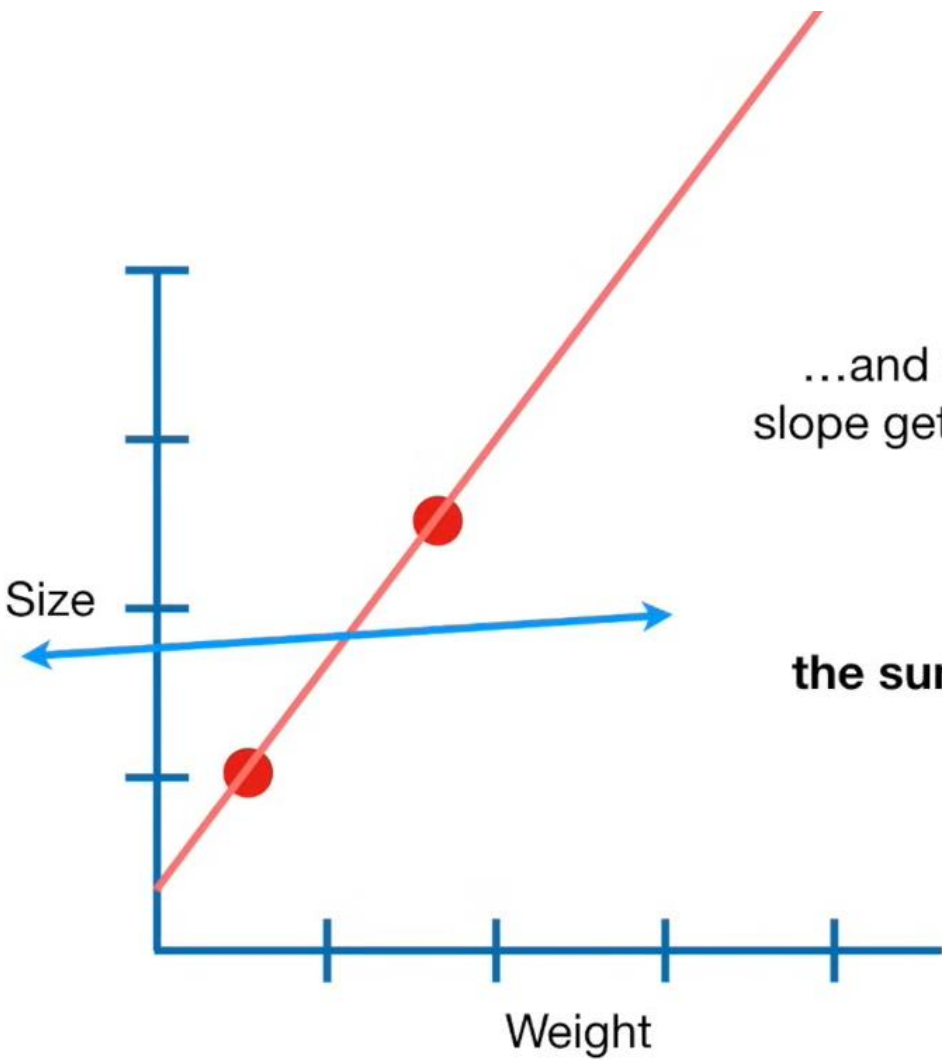
...and the larger we make λ , the slope gets asymptotically close to **0**.

the sum of the squared residuals
+
 $4 \times \text{the slope}^2$



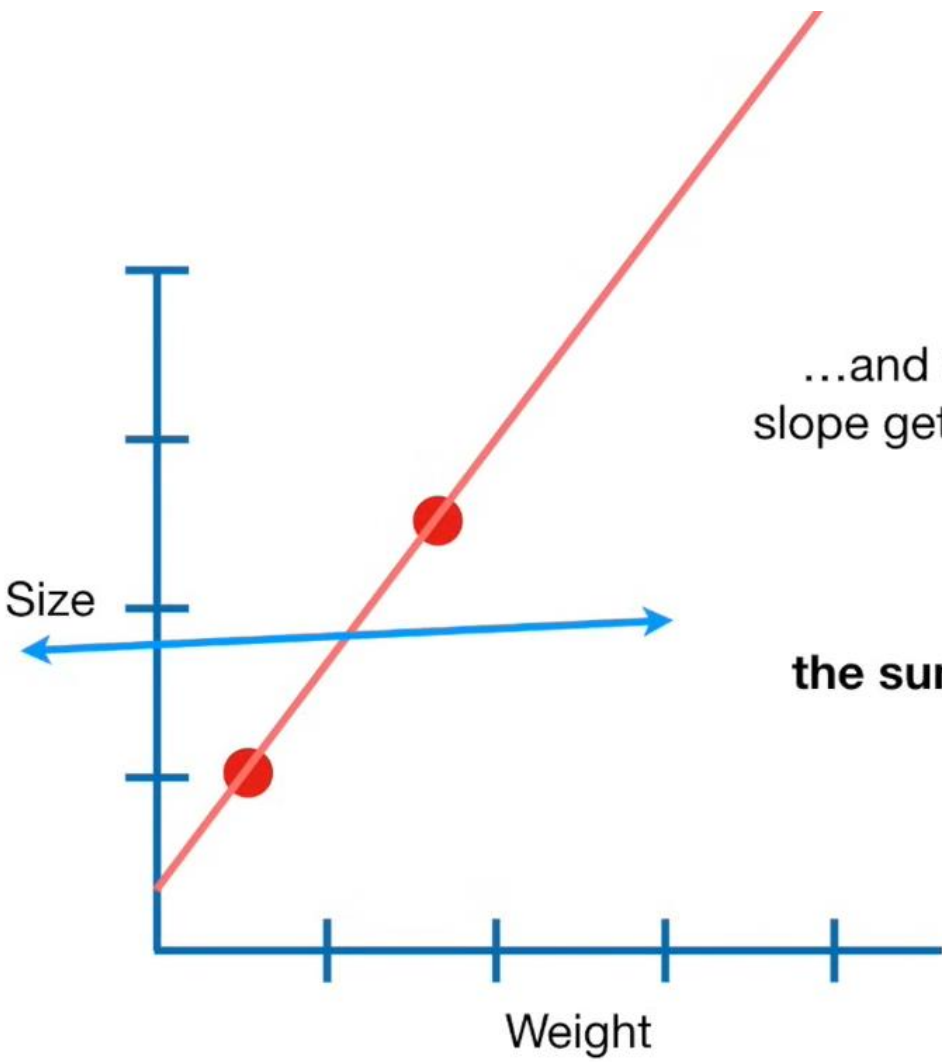
...and the larger we make λ , the slope gets asymptotically close to **0**.

the sum of the squared residuals
+
10 x the slope²



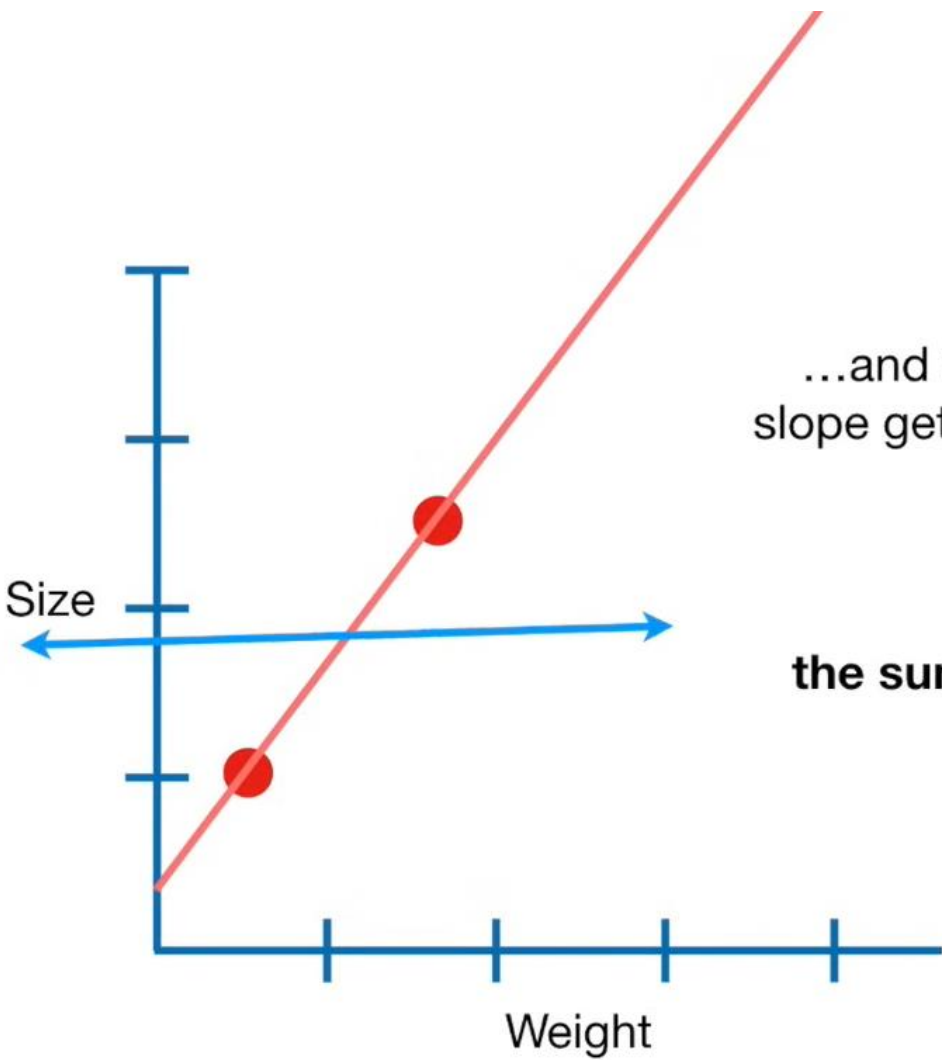
...and the larger we make λ , the slope gets asymptotically close to **0**.

the sum of the squared residuals
+
 $100 \times \text{the slope}^2$



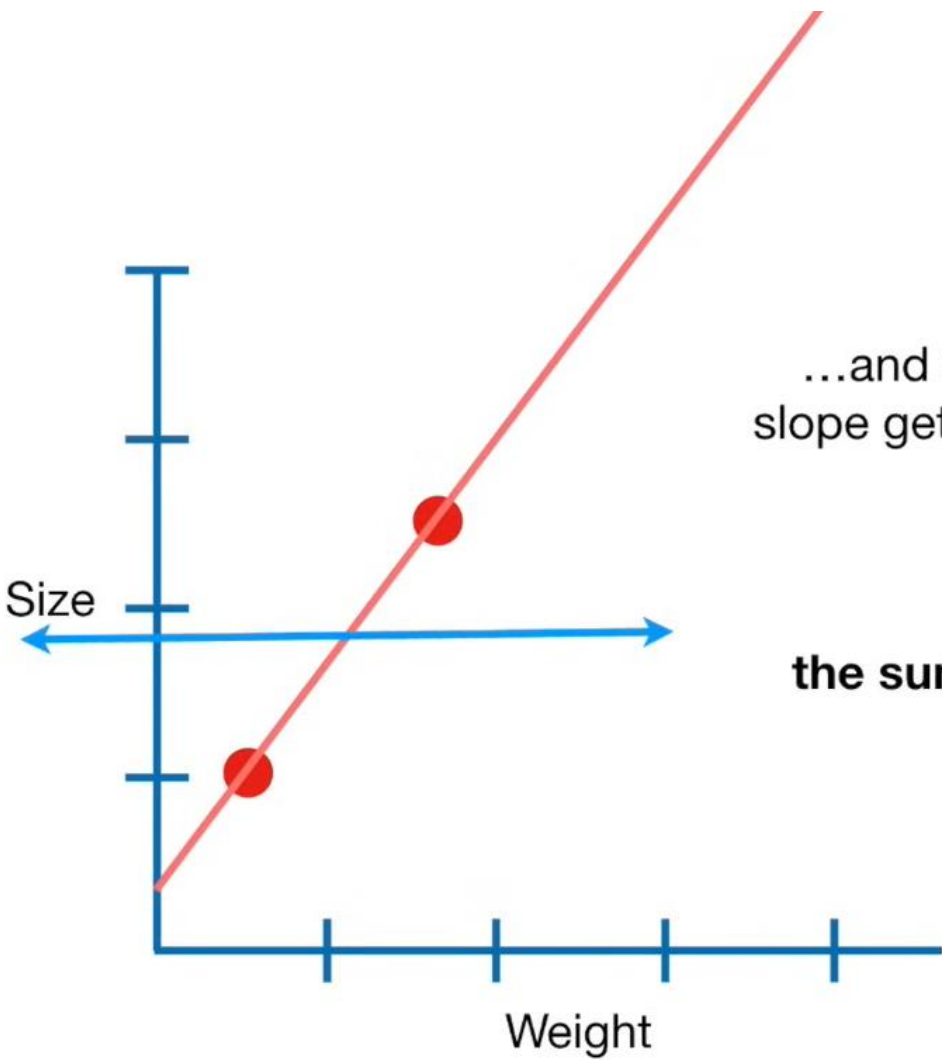
...and the larger we make λ , the slope gets asymptotically close to **0**.

the sum of the squared residuals
+
 $1000 \times \text{the slope}^2$



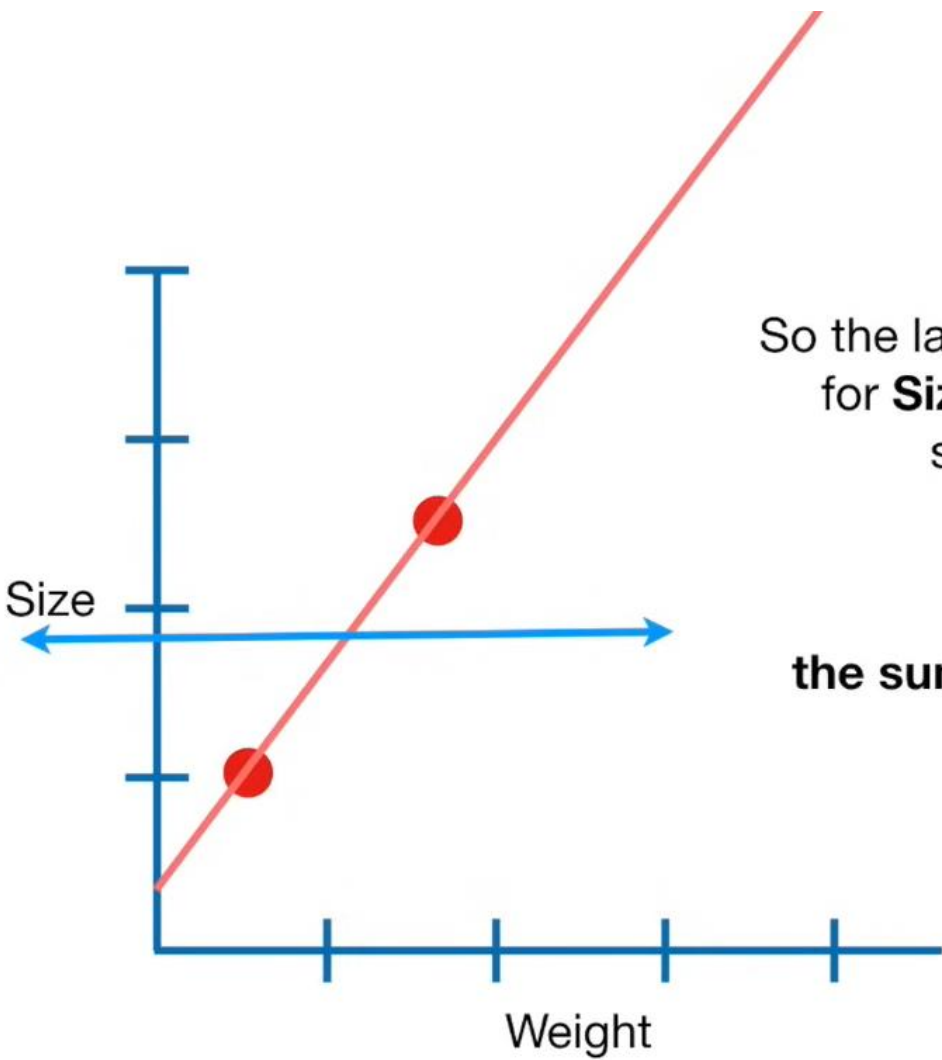
...and the larger we make λ , the slope gets asymptotically close to **0**.

the sum of the squared residuals
+
 $10000 \times \text{the slope}^2$



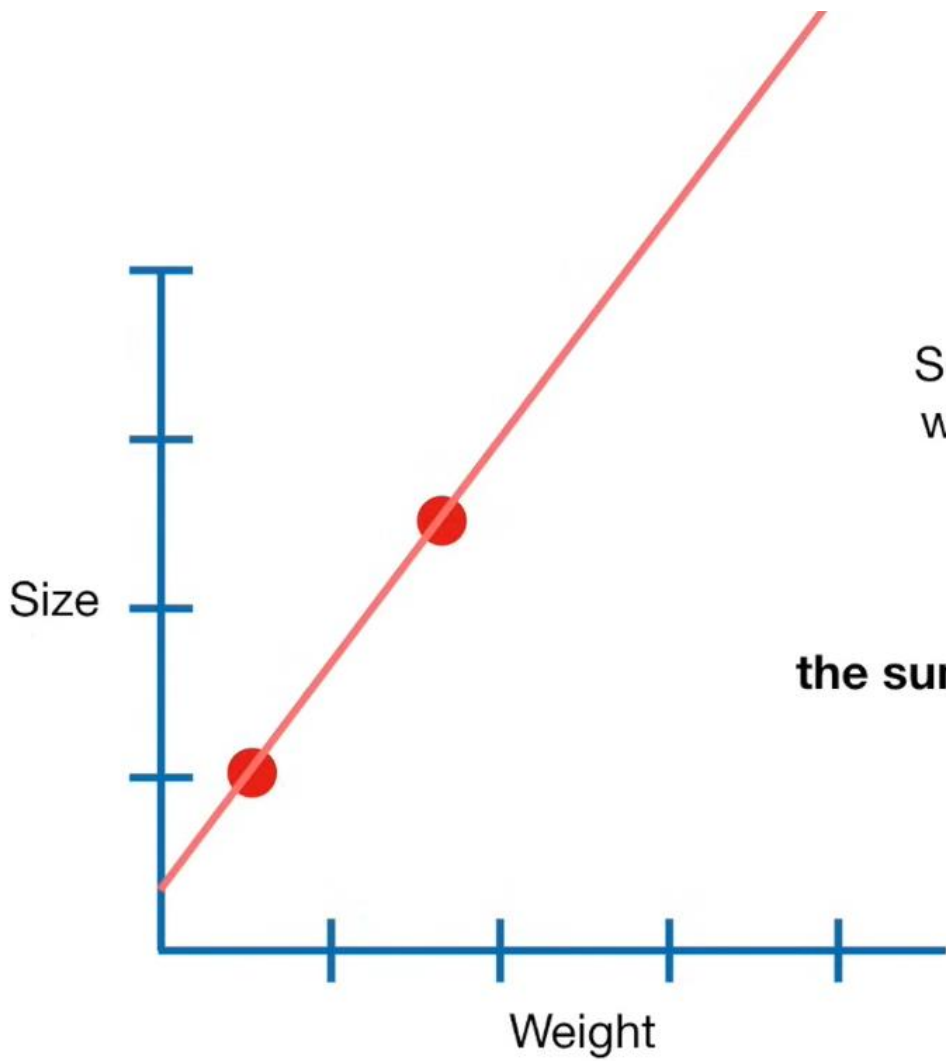
...and the larger we make λ , the slope gets asymptotically close to **0**.

the sum of the squared residuals
+
 $100000 \times \text{the slope}^2$



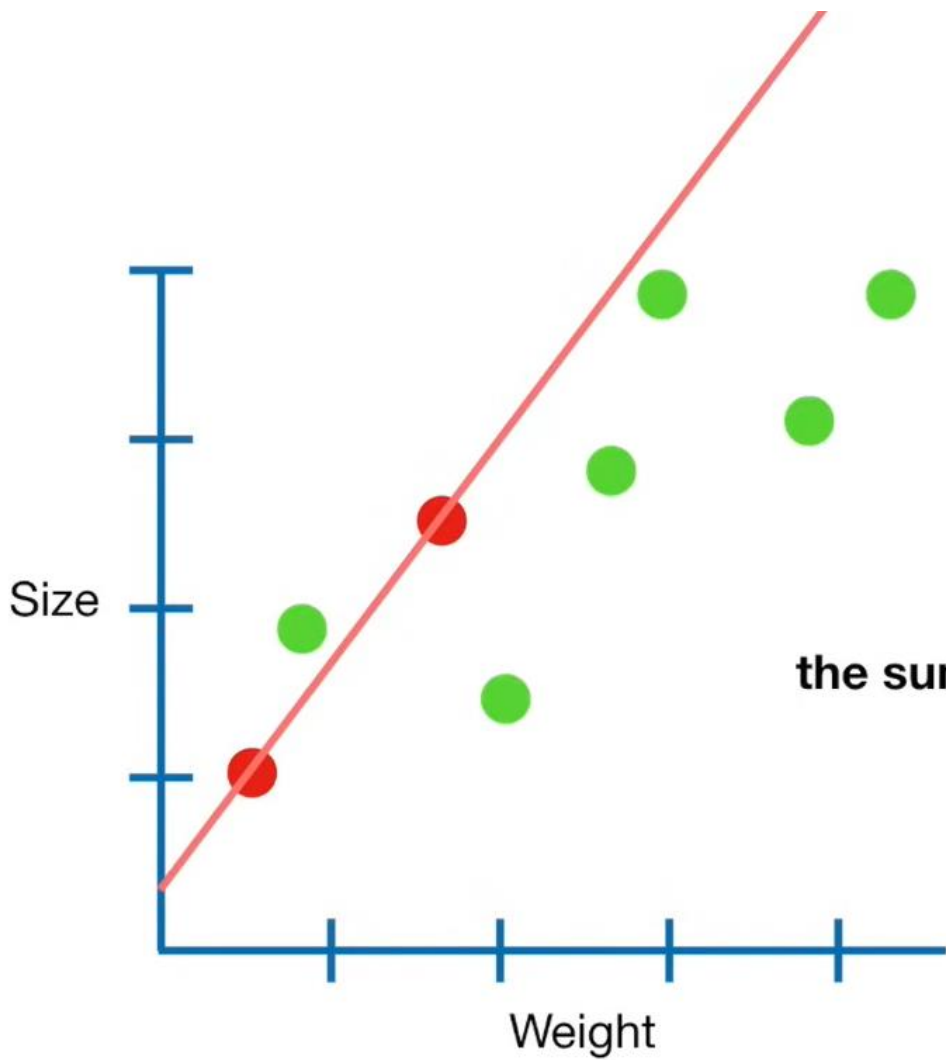
So the larger λ gets, our predictions for **Size** become less and less sensitive to **Weight**.

the sum of the squared residuals
+
 $100000 \times \text{the slope}^2$



So how do we decide
what value to give λ ?

the sum of the squared residuals
+
 $\lambda \times \text{the slope}^2$

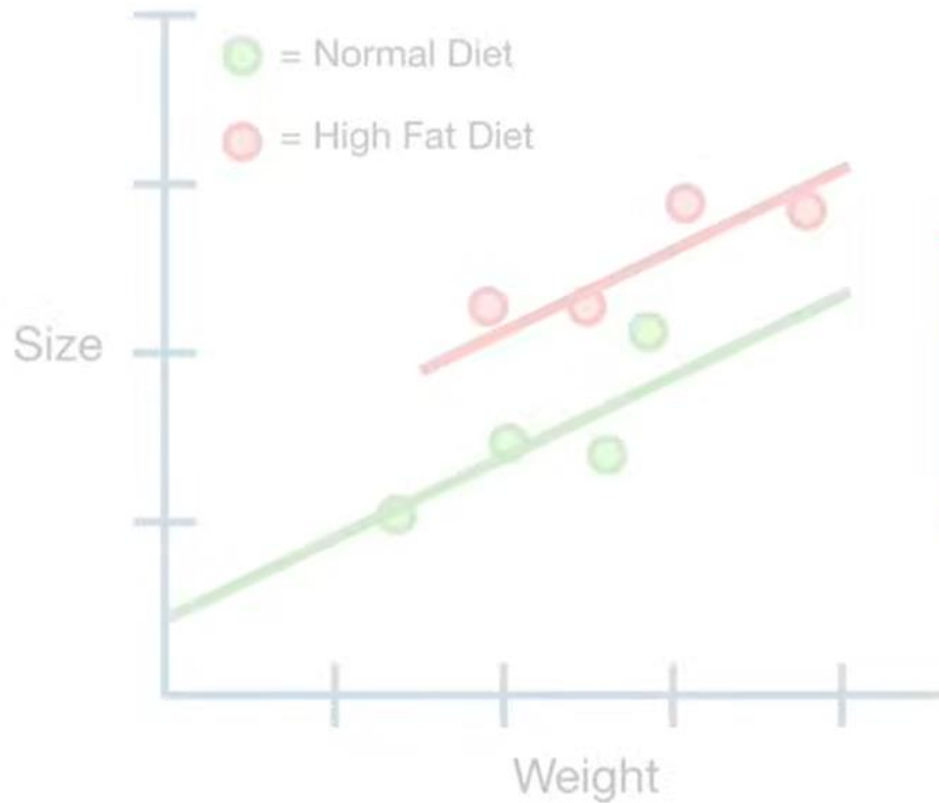


We just try a bunch of values for λ and use **Cross Validation**, typically **10-fold Cross Validation**, to determine which one results in the lowest **Variance**.

the sum of the squared residuals
+
 $\lambda \times \text{the slope}^2$

So far we've seen simple examples of how
Ridge Regression helps **reduce Variance** by
shrinking parameters and
making our predictions less sensitive to them...

Size = y-intercept + slope × **Weight** + diet difference × **High Fat Diet**

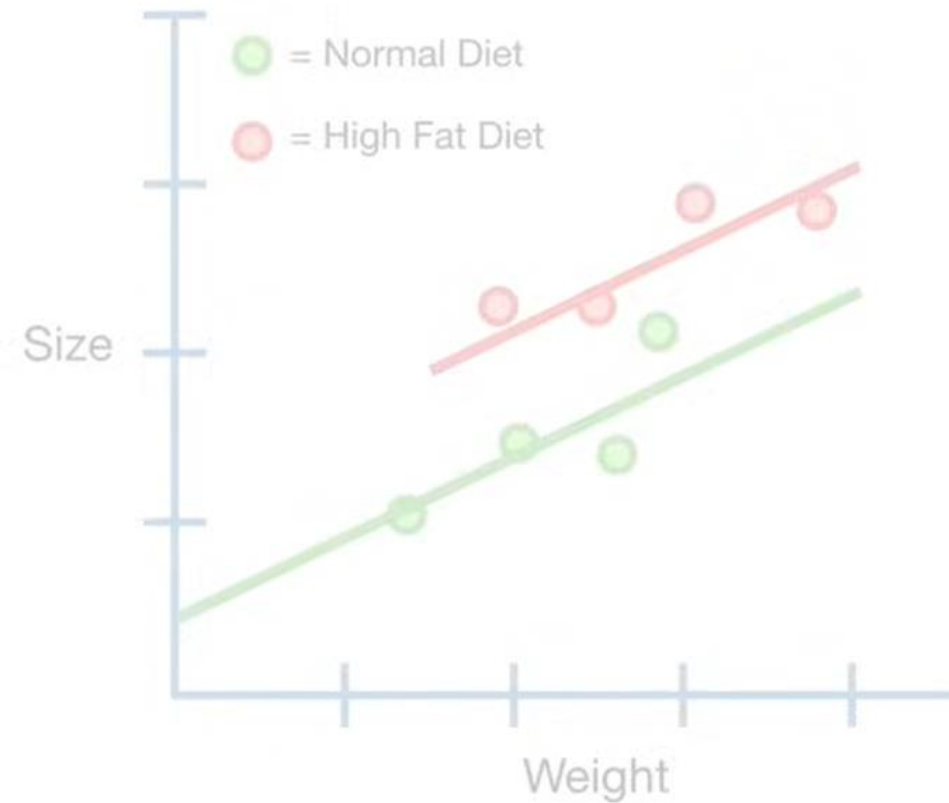


...and **Ridge Regression** tries to minimize this.



the sum of the squared residuals
+
 $\lambda \times (\text{slope}^2 + \text{diet difference}^2)$

Size = y-intercept + slope × **Weight** + diet difference × **High Fat Diet**



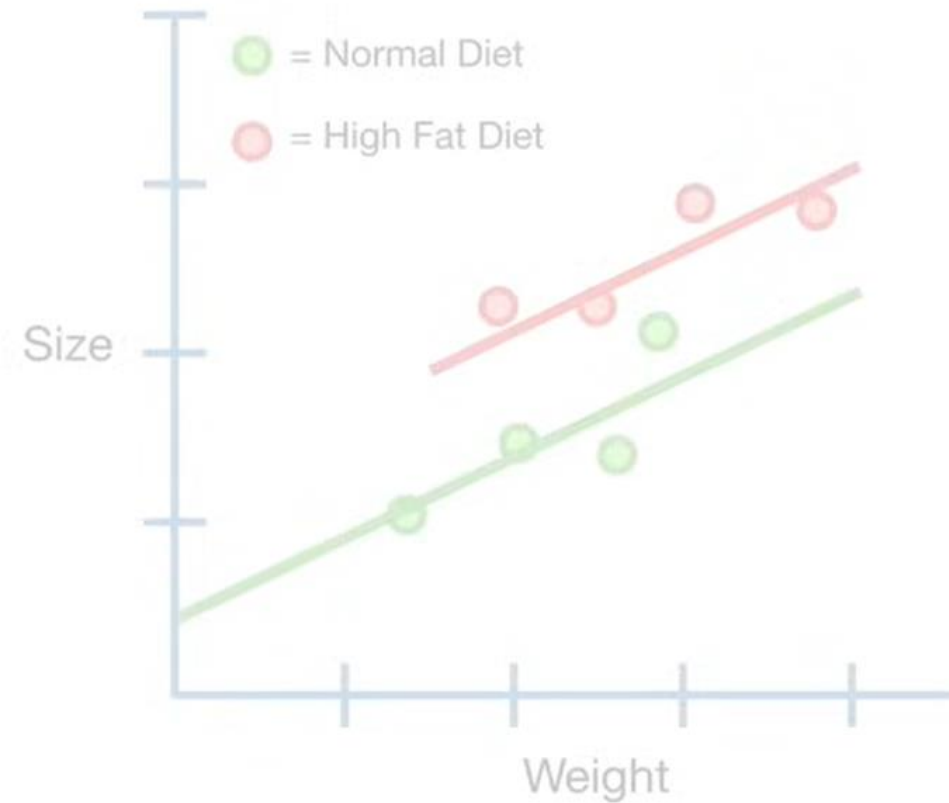
Now the **Ridge Regression Penalty** contains the parameters for the **slope** and the **difference between diets**.

the sum of the squared residuals

+

$$\lambda \times (\text{slope}^2 + \text{diet difference}^2)$$

Size = y-intercept + slope × **Weight** + diet difference × **High Fat Diet**



In general, the **Ridge Regression Penalty** contains all of the parameters except for the **y-intercept**.

the sum of the squared residuals

+

$\lambda \times (\text{slope}^2 + \text{diet difference}^2)$

Size = y-intercept + slope × **Weight** + diet difference × **High Fat Diet**

+ astrological offset × **Sign** + airspeed scalar × **Airspeed of Swallow...**



If we had a big, huge, crazy
equation with terms for
astrological sign, the airspeed
of a swallow and other stuff....

Size = y-intercept + slope × **Weight** + diet difference × **High Fat Diet**
+ astrological offset × **Sign** + airspeed scalar × **Airspeed of Swallow...**

Then the **Ridge Regression Penalty** would have all those parameters squared, *except* for the **y-intercept**.

$$\lambda \text{ (slope}^2 + \text{diet difference}^2 + \text{astrological offset}^2 + \text{airspeed scalar}^2 + \dots)$$

$$\text{Size} = \text{y-intercept} + \text{slope} \times \text{Weight} + \text{diet difference} \times \text{High Fat Diet} \\ + \text{astrological offset} \times \text{Sign} + \text{airspeed scalar} \times \text{Airspeed of Swallow} \dots$$



Every parameter, *except* for the **y-intercept**, is scaled by the measurements...

Size = y-intercept + slope × **Weight** + diet difference × **High Fat Diet**
+ astrological offset × **Sign** + airspeed scalar × **Airspeed of Swallow...**

...and that's why the **y-intercept**
is not included in the **Ridge**
Regression Penalty...



$$\lambda \quad (\text{slope}^2 + \text{diet difference}^2 + \text{astrological offset}^2 + \text{airspeed scalar}^2 + \dots)$$

Ok, now the next thing we're going to talk about is going to sound totally random, but trust me, it will lead to the coolest thing about **Ridge Regression**.

$$\mathbf{Size} = \text{y-intercept} + \text{slope} \times \mathbf{Weight}$$



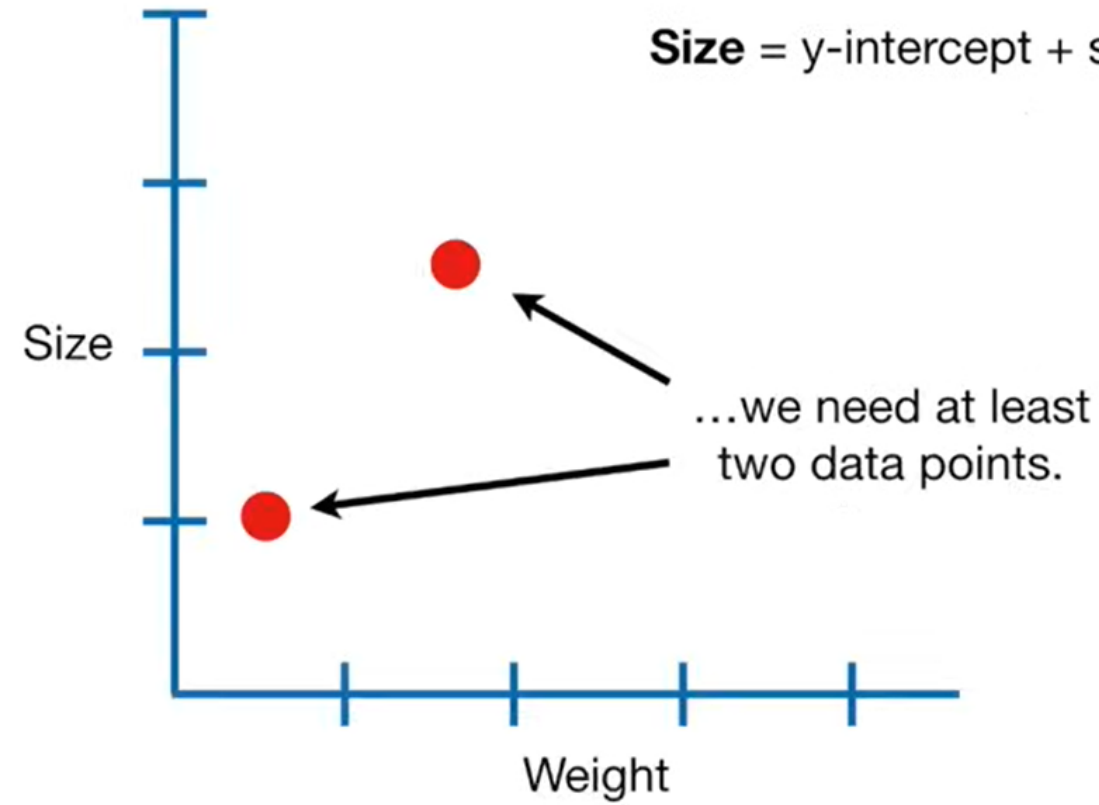
We all know that this is the
equation for a line...

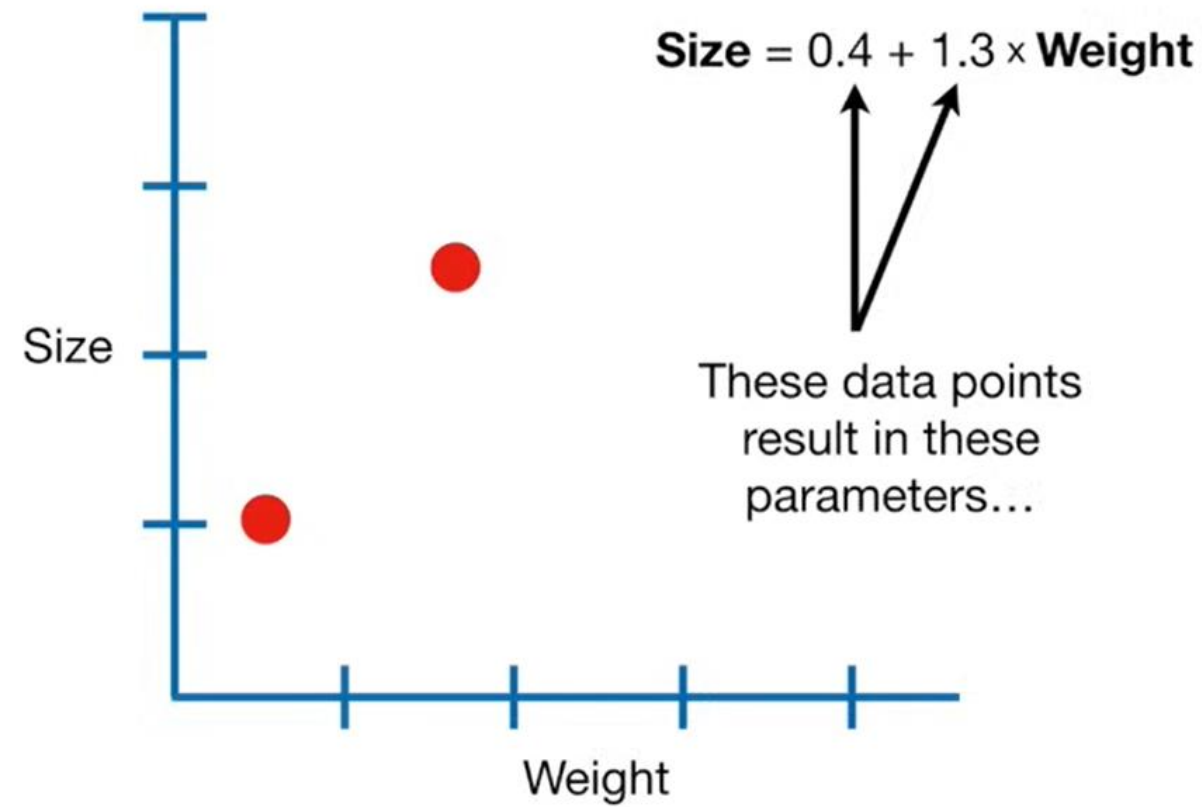
$$\text{Size} = \text{y-intercept} + \text{slope} \times \text{Weight}$$

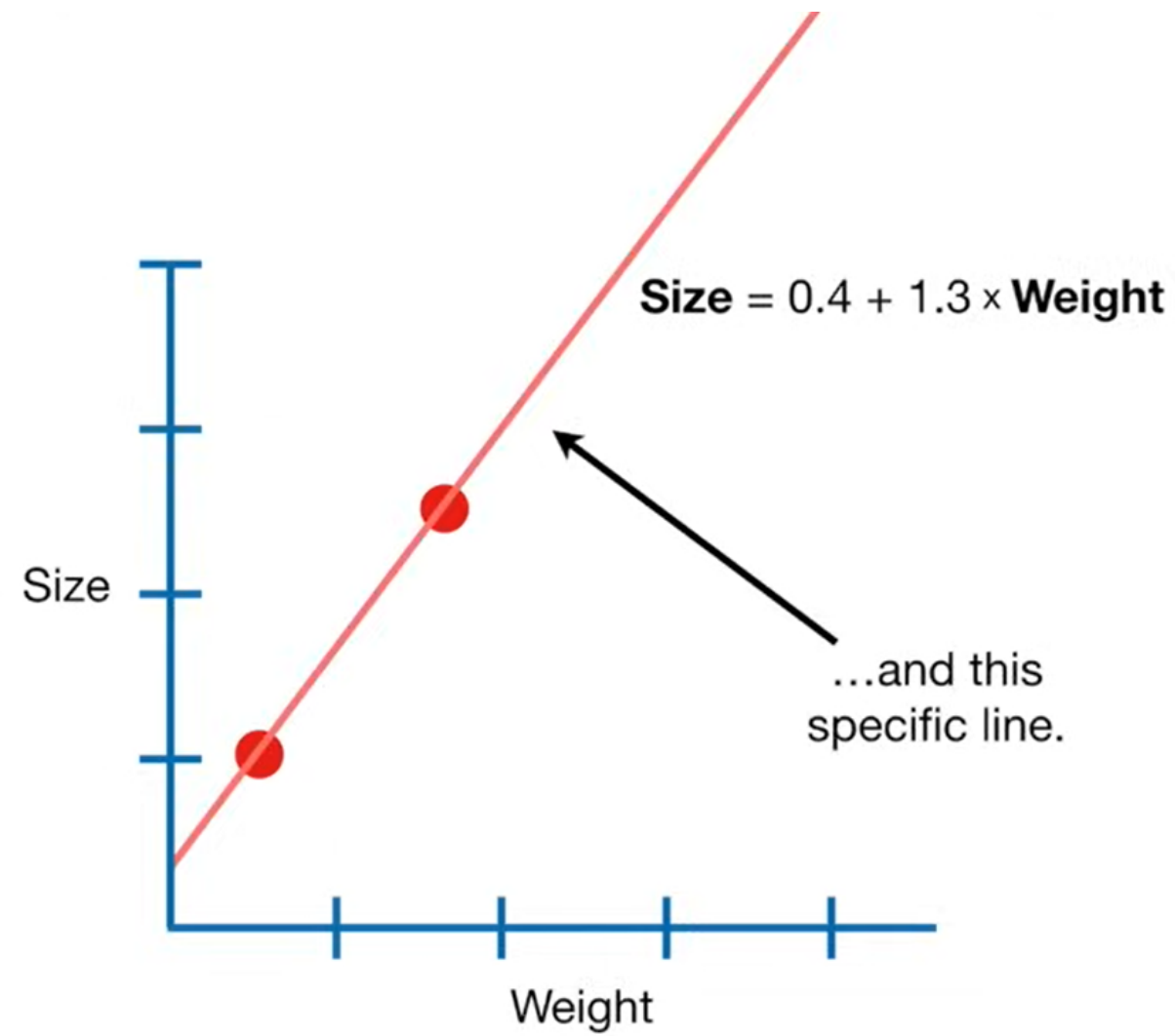


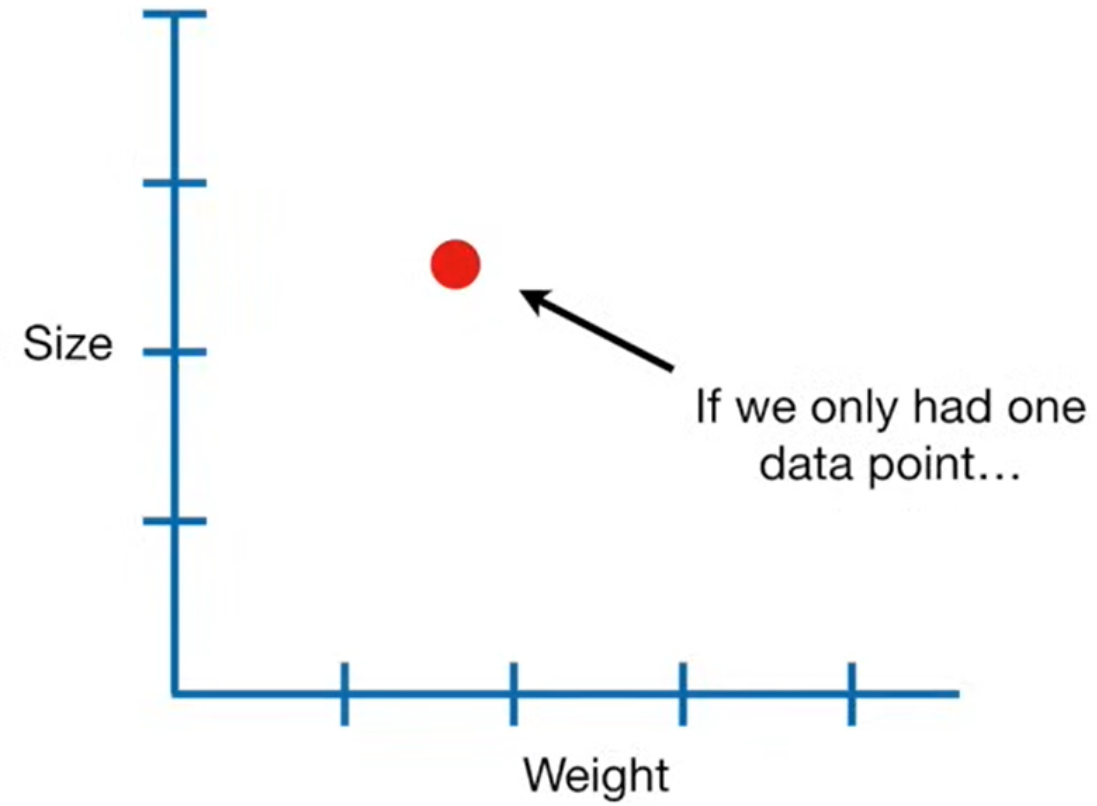
...and in order for **Least Squares** to solve for the parameters, the **y-intercept** and **slope**,...

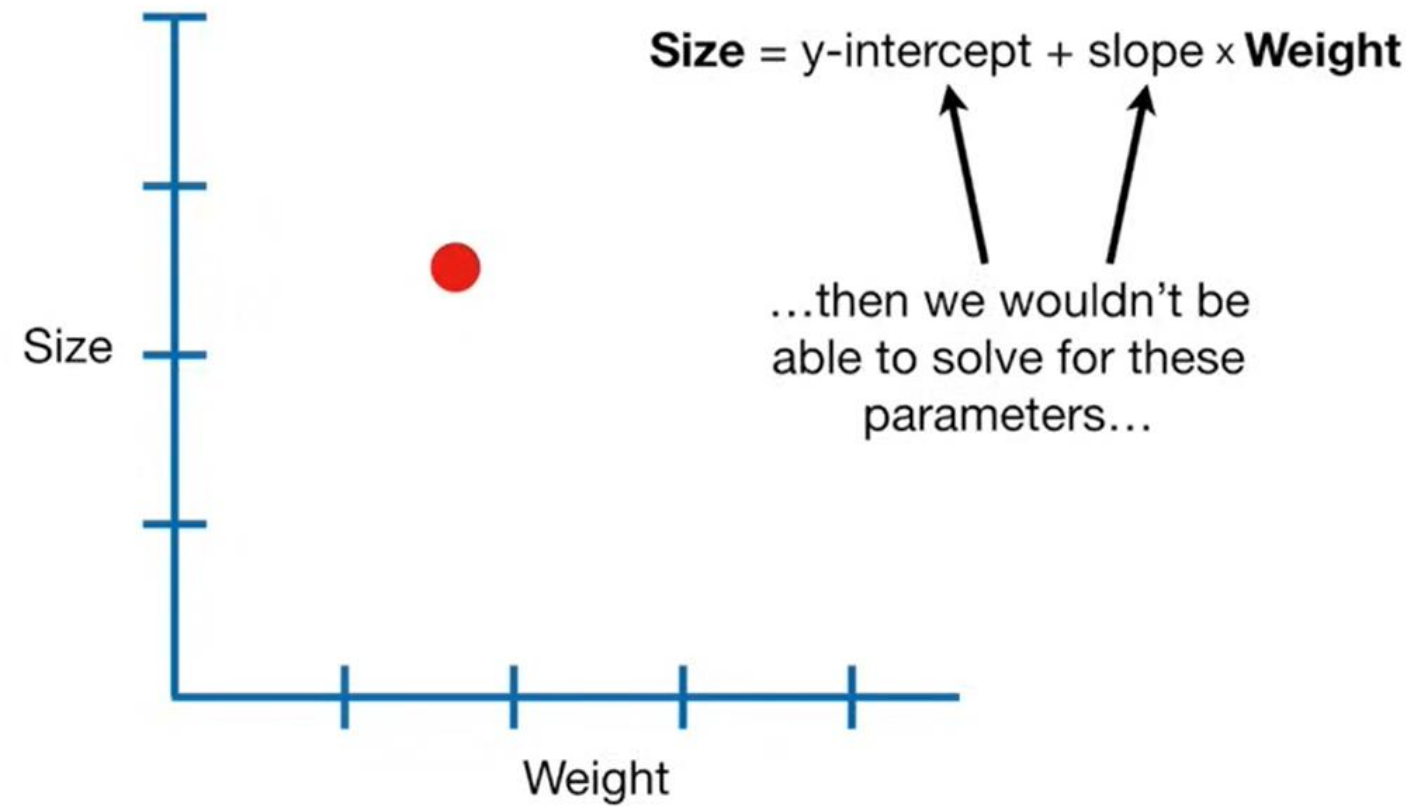
$$\text{Size} = \text{y-intercept} + \text{slope} \times \text{Weight}$$

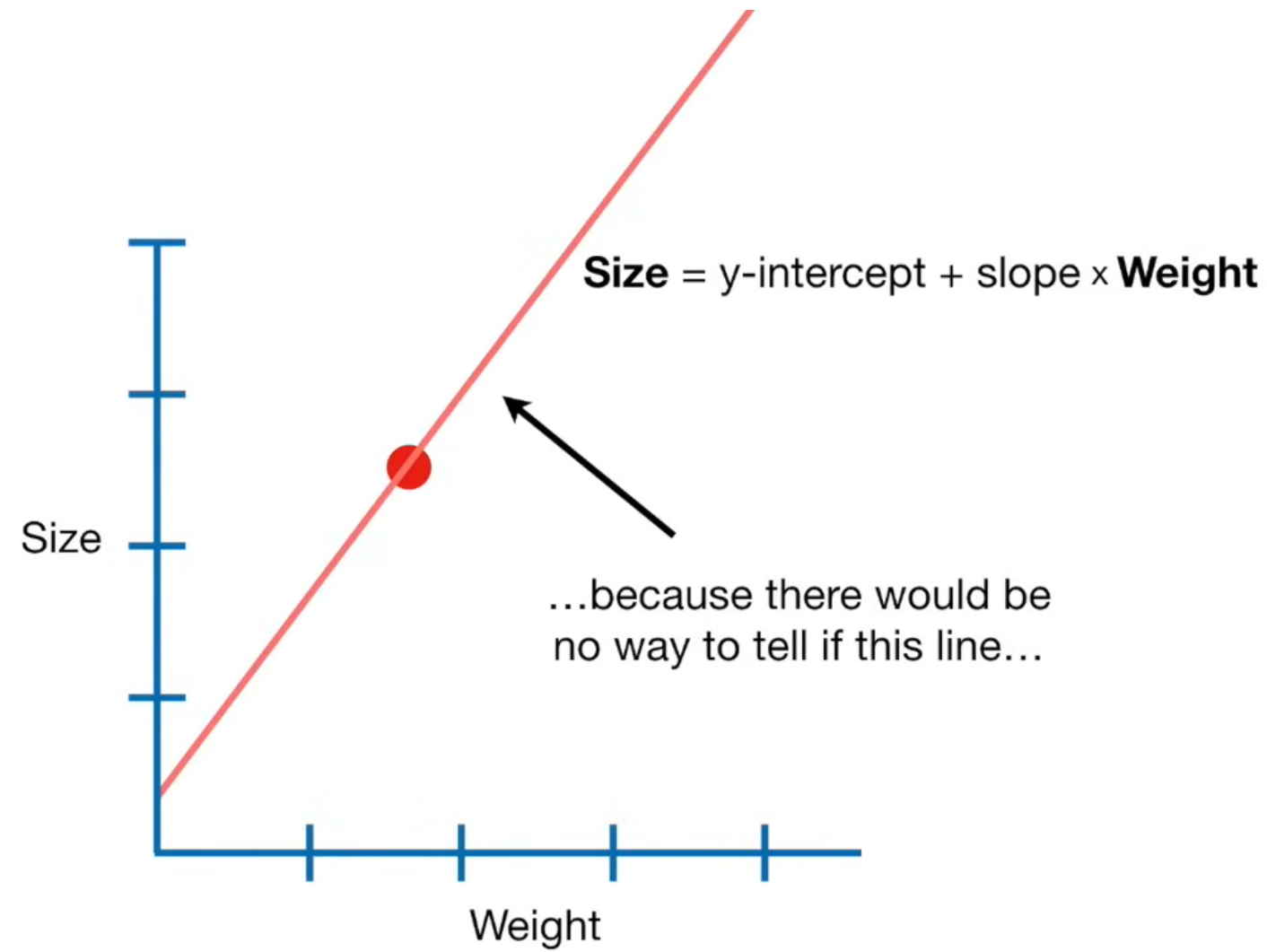


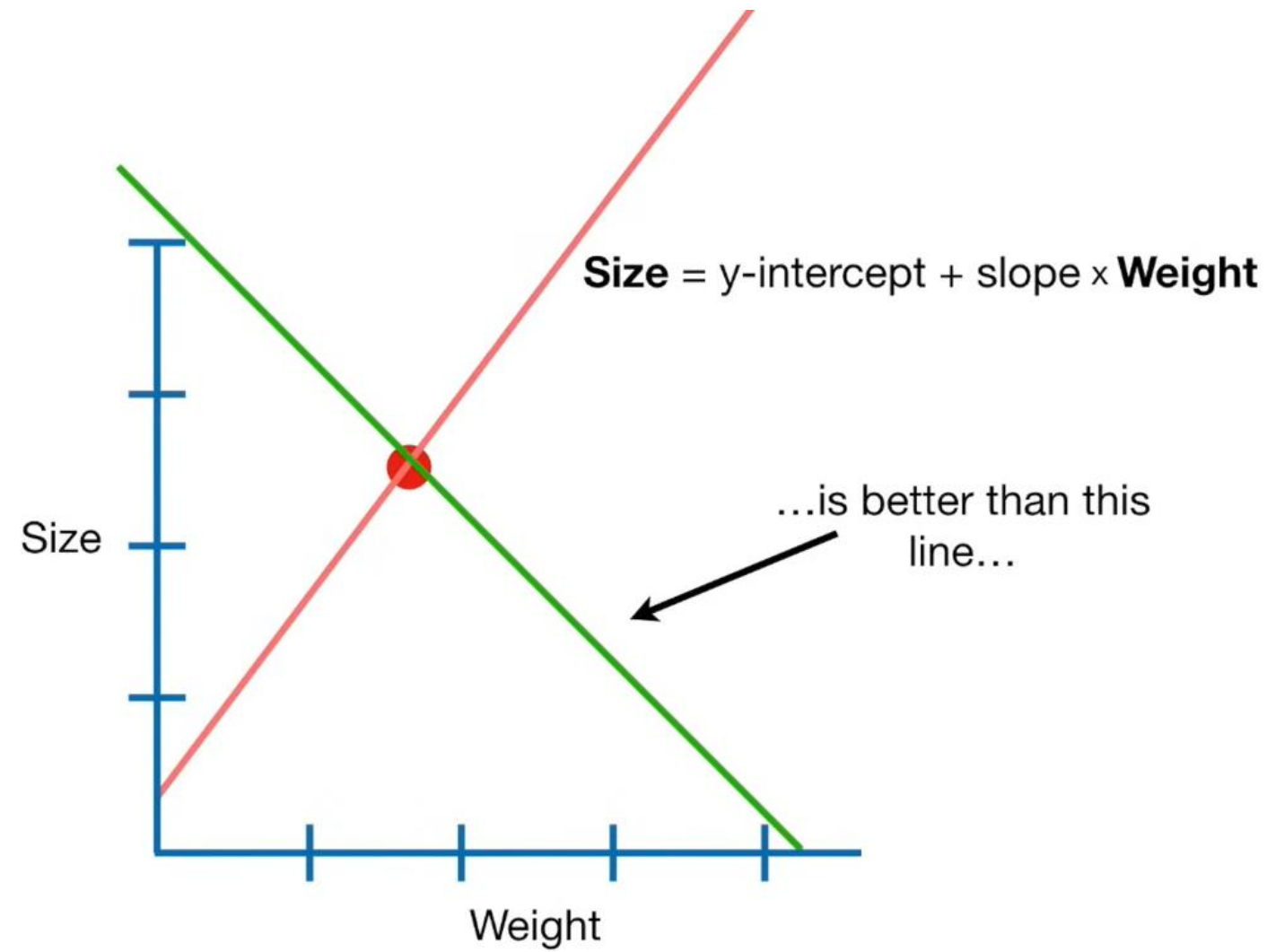


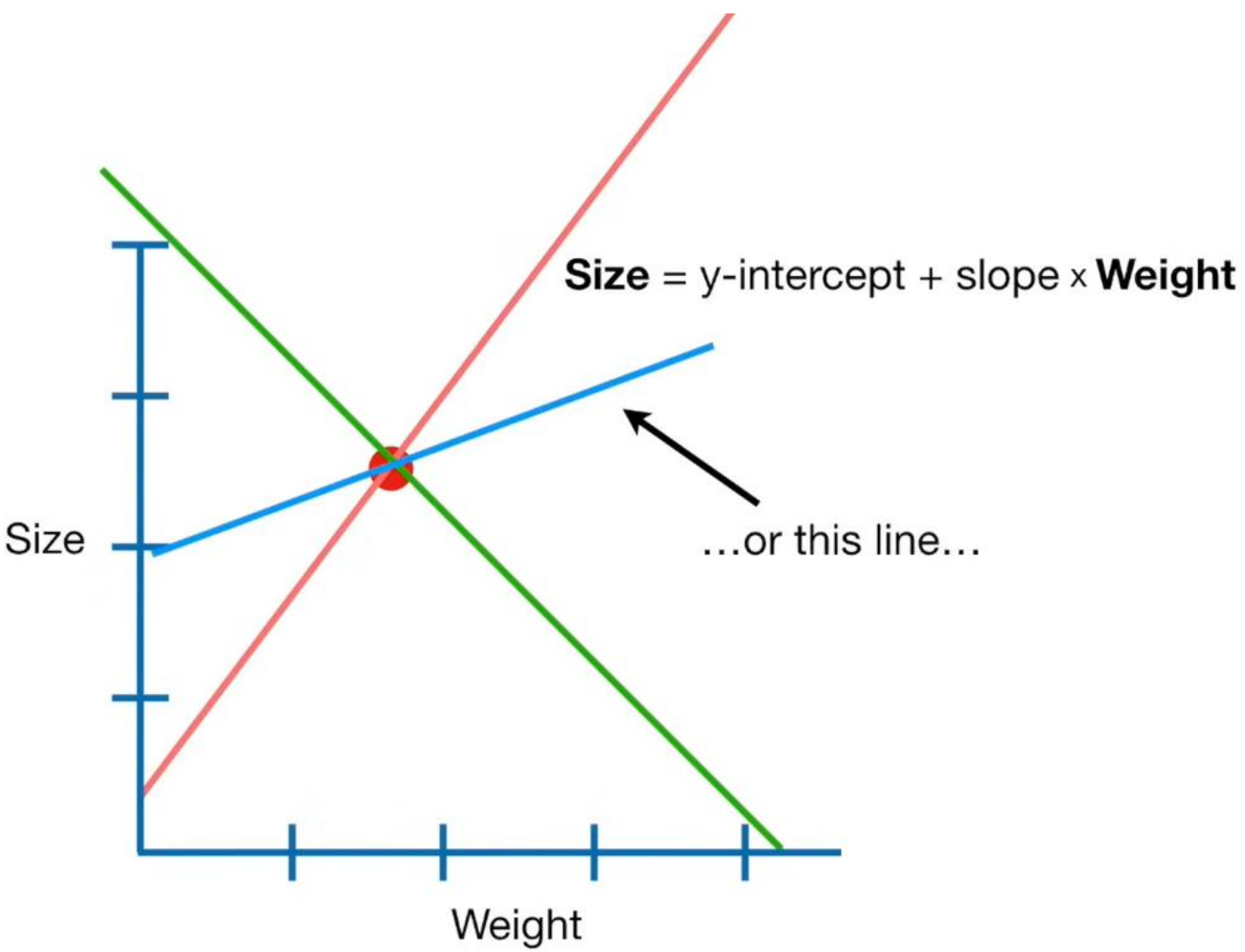


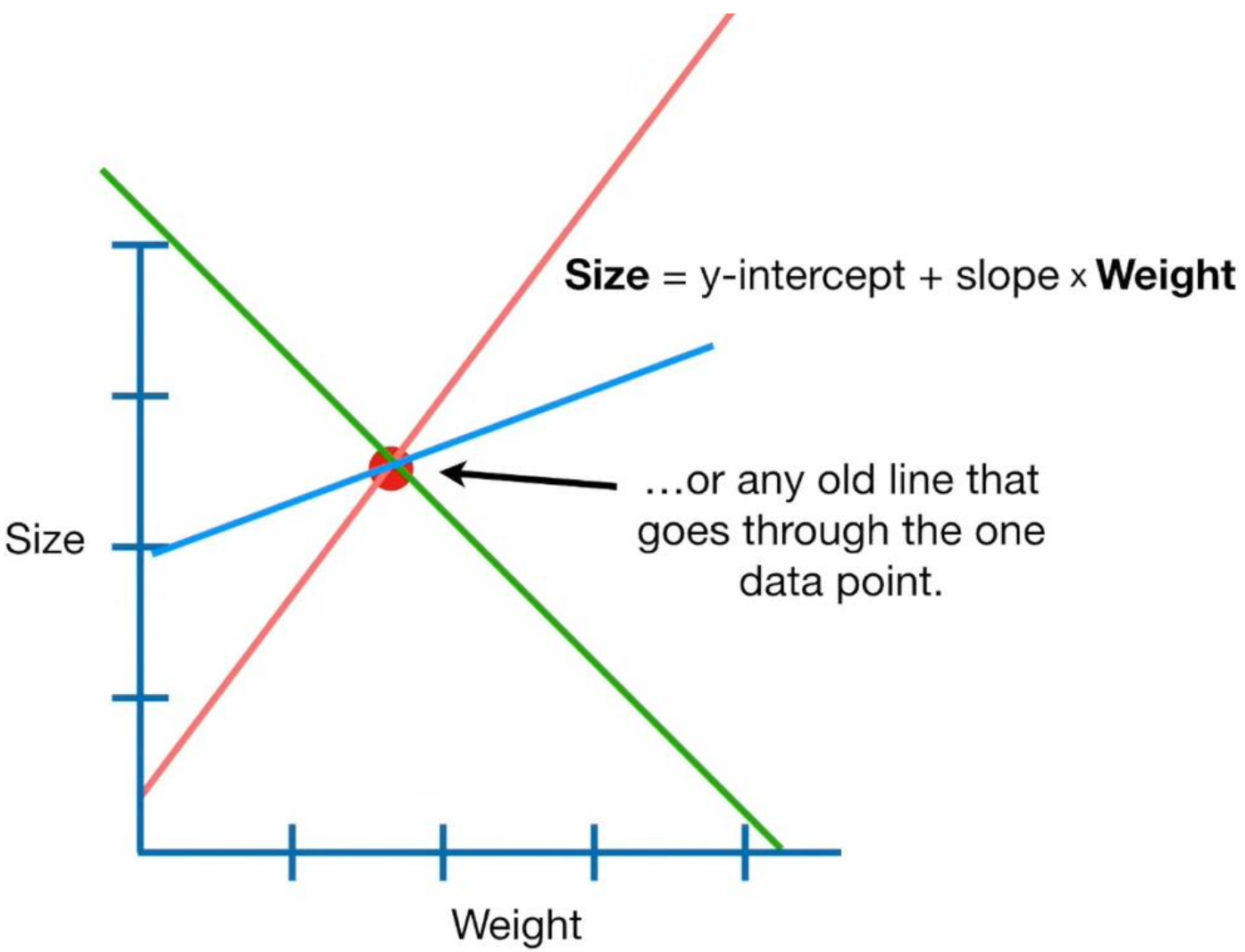


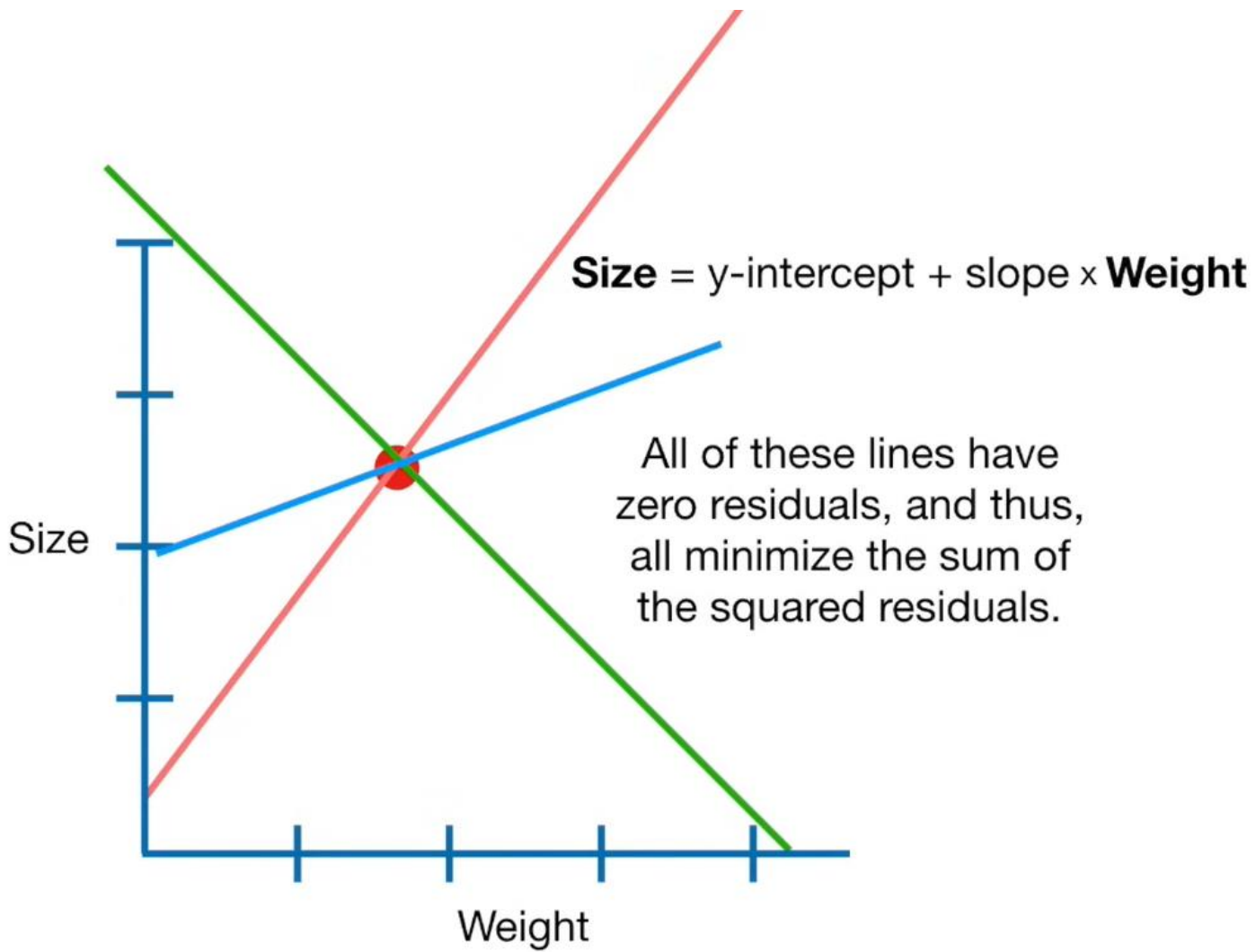


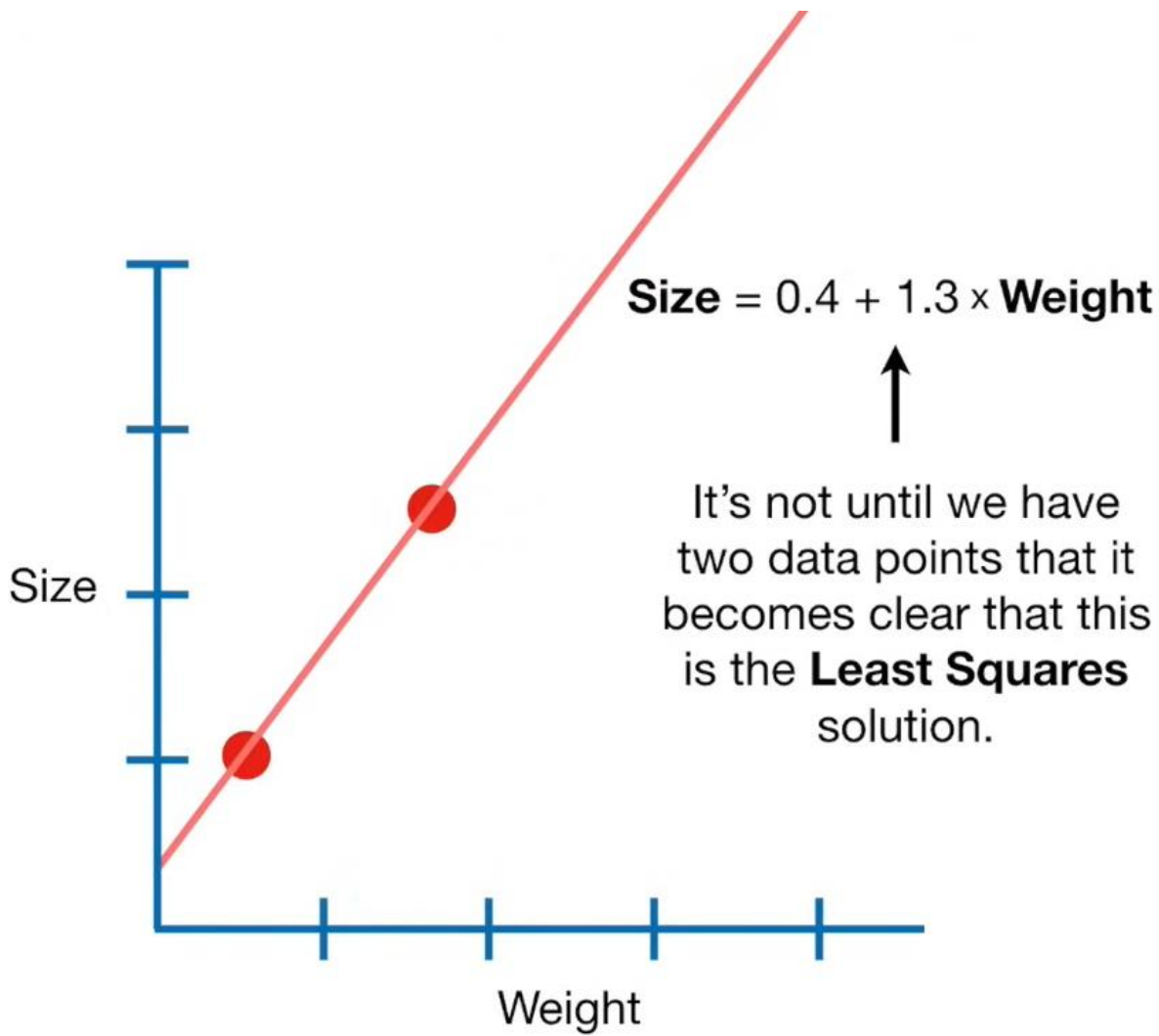






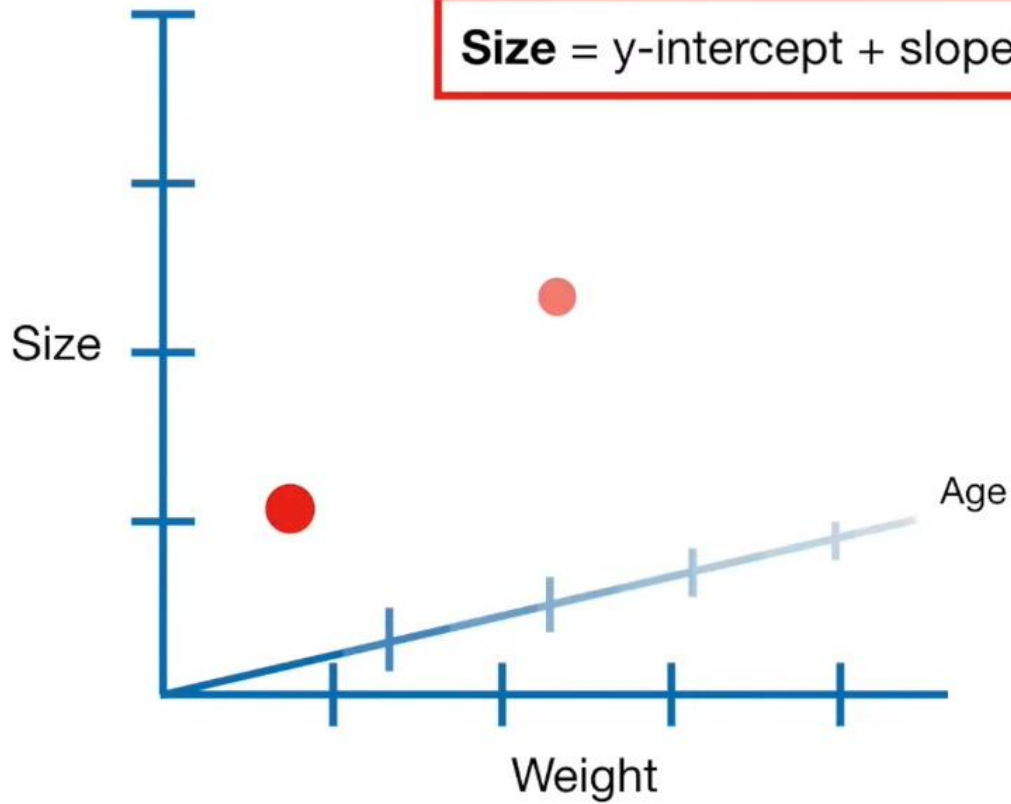






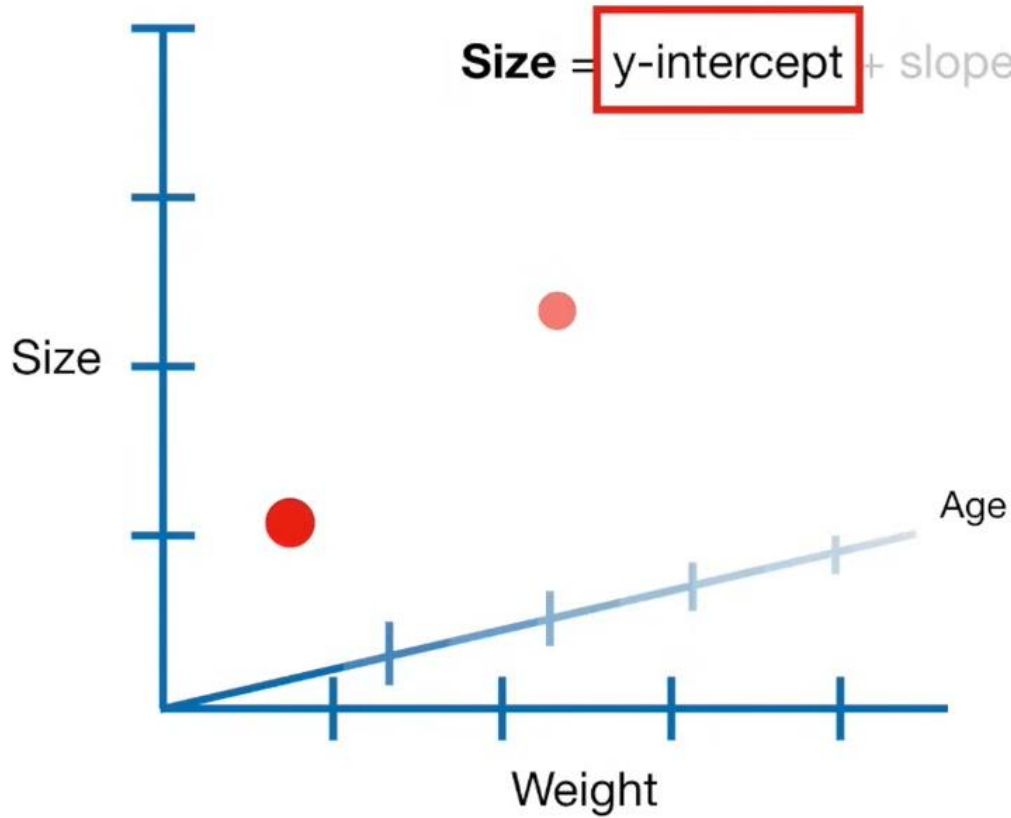
Now let's look at an equation that has three parameters to estimate.

$$\mathbf{Size} = \text{y-intercept} + \text{slope1} \times \mathbf{Weight} + \text{slope2} \times \mathbf{Age}$$



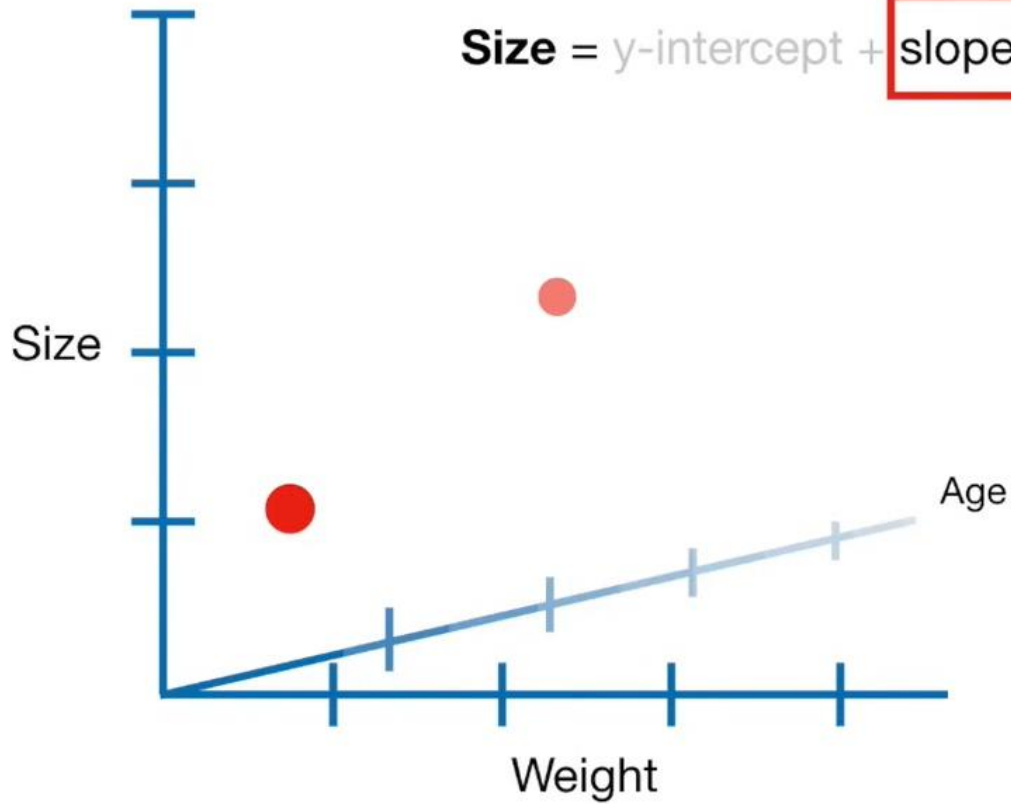
We need to estimate a **y-intercept**...

$$\text{Size} = \boxed{\text{y-intercept}} + \text{slope1} \times \text{Weight} + \text{slope2} \times \text{Age}$$



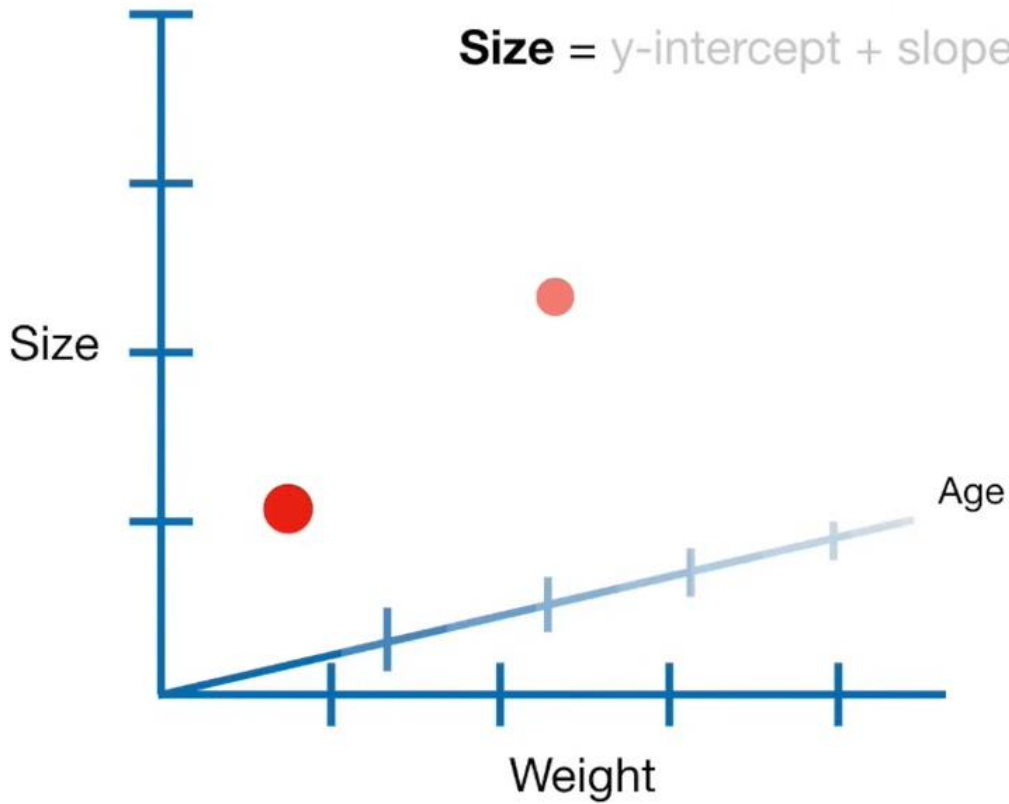
...a slope that reflects how **Weight** contributes to the prediction of **Size**...

$$\text{Size} = \text{y-intercept} + \boxed{\text{slope1} \times \text{Weight}} + \text{slope2} \times \text{Age}$$



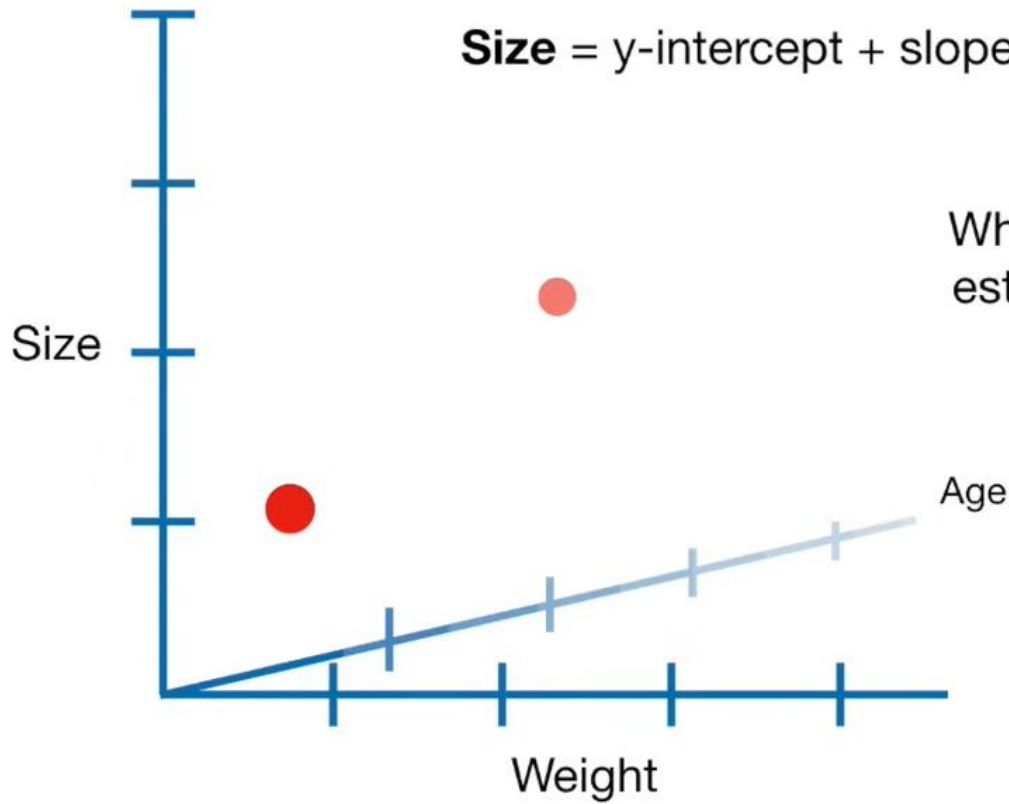
...and a **slope** that reflects how **Age** contributes to the prediction of **Size**.

$$\text{Size} = \text{y-intercept} + \text{slope1} \times \text{Weight} + \text{slope2} \times \text{Age}$$



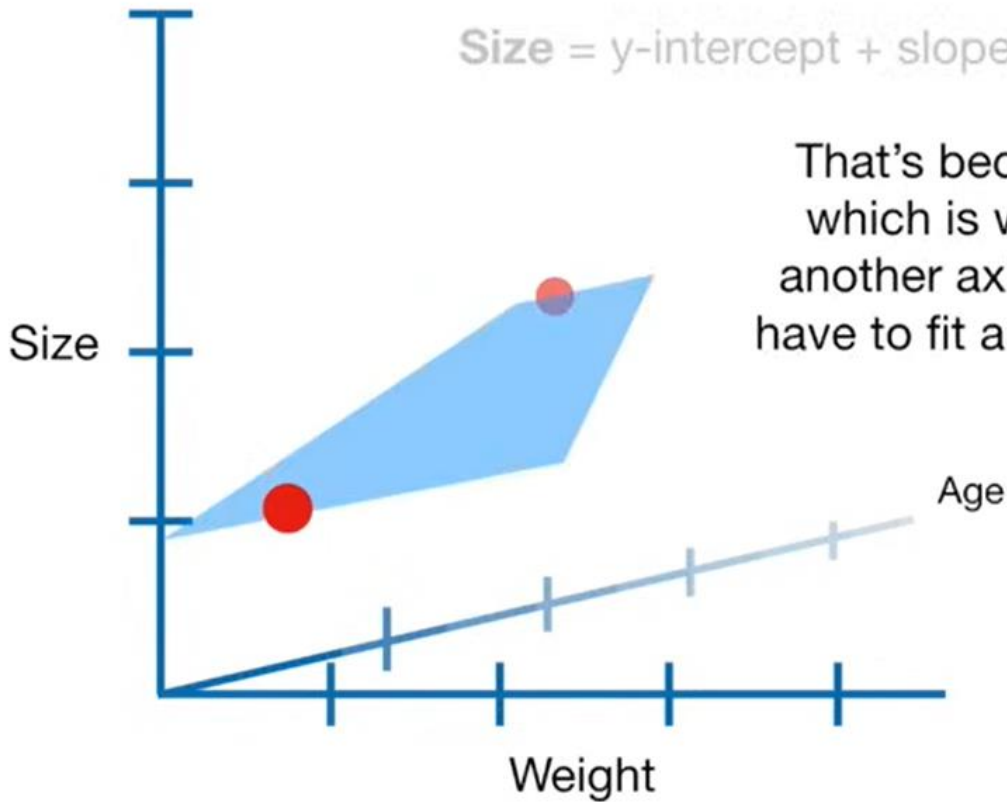
$$\text{Size} = \text{y-intercept} + \text{slope1} \times \text{Weight} + \text{slope2} \times \text{Age}$$

When we have three parameters to estimate, then just two data points isn't going to cut it.



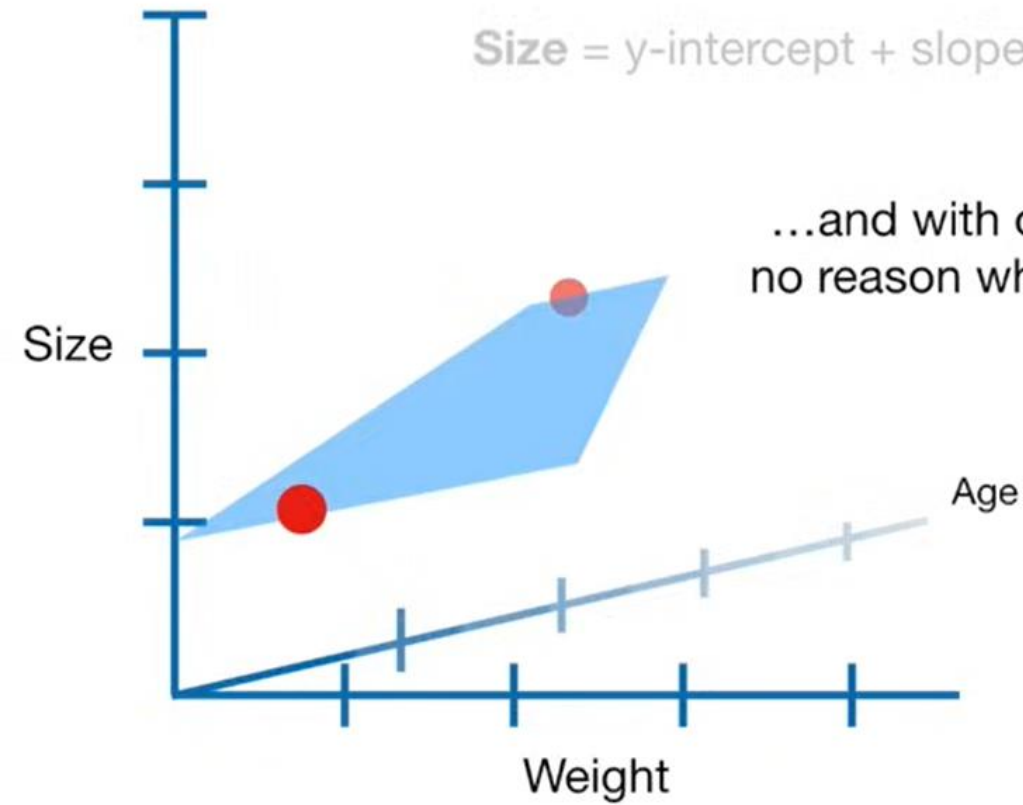
$$\text{Size} = \text{y-intercept} + \text{slope1} \times \text{Weight} + \text{slope2} \times \text{Age}$$

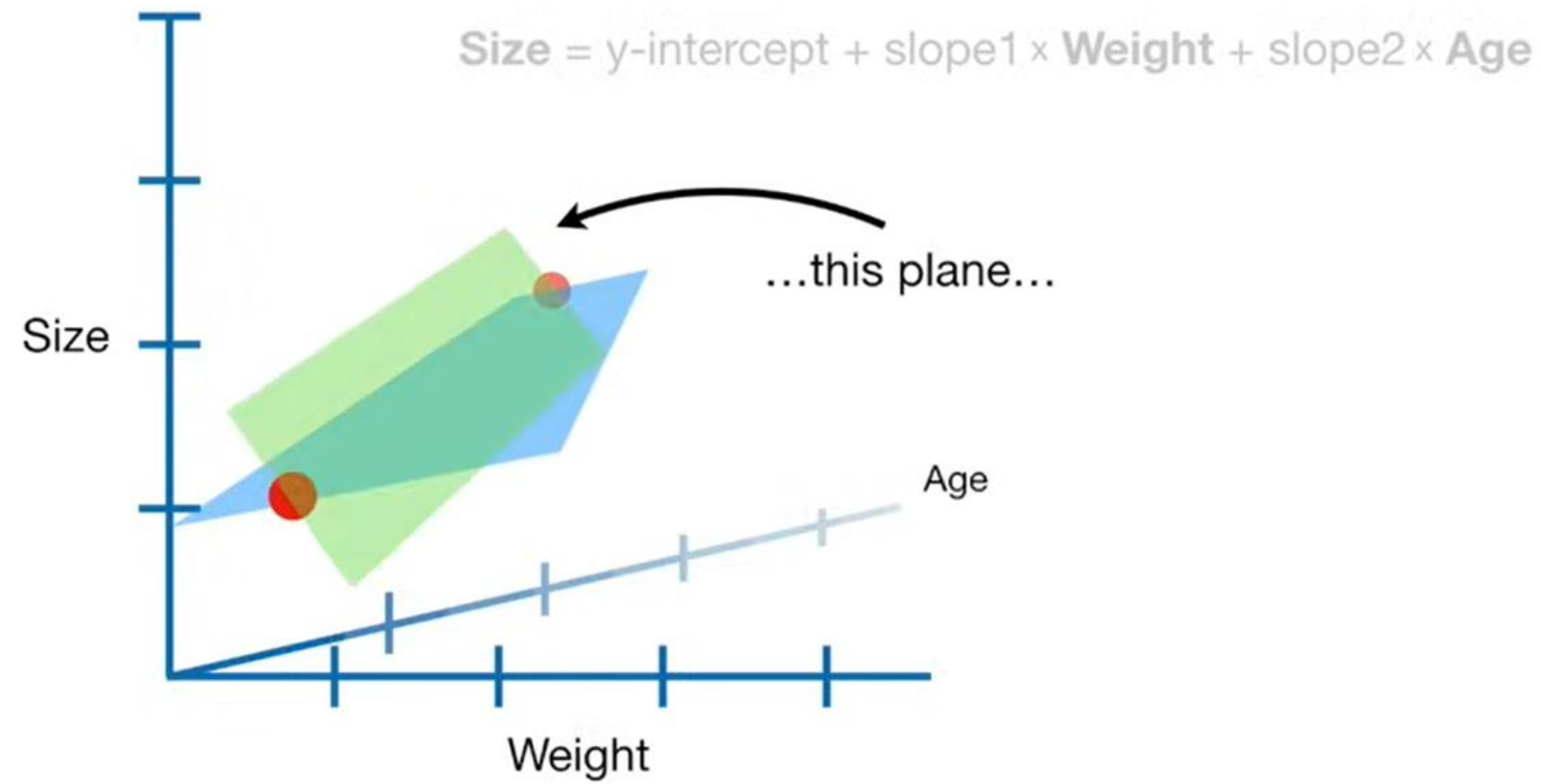
That's because in three dimensions, which is what we get when we add another axis to our graph for **Age**, we have to fit a plane to the data instead of just a line...

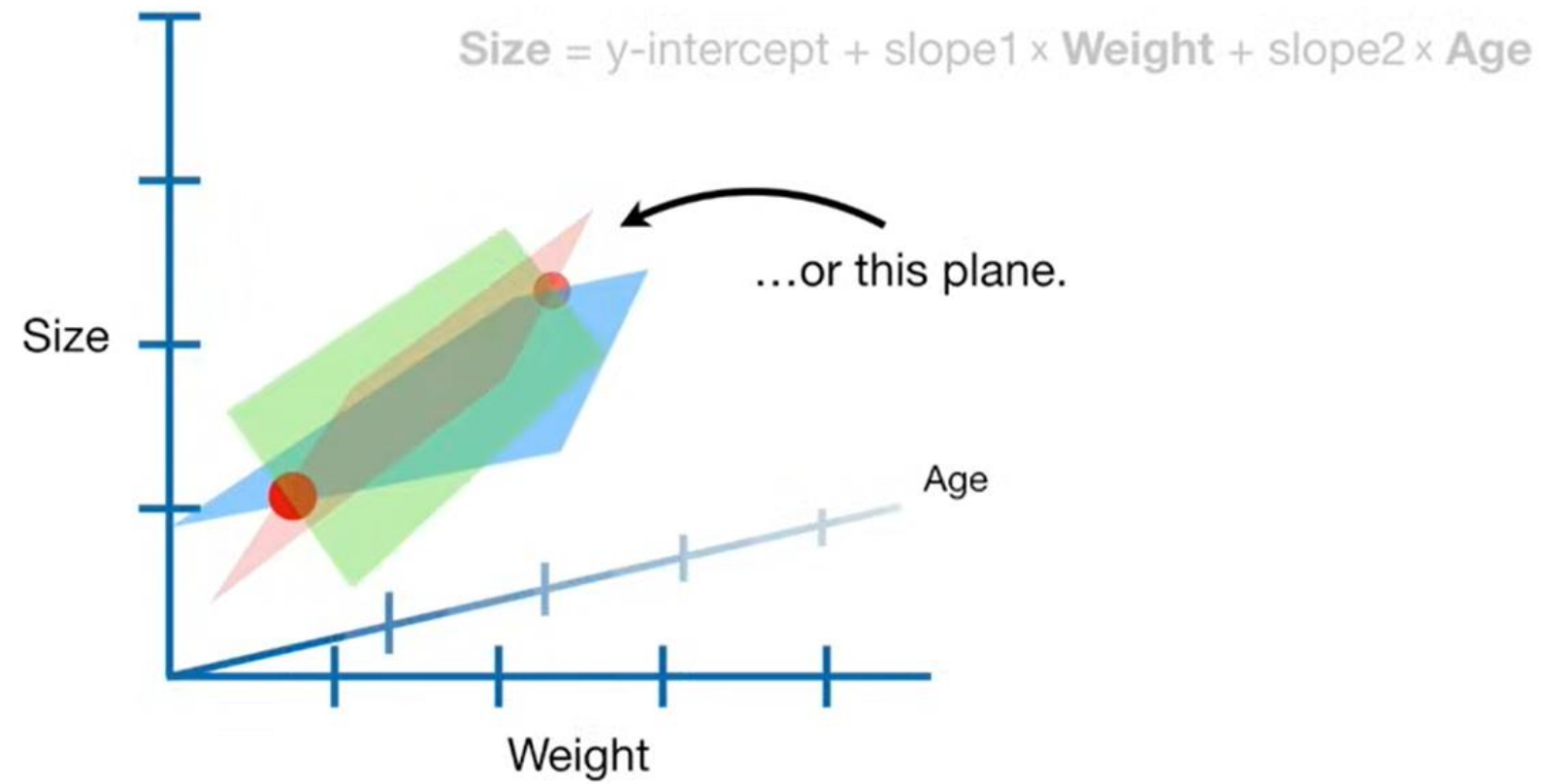


$$\text{Size} = \text{y-intercept} + \text{slope1} \times \text{Weight} + \text{slope2} \times \text{Age}$$

...and with only two data points, there's no reason why this plane fits the data any better than...

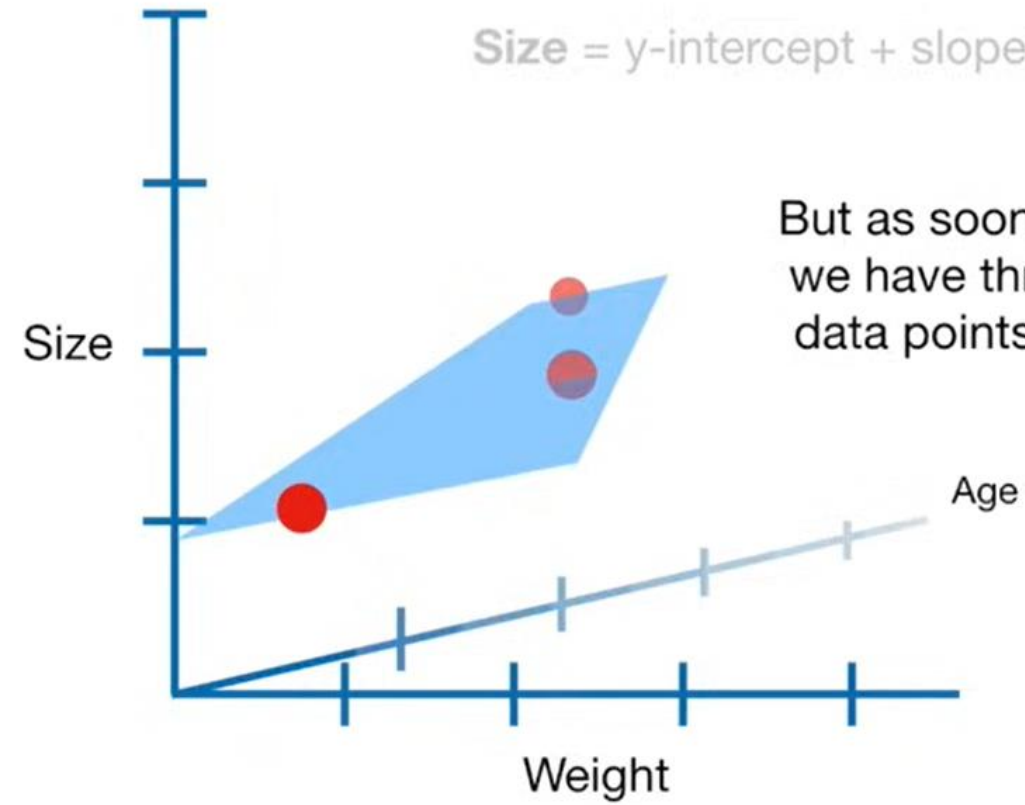


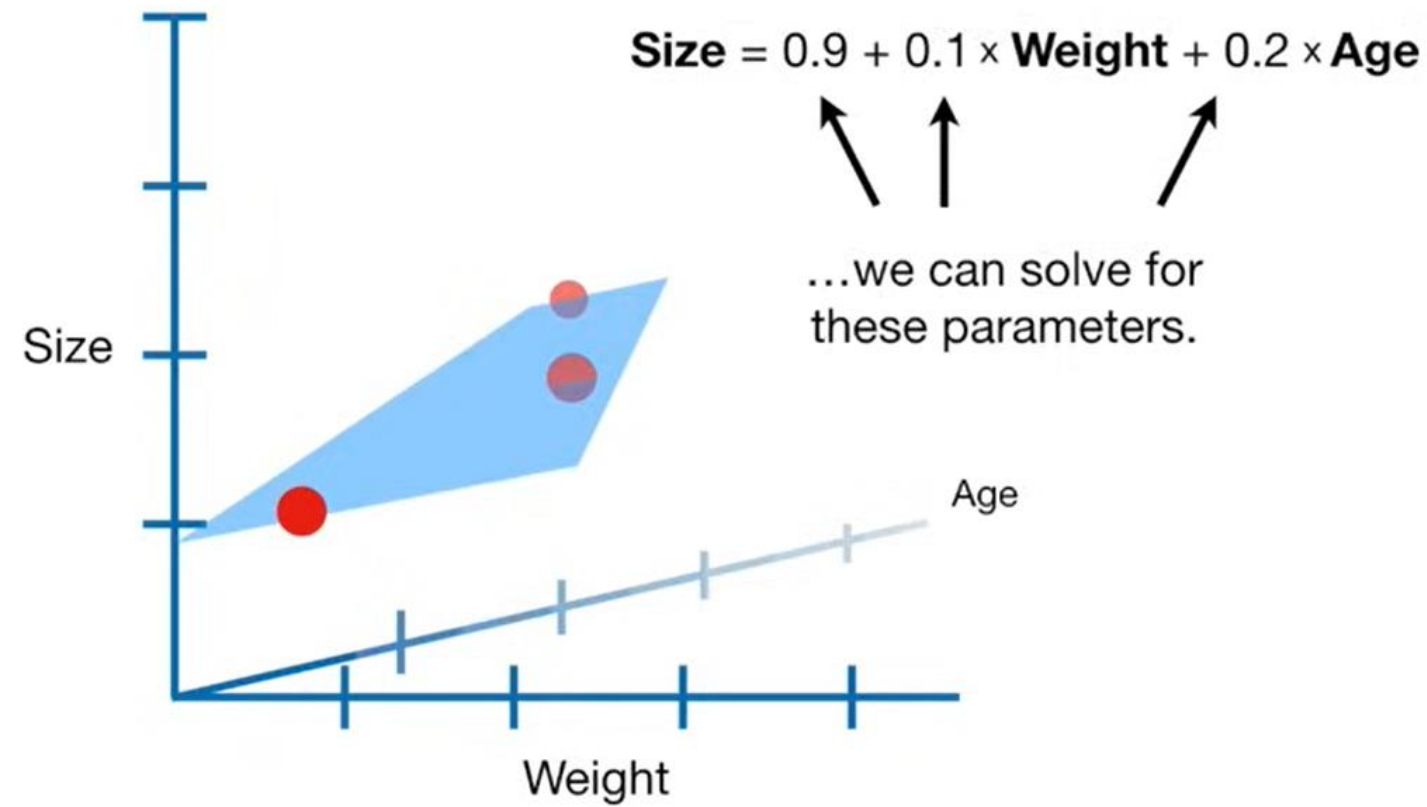




$$\text{Size} = \text{y-intercept} + \text{slope1} \times \text{Weight} + \text{slope2} \times \text{Age}$$

But as soon as
we have three
data points...





$$\text{Size} = \text{y-intercept} + \text{slope1} \times \text{Weight} + \text{slope2} \times \text{Age} + \text{slope3} \times \text{fur color}$$



If we have an equation with four
parameters...

$$\text{Size} = \text{y-intercept} + \text{slope1} \times \text{Weight} + \text{slope2} \times \text{Age} + \text{slope3} \times \text{fur color}$$



If we have an equation with four
parameters...

...then **Least Squares** needs at least four
data points to estimate all four
parameters.

$$\text{Size} = \text{y-intercept} + \text{slope1} \times \text{Weight} + \text{slope2} \times \text{Age} + \text{slope3} \times \text{fur color} + \dots$$



And if we have an equation with 10,001
parameters...

$$\text{Size} = \text{y-intercept} + \text{slope1} \times \text{Weight} + \text{slope2} \times \text{Age} + \text{slope3} \times \text{fur color} + \dots$$



And if we have an equation with 10,001
parameters...

...then we need at least 10,001 data
points to estimate all of the parameters.

Size = y-intercept + slope1 × **Weight** + slope2 × **Age** + slope3 × **fur color** + ...

An equation with 10,001 parameters might sound bonkers, but it's more common than you might expect.

$$\mathbf{Size} = \text{y-intercept} + \text{slope1} \times \mathbf{Gene1} + \text{slope2} \times \mathbf{Gene2} + \dots + \text{slope10000} \times \mathbf{Gene10000}$$

For example, we might use gene expression measurements from 10,000 genes to predict **Size**...

$$\mathbf{Size} = \text{y-intercept} + \text{slope1} \times \mathbf{Gene1} + \text{slope2} \times \mathbf{Gene2} + \dots + \text{slope10000} \times \mathbf{Gene10000}$$

Unfortunately, collecting gene expression measurements from 10,001 mice is crazy expensive and time consuming right now.

$$\mathbf{Size} = \text{y-intercept} + \text{slope1} \times \mathbf{Gene1} + \text{slope2} \times \mathbf{Gene2} + \dots + \text{slope10000} \times \mathbf{Gene10000}$$

Unfortunately, collecting gene expression measurements from 10,001 mice is crazy expensive and time consuming right now.

In practice, a “huge” dataset might have measurements from 500 mice.

$$\mathbf{Size} = \text{y-intercept} + \text{slope1} \times \mathbf{Gene1} + \text{slope2} \times \mathbf{Gene2} + \dots + \text{slope10000} \times \mathbf{Gene10000}$$

So, what do we do if we have an equation with 10,001 parameters and only 500 data points???

$$\mathbf{Size} = \text{y-intercept} + \text{slope1} \times \mathbf{Gene1} + \text{slope2} \times \mathbf{Gene2} + \dots + \text{slope10000} \times \mathbf{Gene10000}$$

So, what do we do if we have an equation with 10,001 parameters and only 500 data points???

We use **Ridge Regression!!!**

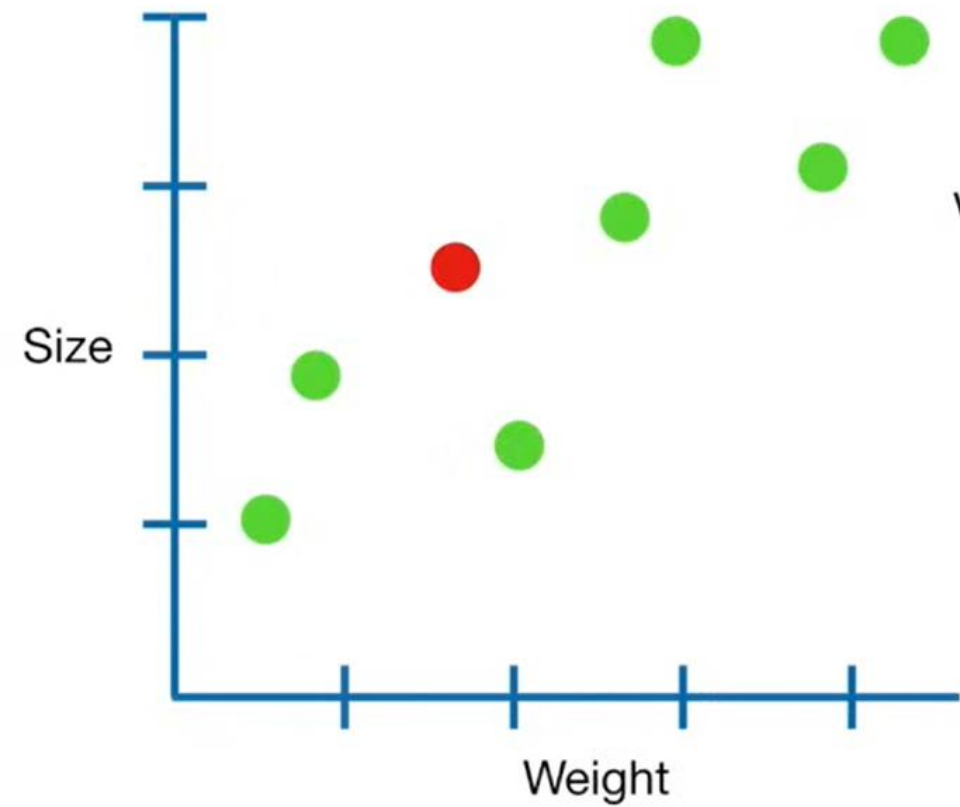
$$\text{Size} = \text{y-intercept} + \text{slope1} \times \text{Gene1} + \text{slope2} \times \text{Gene2} + \dots + \text{slope10000} \times \text{Gene10000}$$

It turns out that by adding the **Ridge Regression Penalty**,
we can solve for all 10,001 parameters with only 500 (or
even fewer) samples.

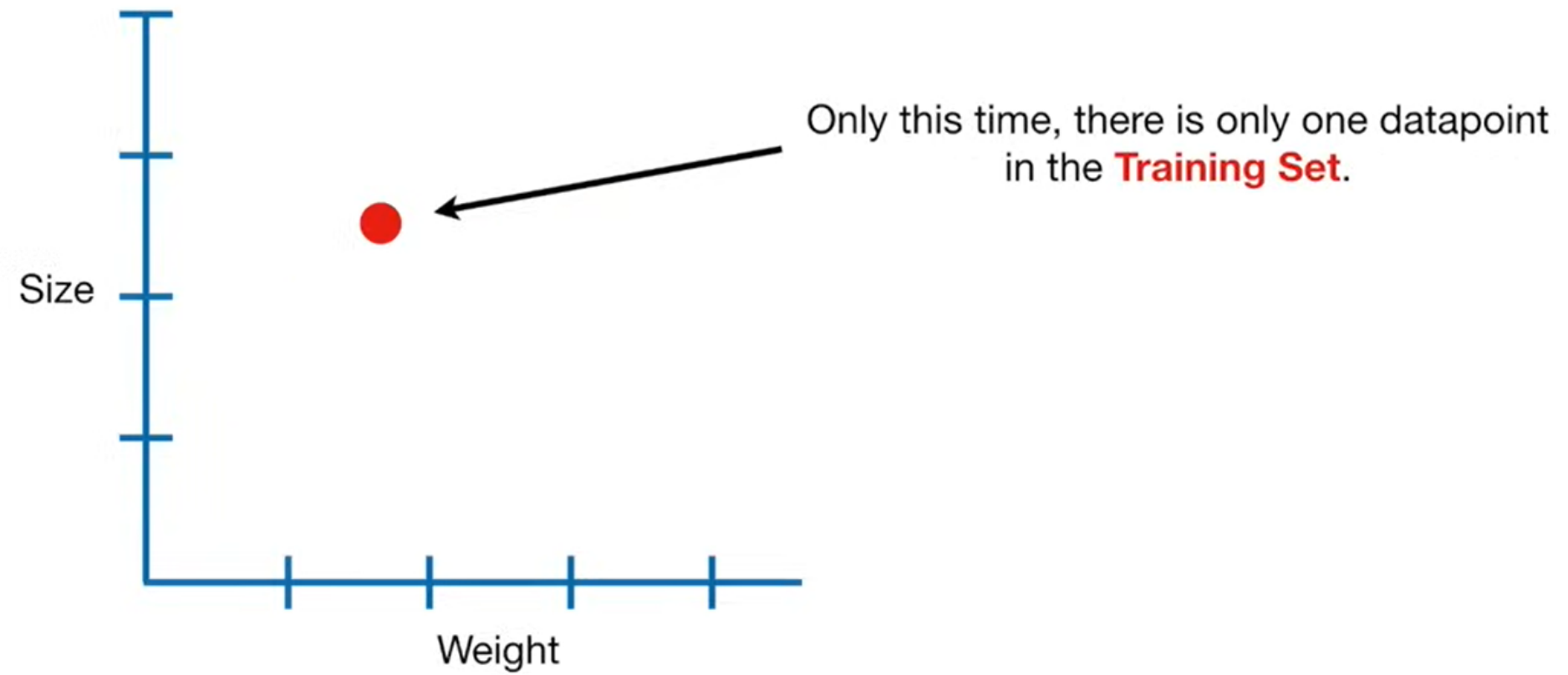
the sum of the squared residuals

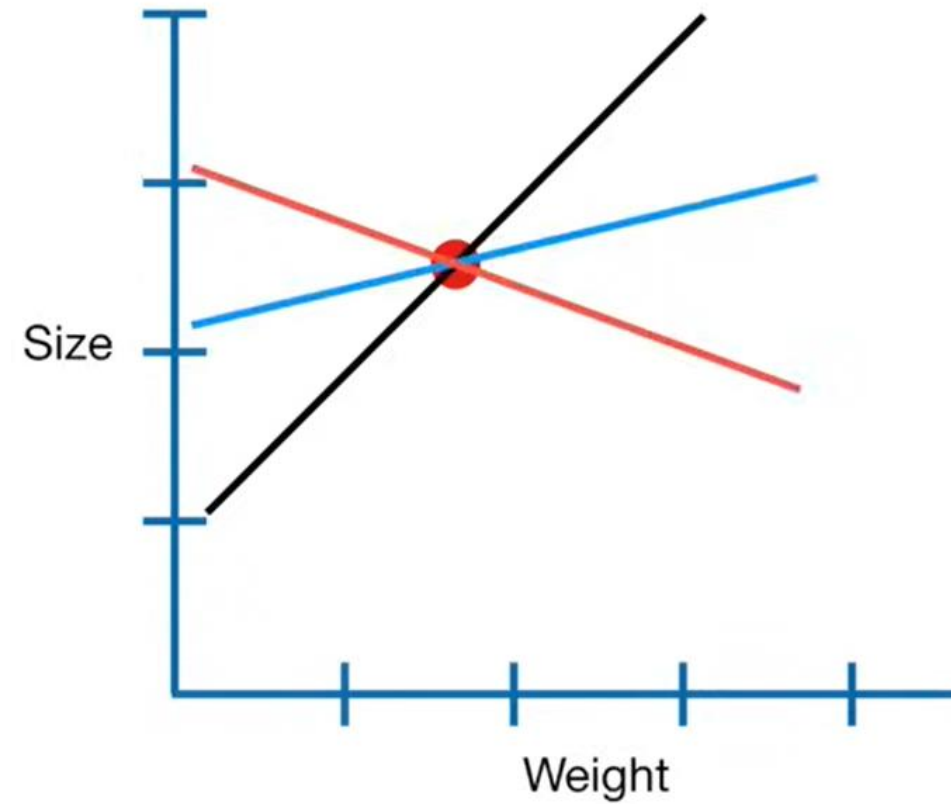
+

$$\lambda \times (\text{slope1}^2 + \text{slope2}^2 + \text{slope3}^2 + \dots + \text{slope10000}^2)$$

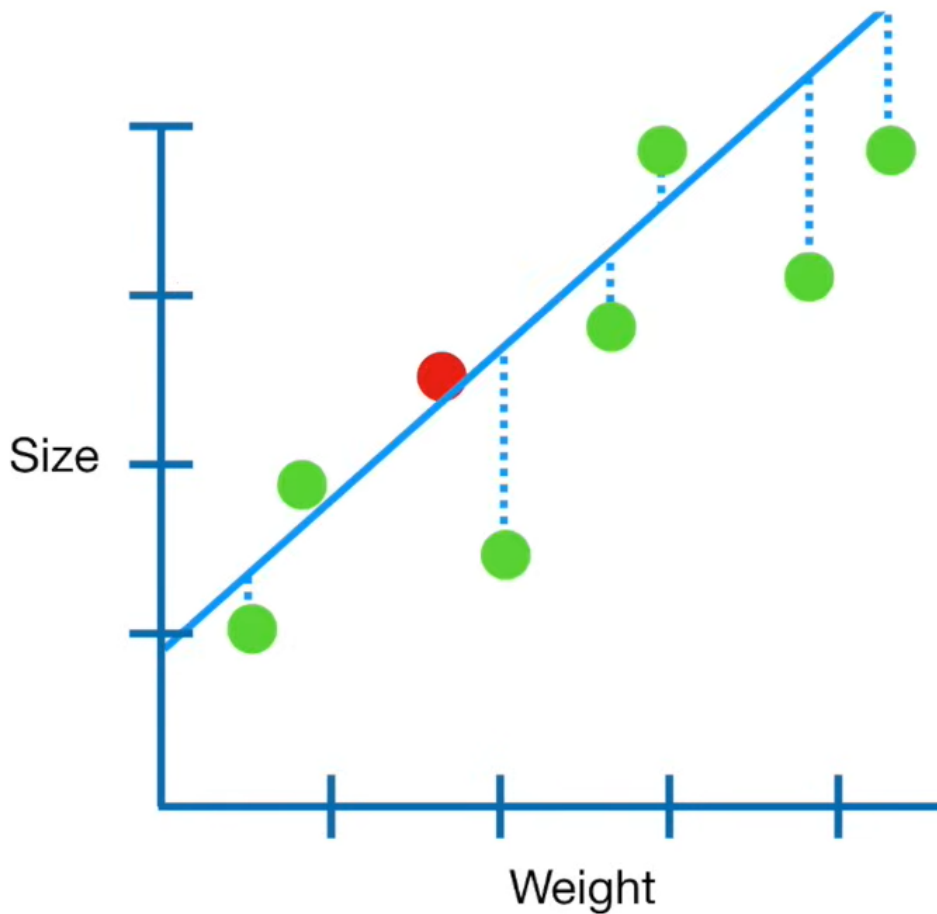


One way to think about how **Ridge Regression** can solve for parameters when there isn't enough data is to go back to our original **Size** vs **Weight** example.





Least Squares can't find a single optimal solution, since any line that goes through the dot will minimize the sum of the squared residuals...



...but **Ridge Regression** can find a solution with **Cross Validation** and the **Ridge Regression Penalty** that favors smaller parameter values.

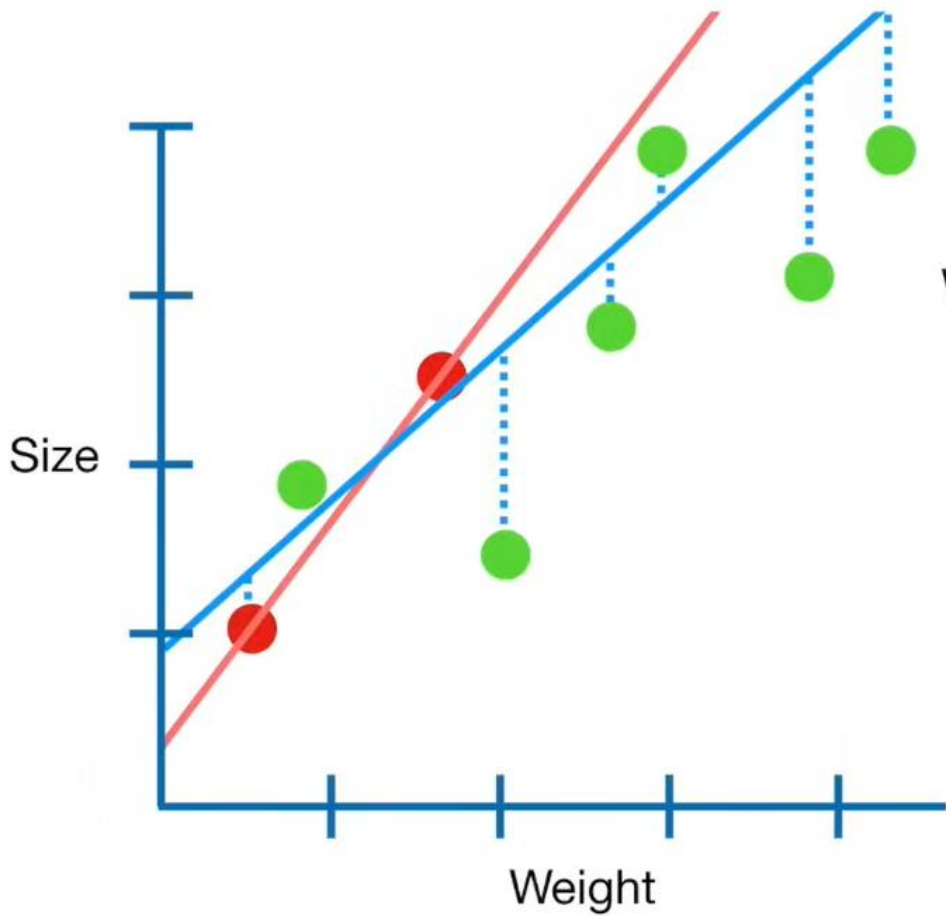


the sum of the squared residuals

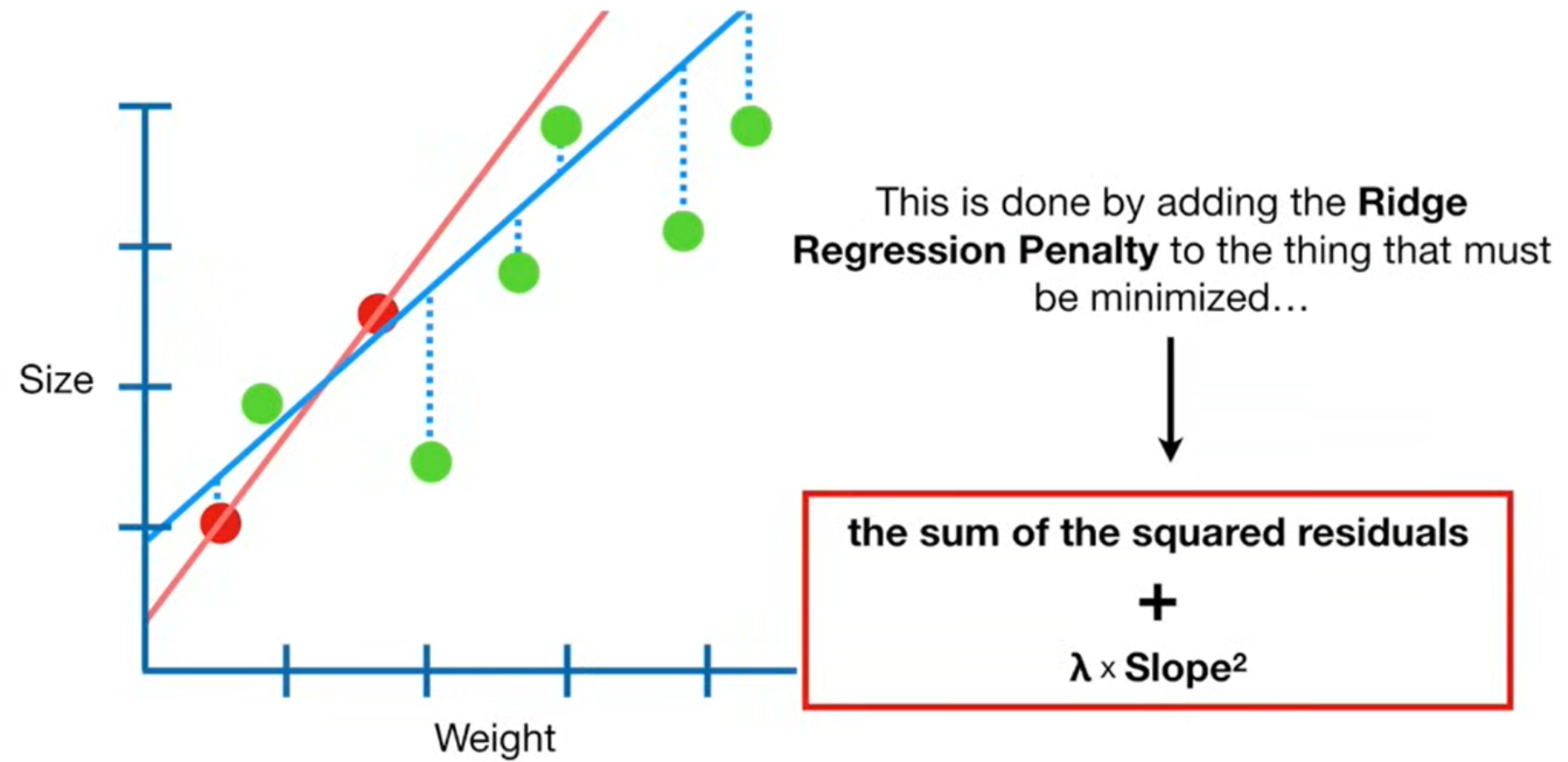
+

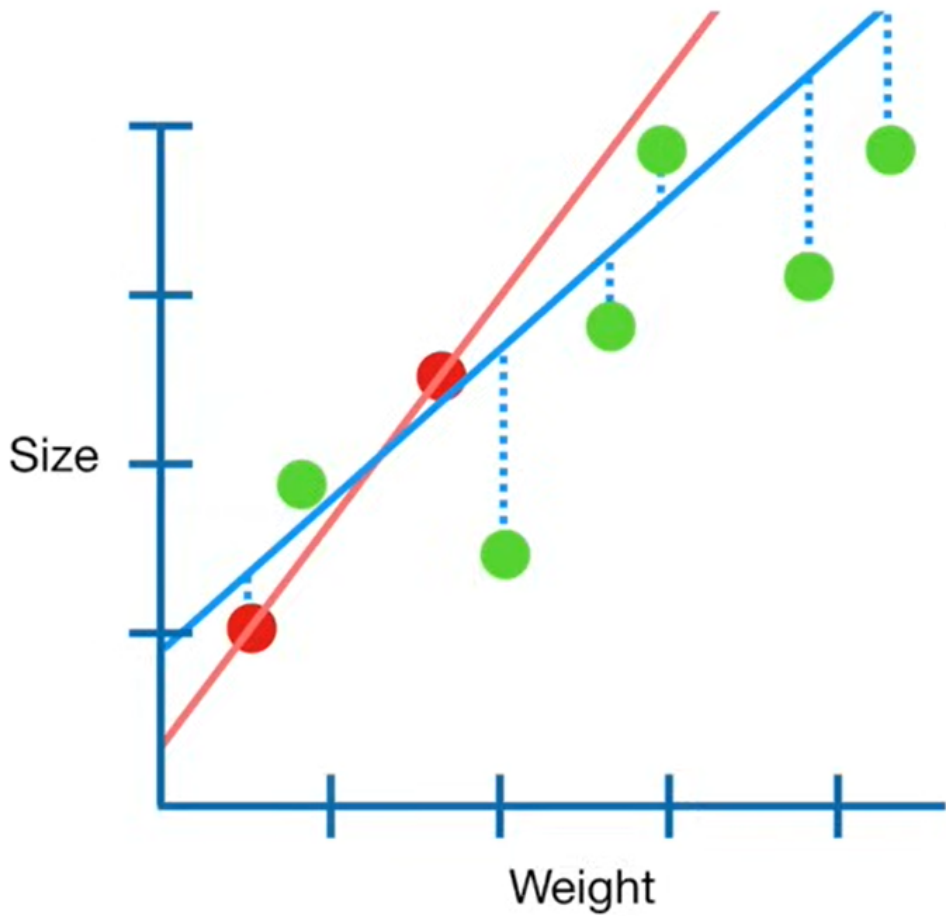
$$\lambda \times \text{Slope}^2$$

In Summary:



When the sample sizes are relatively small, then **Ridge Regression** can improve predictions made from new data (i.e. reduce **Variance**) by making the predictions less sensitive to the **Training Data**.





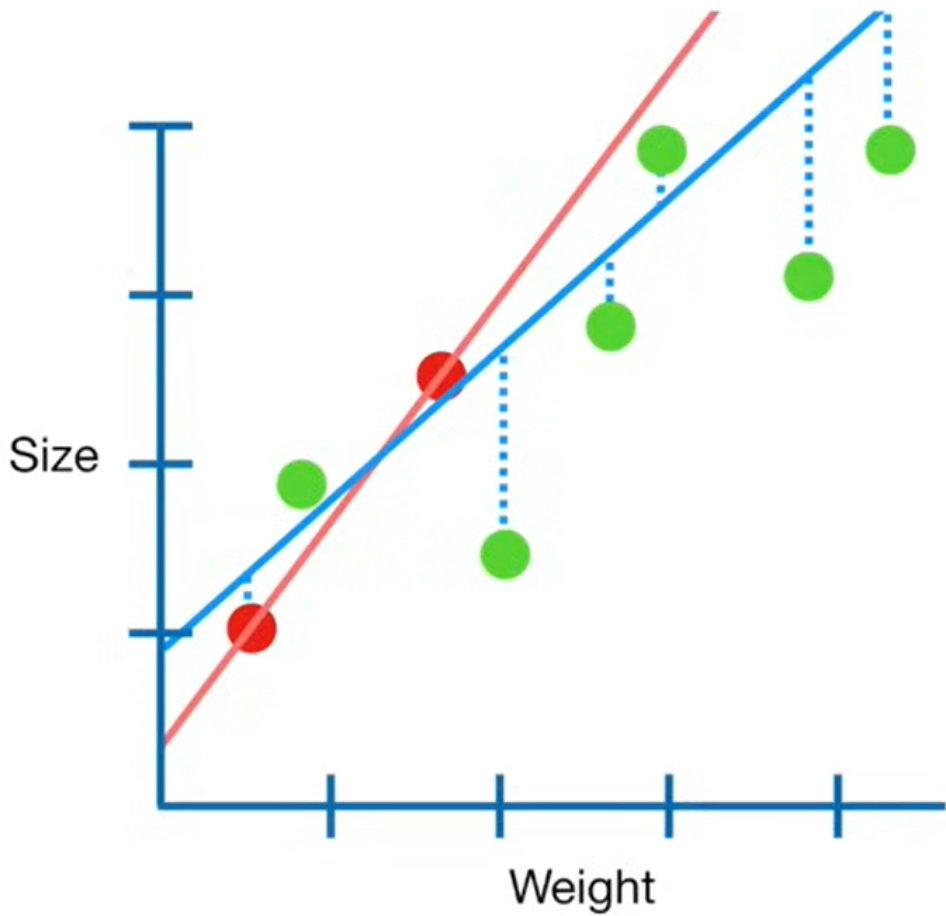
The **Ridge Regression Penalty** itself is λ times the sum of all squared parameters, except for the y-intercept...



the sum of the squared residuals

+

$\lambda \times \text{Slope}^2$



...and λ is determined using **Cross Validation**.



the sum of the squared residuals

+

$\lambda \times \text{Slope}^2$

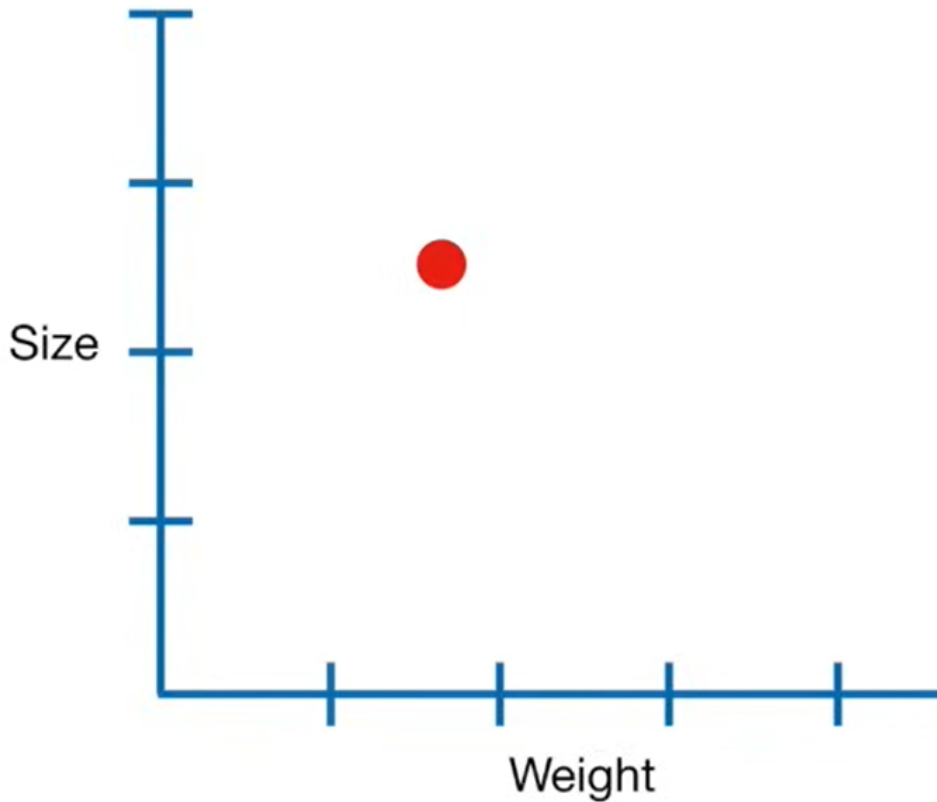
Lastly, even when there isn't enough data to find the **Least Squares** parameter estimates, **Ridge Regression** can still find a solution with **Cross Validation** and the **Ridge Regression Penalty**.

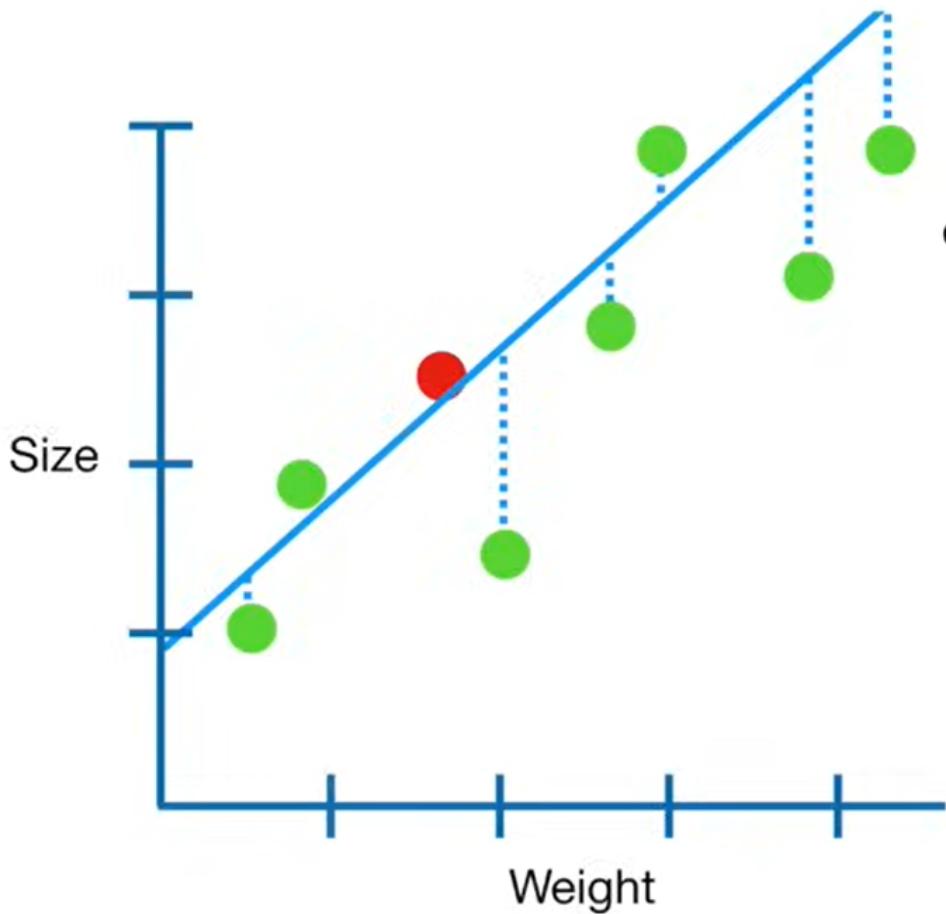


the sum of the squared residuals

+

$\lambda \times \text{Slope}^2$





Lastly, even when there isn't enough data to find the **Least Squares** parameter estimates, **Ridge Regression** can still find a solution with **Cross Validation** and the **Ridge Regression Penalty**.



the sum of the squared residuals

+

$\lambda \times \text{Slope}^2$